

Depression Connect Forum

Master Analysis Report

Dataset variant: Combined Dataset

EDA · Time series · CDS prevalence · LIWC · Longitudinal trends · LIWC-22 validation

1. EDA - All Users

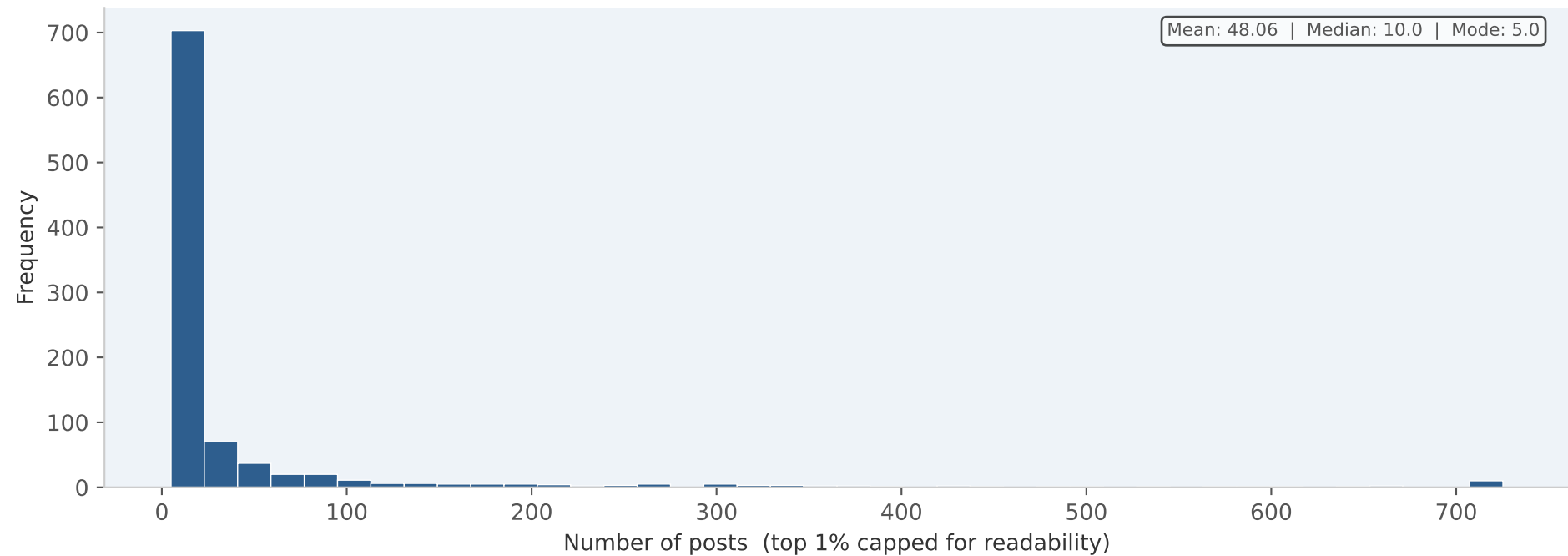
Depression Connect Forum

Exploratory Data Analysis Report — All Users

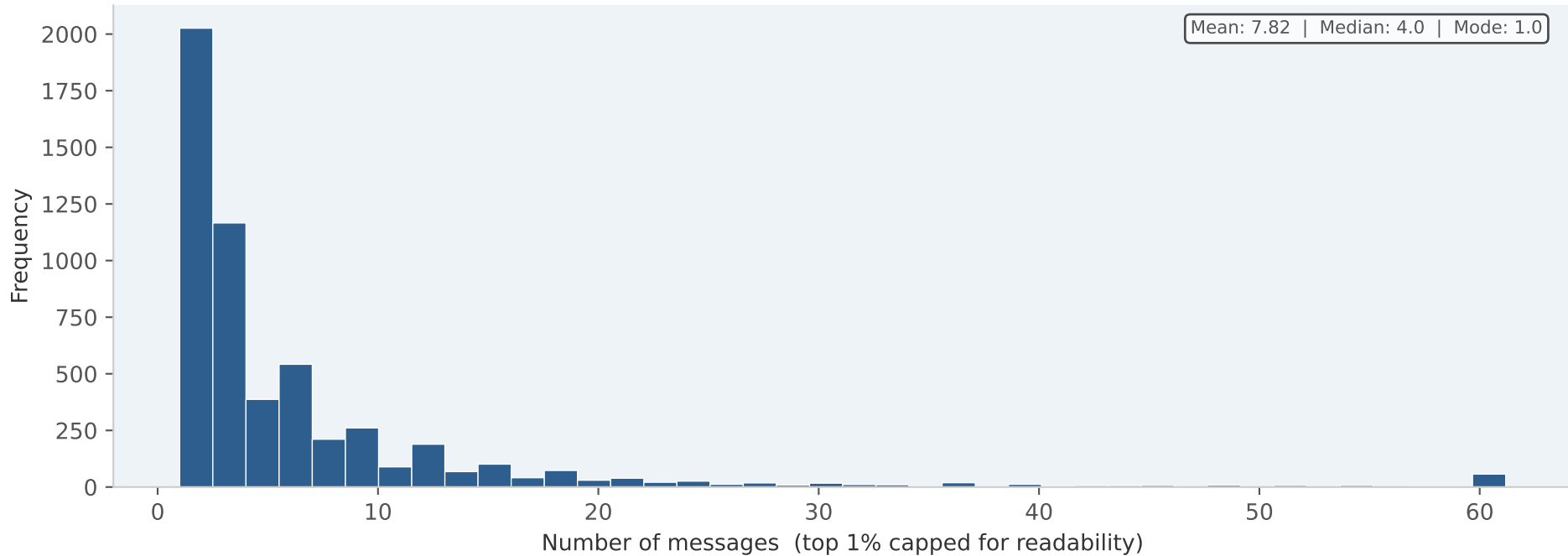
Summary Statistics

Metric	Mean	Median	Mode	Min	Max
Posts per user	48.06	10.0	5.0	5	1656
Messages per thread (total)	7.82	4.0	1.0	1	605
Replies per thread	8.67	4.0	1.0	1	604
Activity span (days)	202.34	51.0	0.0	0	2493
Active days per user	20.6	5.0	4.0	1	514
Words per post	101.92	65.0	5.0	0	2724
Words per user (total)	4898.54	1099.0	373.0	75	317971
'ik' count per user	210.78	58.0	18.0	0	10088
'ik' % per user	5.23	5.31	5.26	0.0	11.79
'mijn' count per user	44.11	12.0	4.0	0	2451
'mijn' % per user	1.1	1.04	0.0	0.0	4.62

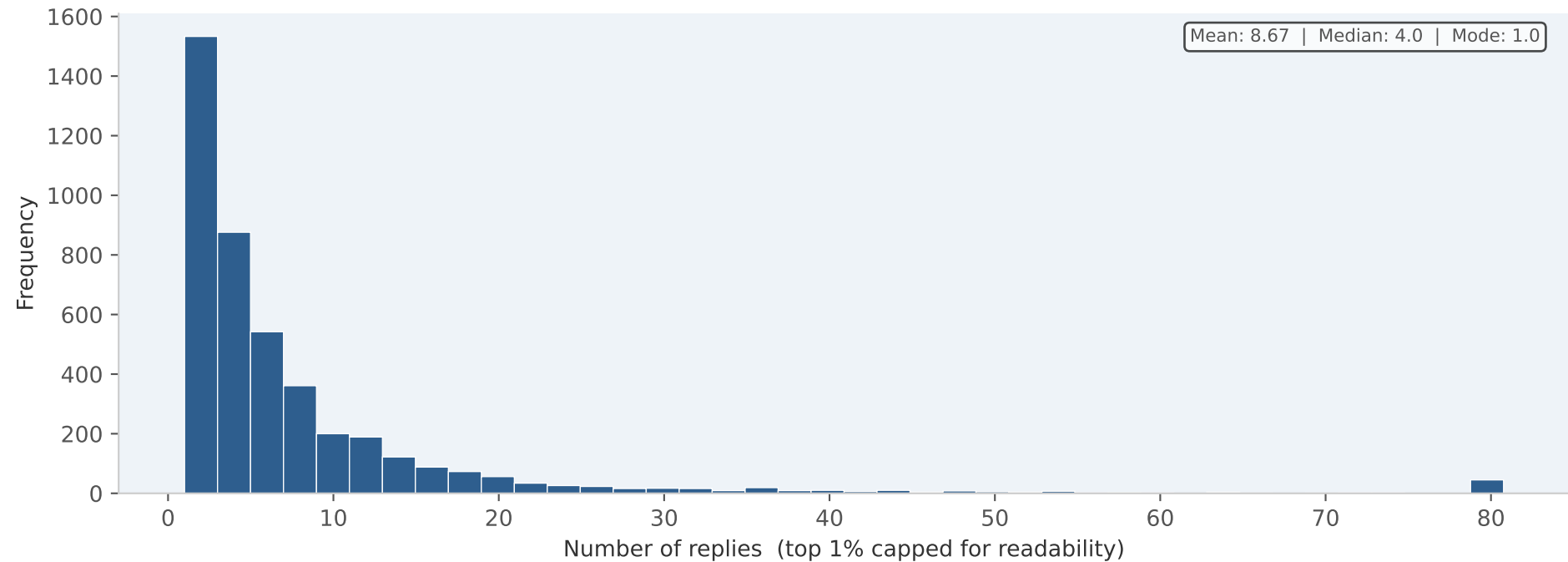
Distribution: Posts per User



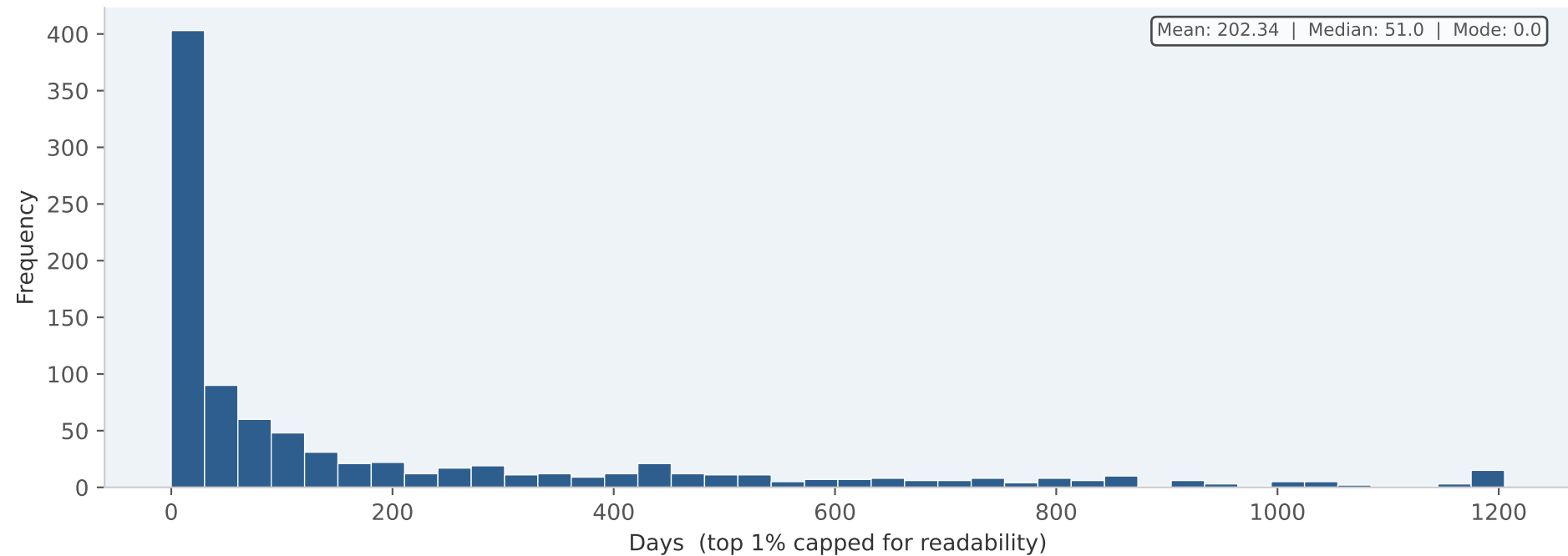
Distribution: Total Messages per Thread



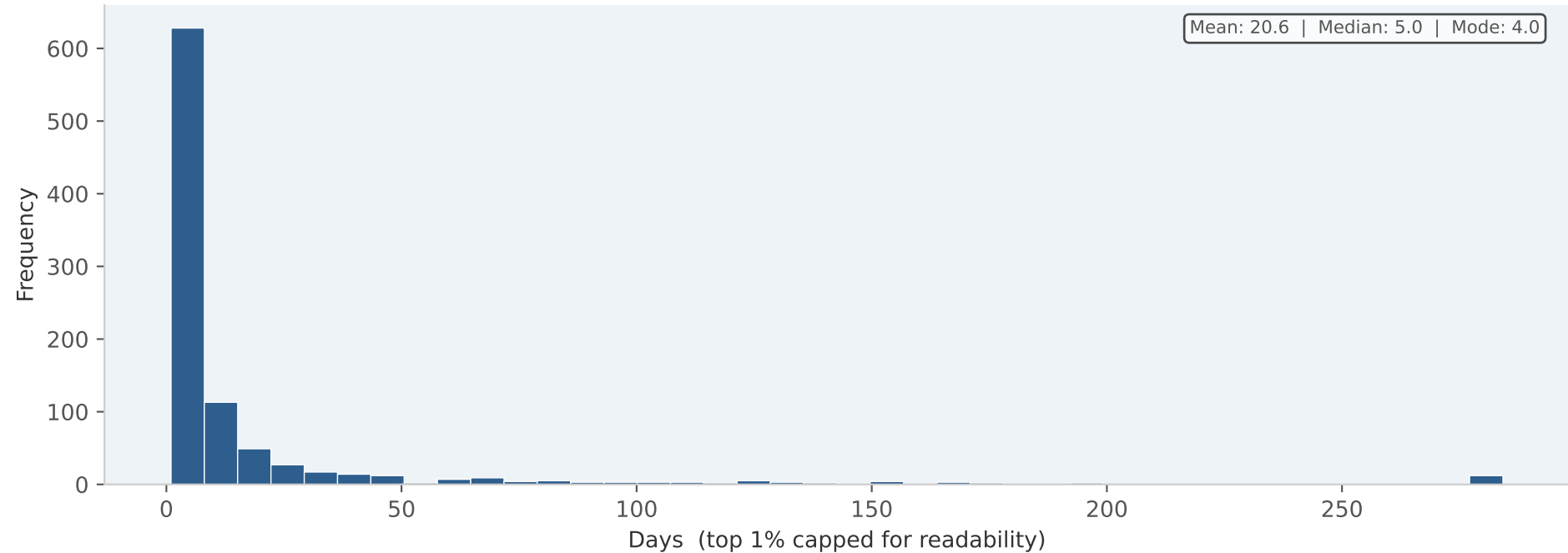
Distribution: Replies per Thread (excluding opening post)



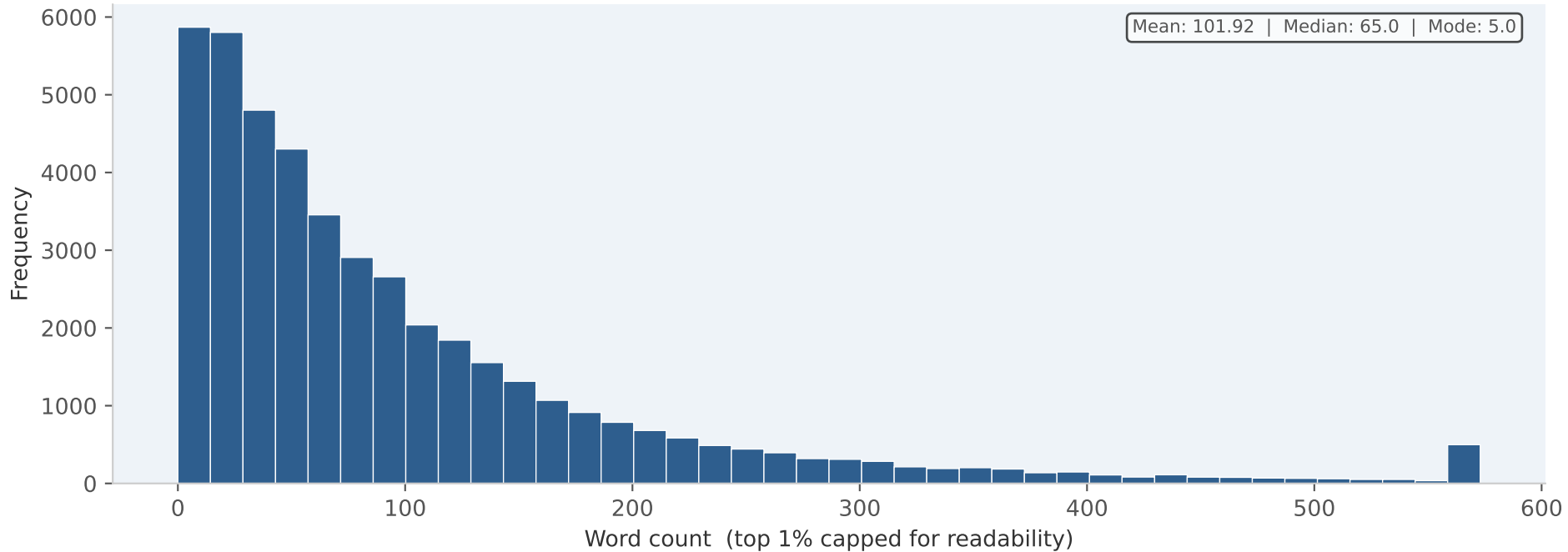
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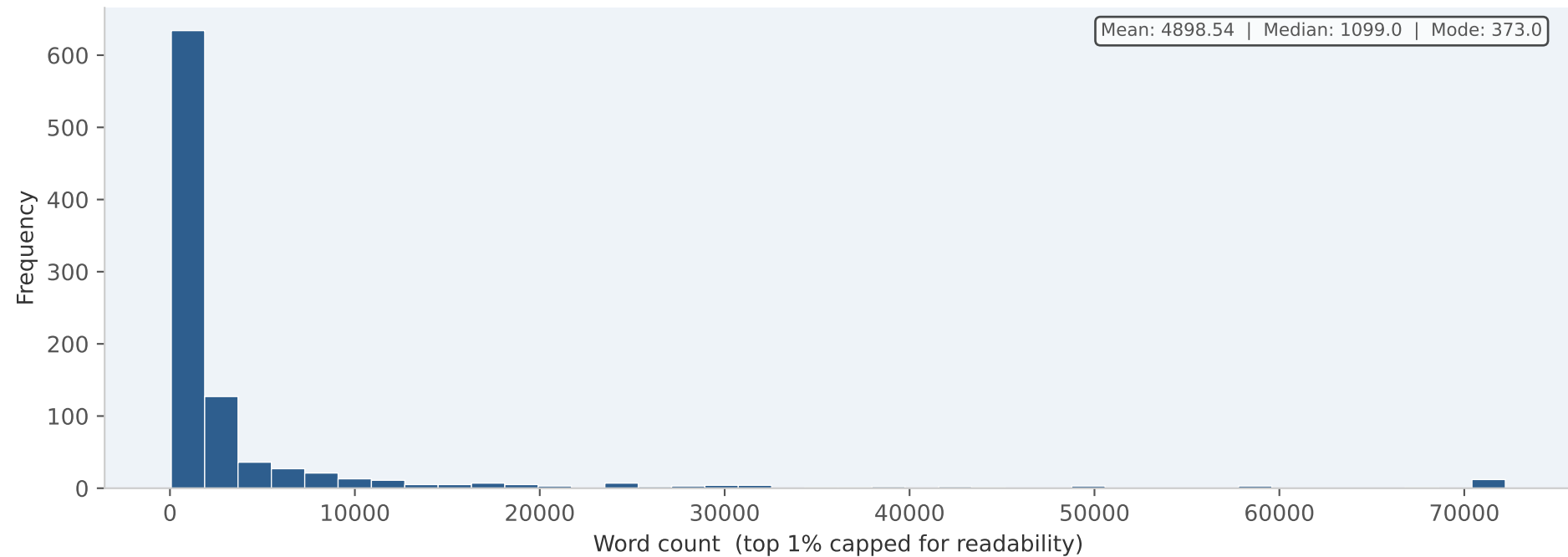
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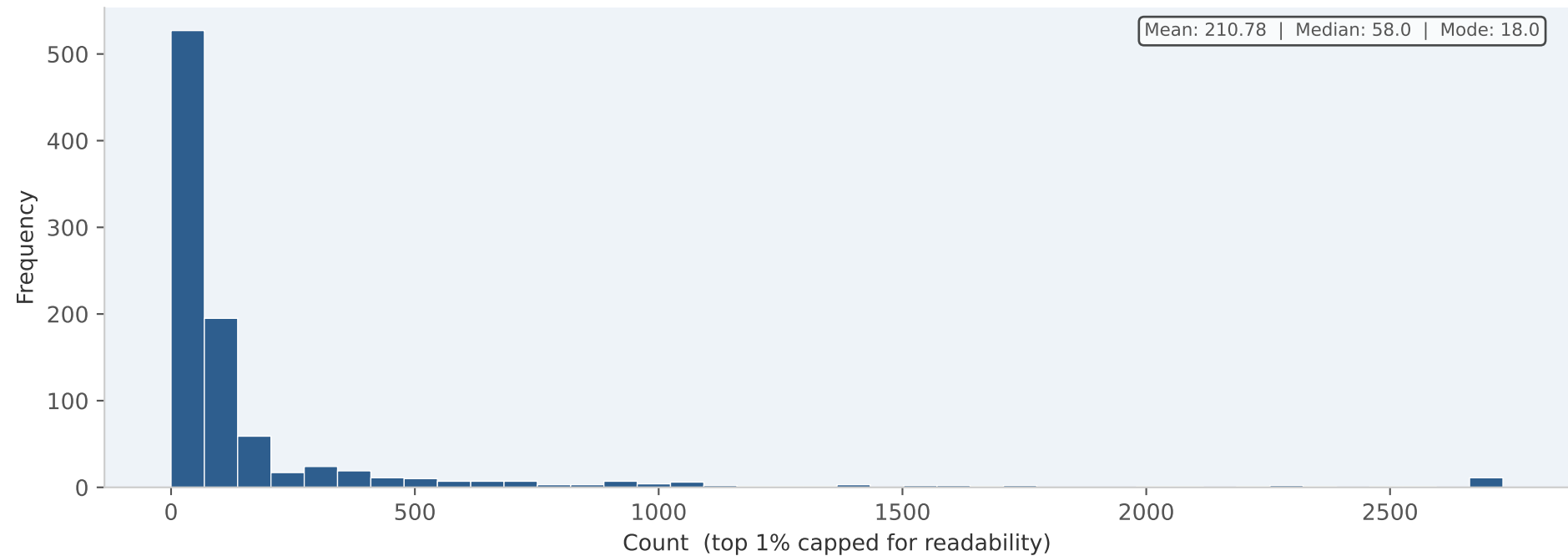
Distribution: Words per Post



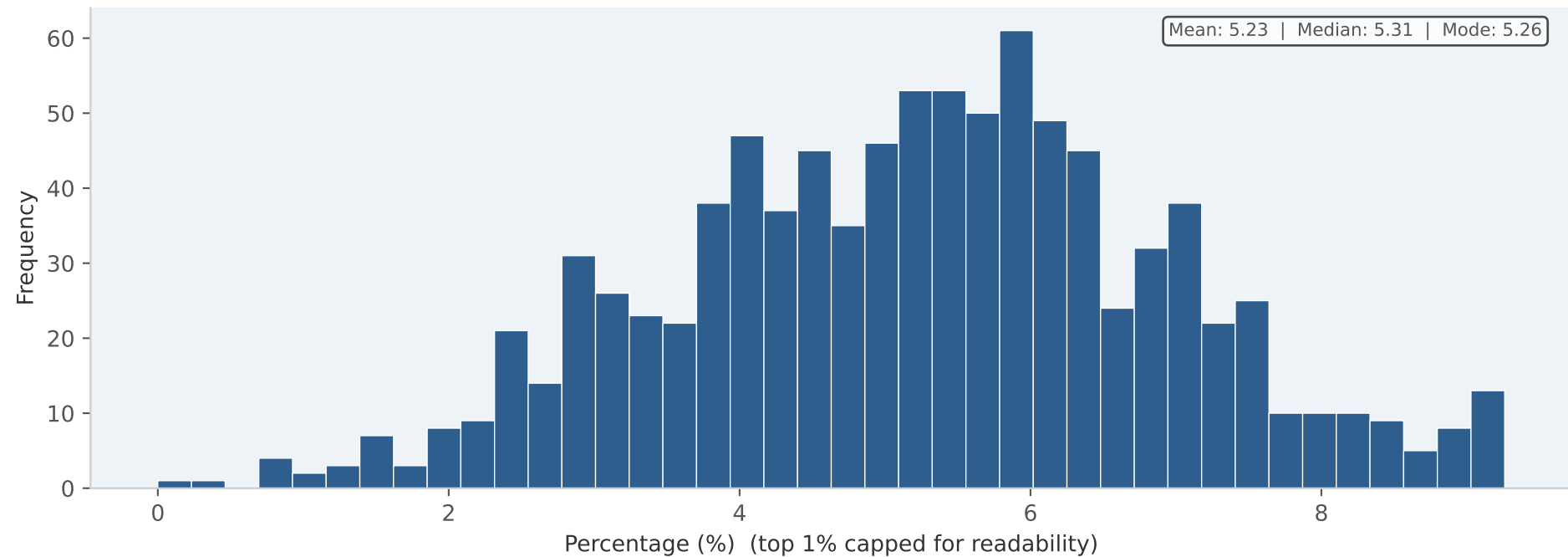
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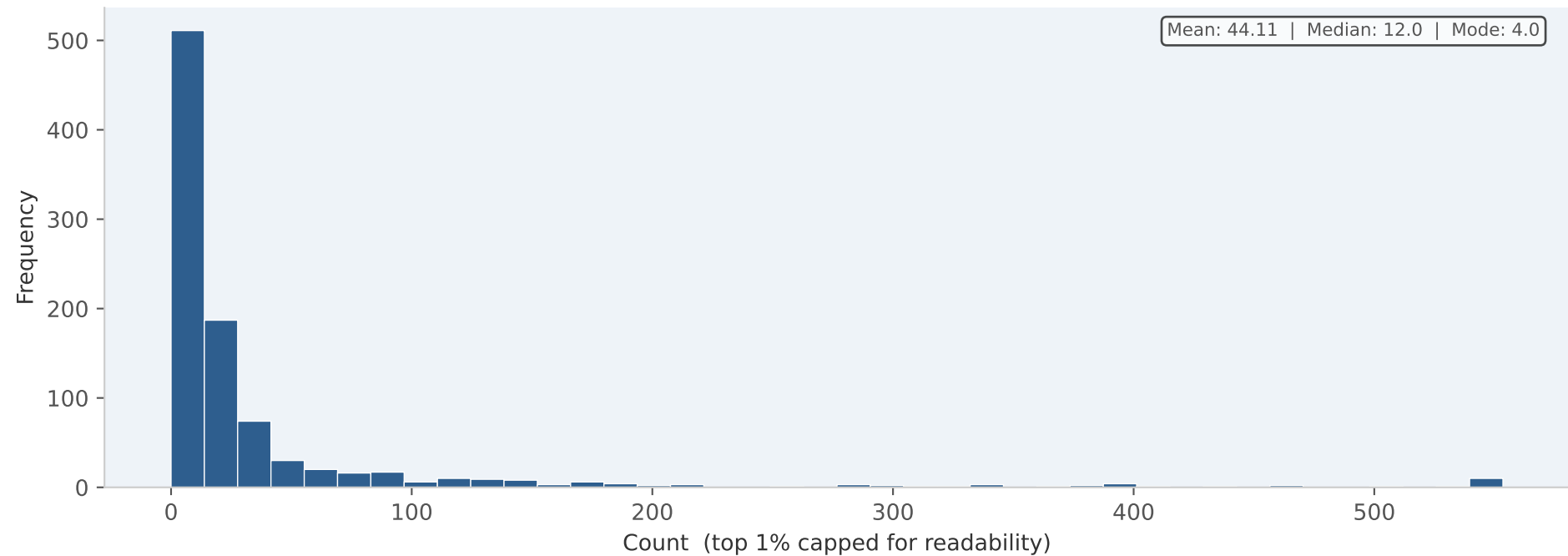
Distribution: 'ik' Count per User



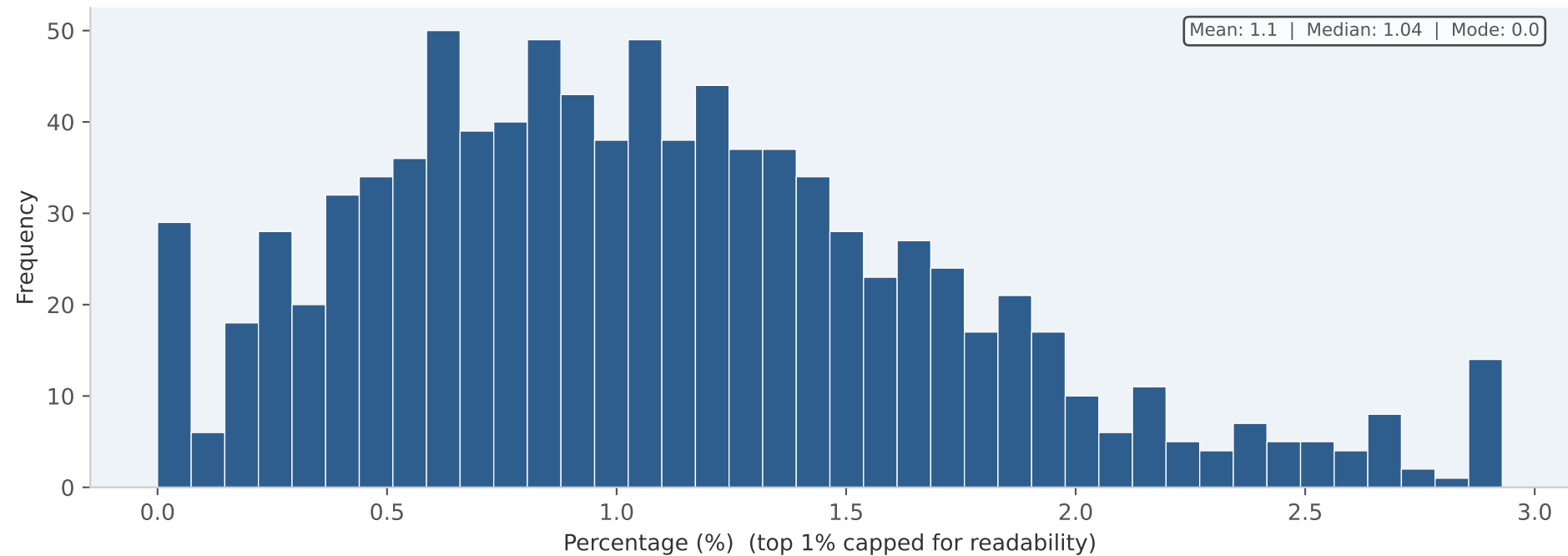
Distribution: 'ik' as % of Total Words per User



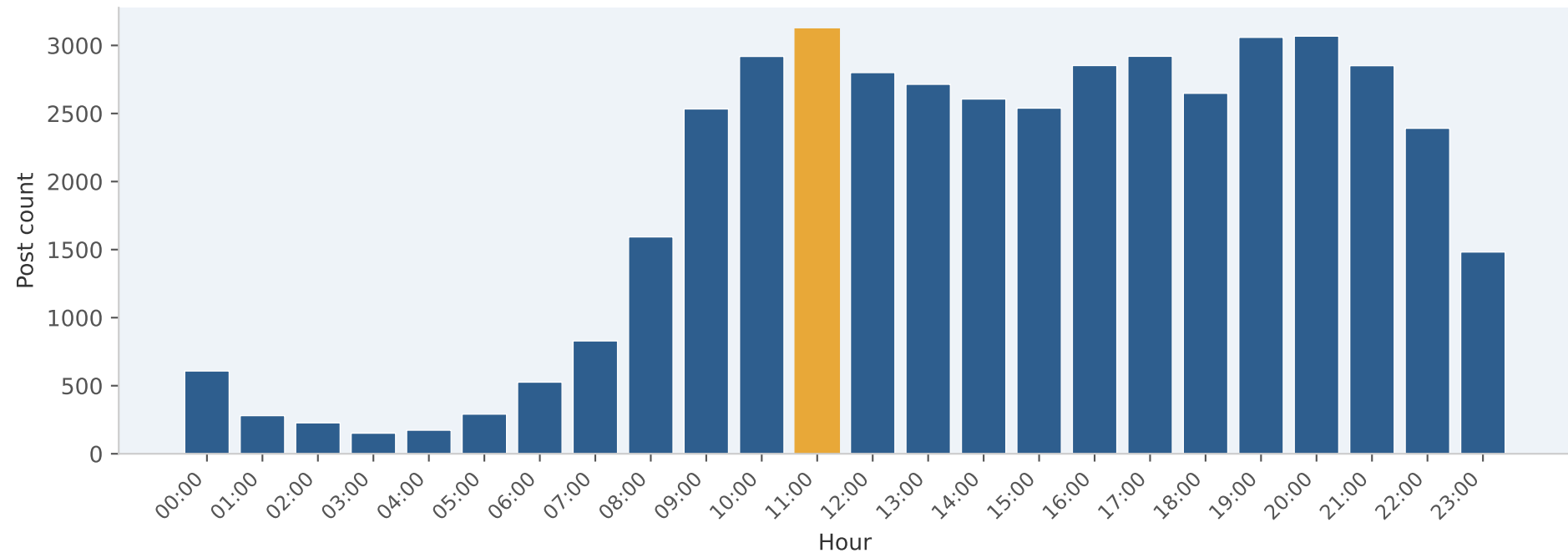
Distribution: 'mijn' Count per User



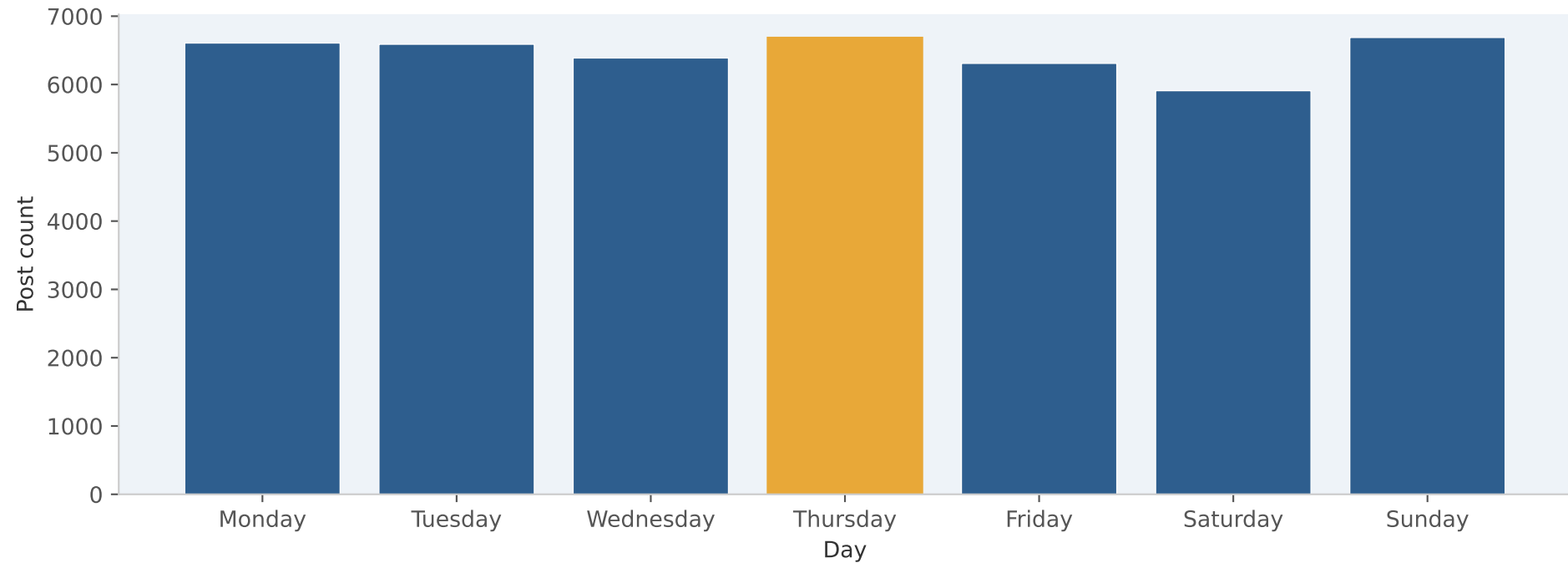
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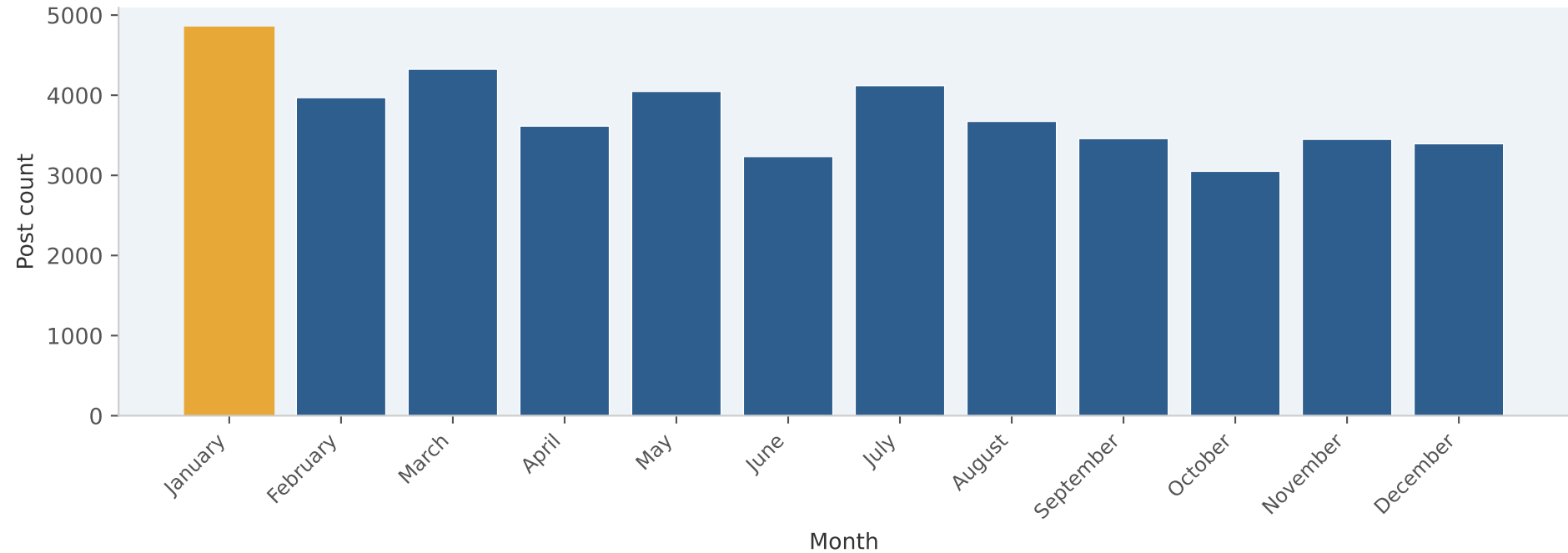
Popular Hours of Day



Popular Days of Week

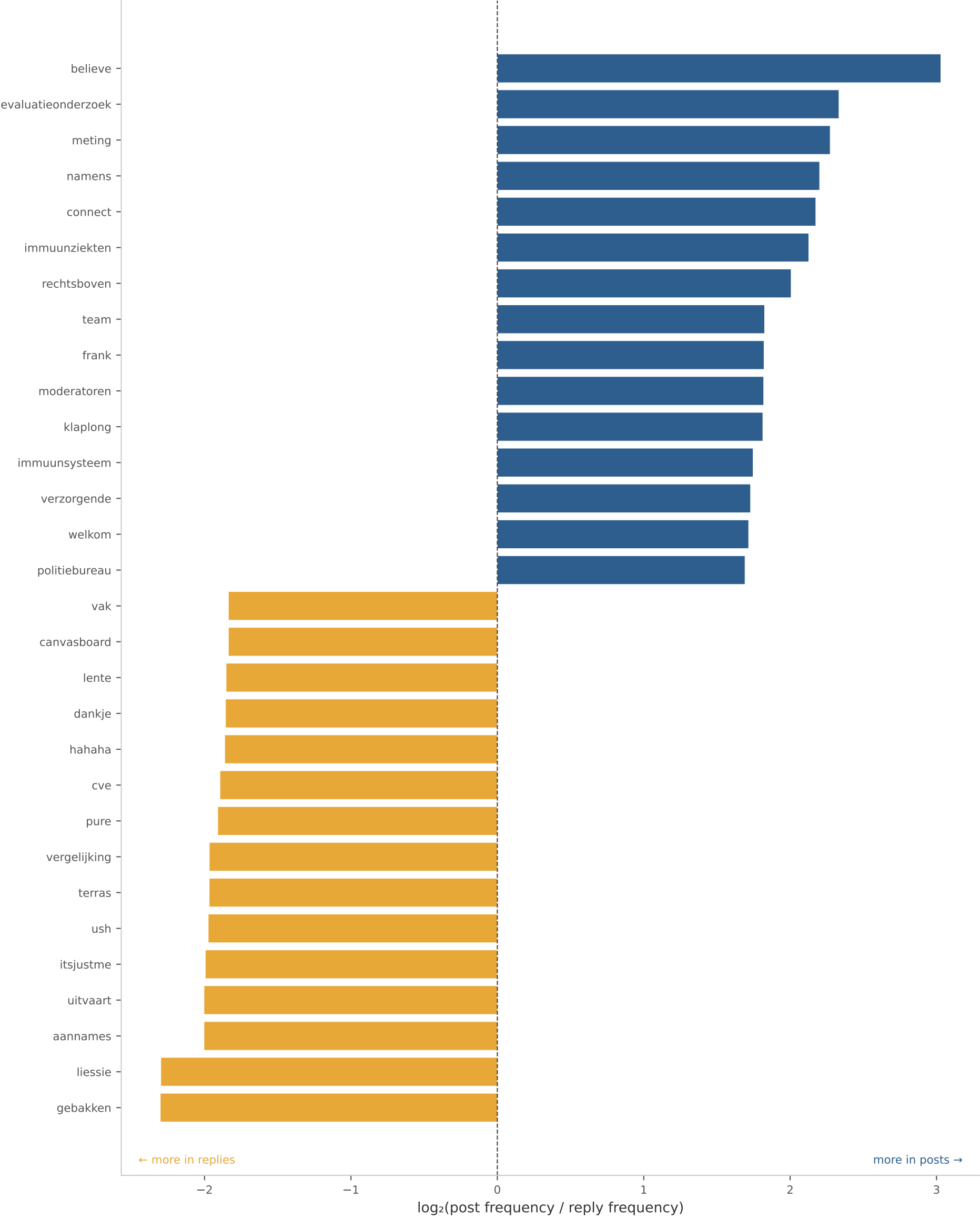


Popular Months



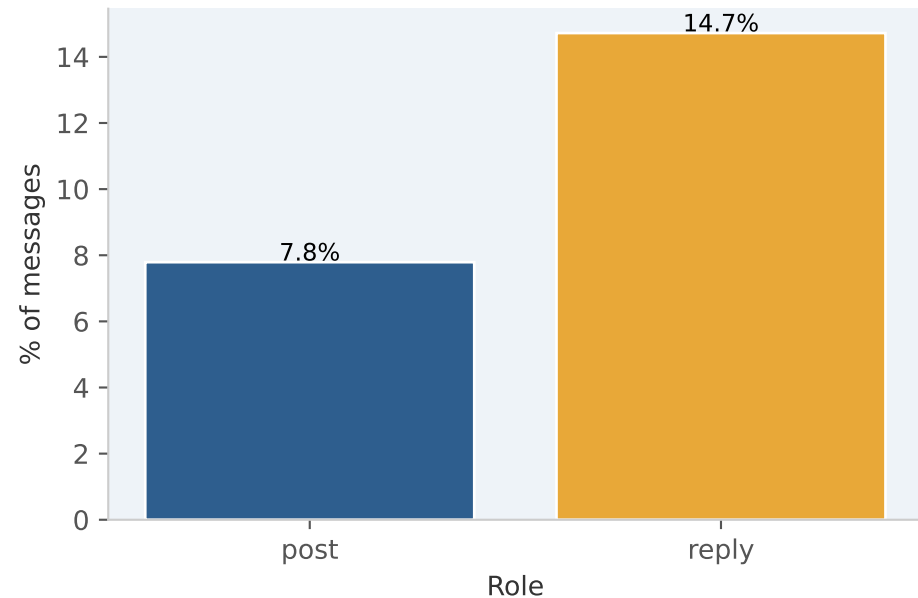
Role-Based Analysis — Posts vs Replies

Word Choice: Initial Posts vs Replies

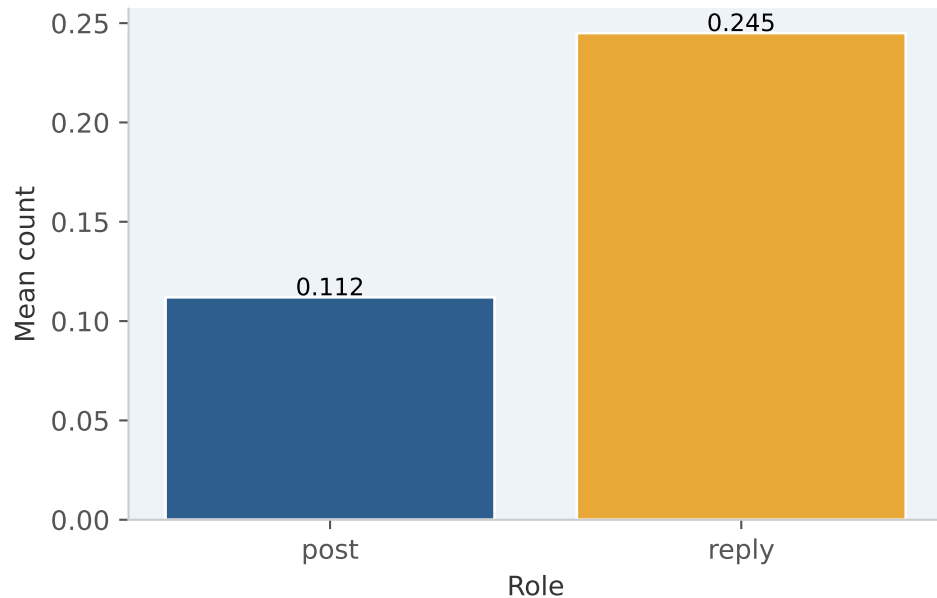


Emoji Use by Message Role

% Messages Containing Emoji



Mean Emoji Count per Message



Top 20 Most Frequent Emojis — Posts vs Replies

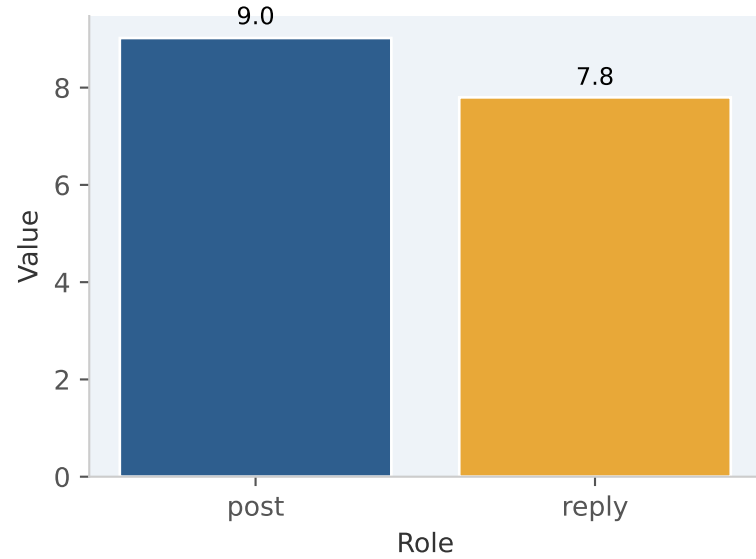
#	Posts emoji	Count	%	Replies emoji	Count	%
1	rose	99	16.1	rose	1005	10.36
2	four leaf clover	58	9.43	four leaf clover	853	8.79
3	folded hands	23	3.74	face blowing a kiss	442	4.55
4	winking face	21	3.41	folded hands	425	4.38
5	people hugging	19	3.09	grinning face with sweat	357	3.68
6	smiling face with smiling eyes	18	2.93	winking face	352	3.63
7	loudly crying face	17	2.76	smiling face with smiling eyes	335	3.45
8	smiling face with open hands	17	2.76	heart suit	332	3.42
9	face blowing a kiss	15	2.44	smiling face with open hands	314	3.24
10	sparkling heart	15	2.44	upside-down face	261	2.69
11	cherry blossom	15	2.44	people hugging	247	2.55
12	thinking face	12	1.95	sparkling heart	193	1.99
13	grinning face with sweat	12	1.95	red heart	169	1.74
14	puzzle piece	12	1.95	thumbs up	164	1.69
15	beaming face with smiling eyes	11	1.79	flexed biceps	153	1.58
16	two hearts	10	1.63	face with tears of joy	148	1.52
17	crying face	9	1.46	cherry blossom	141	1.45
18	pleading face	8	1.3	rolling on the floor laughing	124	1.28
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Top 20 Most Frequent Words — Posts vs Replies

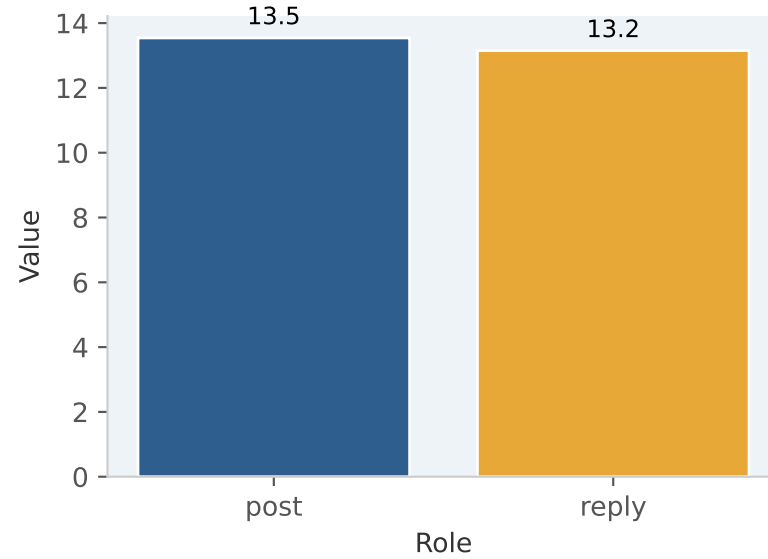
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1	mijn	5904	1.762	mijn	35559	1.726
2	ben	4272	1.275	ben	22775	1.106
3	weer	2850	0.851	weer	18756	0.911
4	goed	2626	0.784	goed	17116	0.831
5	depressie	2616	0.781	heel	16092	0.781
6	veel	2585	0.771	veel	15417	0.749
7	hier	2461	0.734	naar	14674	0.712
8	heel	2400	0.716	over	12021	0.584
9	naar	2381	0.711	hoe	11442	0.556
10	over	1848	0.551	gaat	10958	0.532
11	hoe	1829	0.546	geen	10908	0.53
12	geen	1820	0.543	dus	10867	0.528
13	gaan	1634	0.488	daar	10681	0.519
14	doen	1595	0.476	doen	10344	0.502
15	moet	1564	0.467	depressie	10226	0.496
16	gaat	1564	0.467	echt	10051	0.488
17	hebt	1498	0.447	gaan	9982	0.485
18	mensen	1490	0.445	toch	9915	0.481
19	daar	1438	0.429	moet	9910	0.481
20	iets	1436	0.429	iets	9480	0.46

Sentence Structure — Initial Posts vs Replies

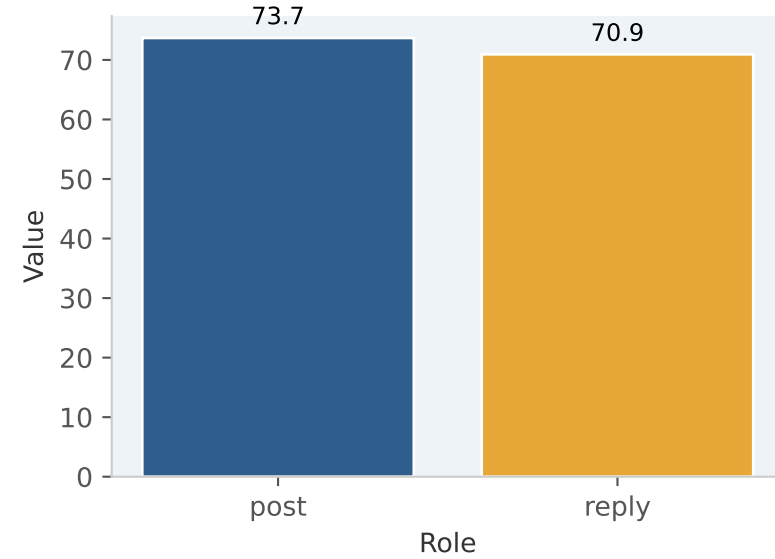
Avg Sentences per Message



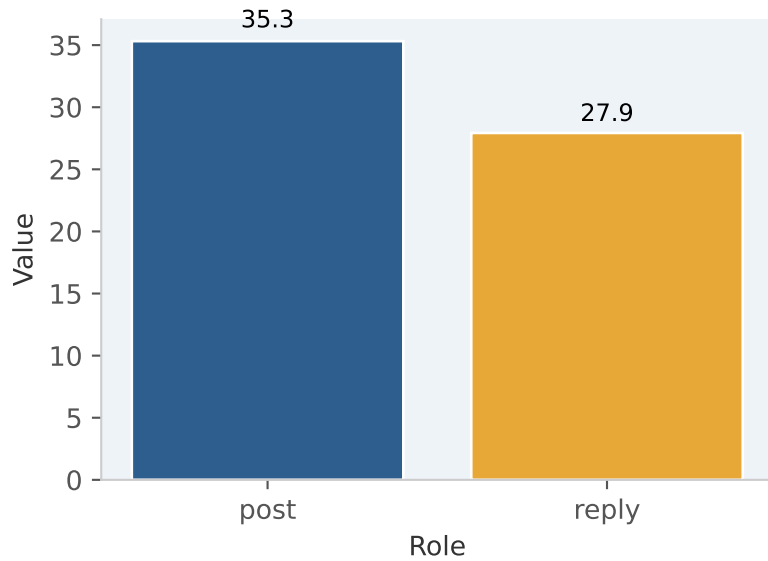
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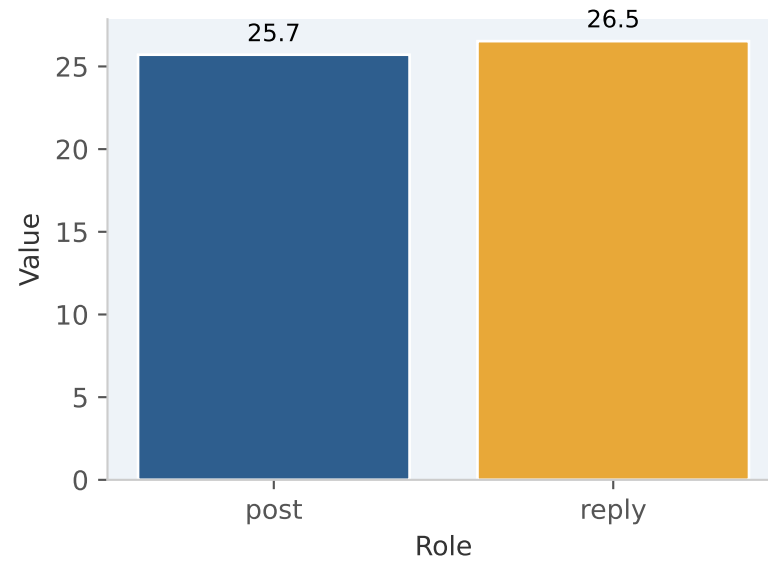
Avg Chars per Sentence



% Messages with '?'



% Messages with '!'



% Messages with '...'



2. EDA - Multi-Posters Only

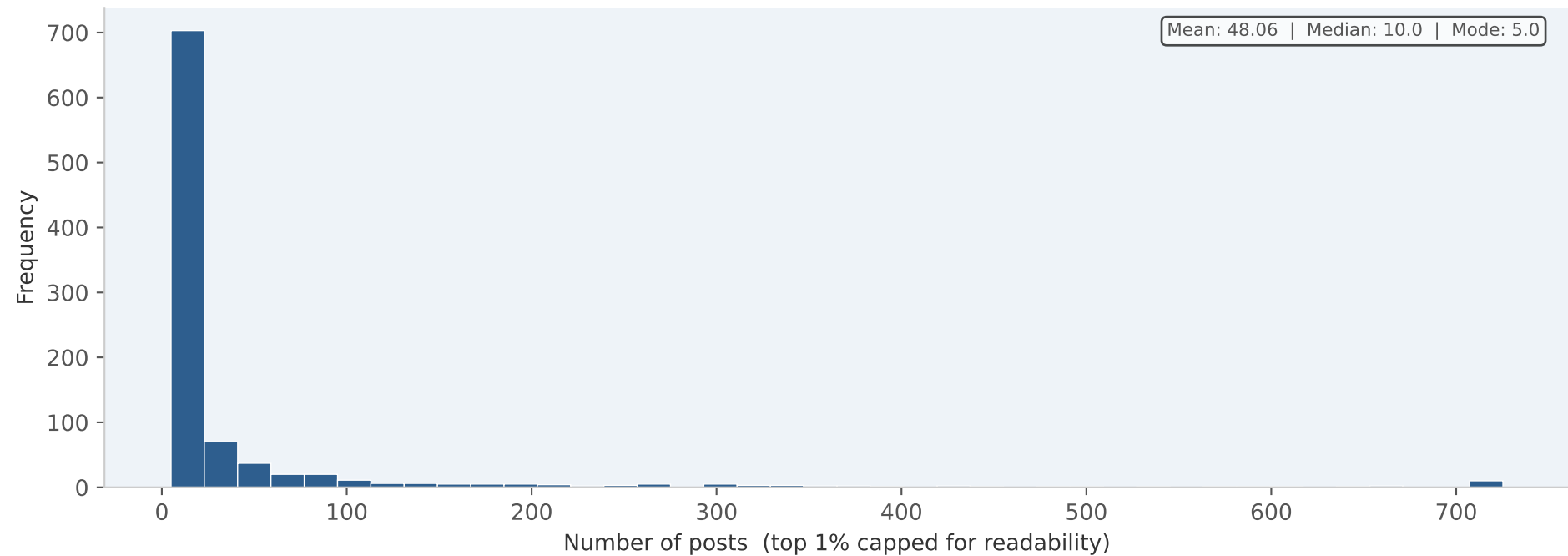
Depression Connect Forum

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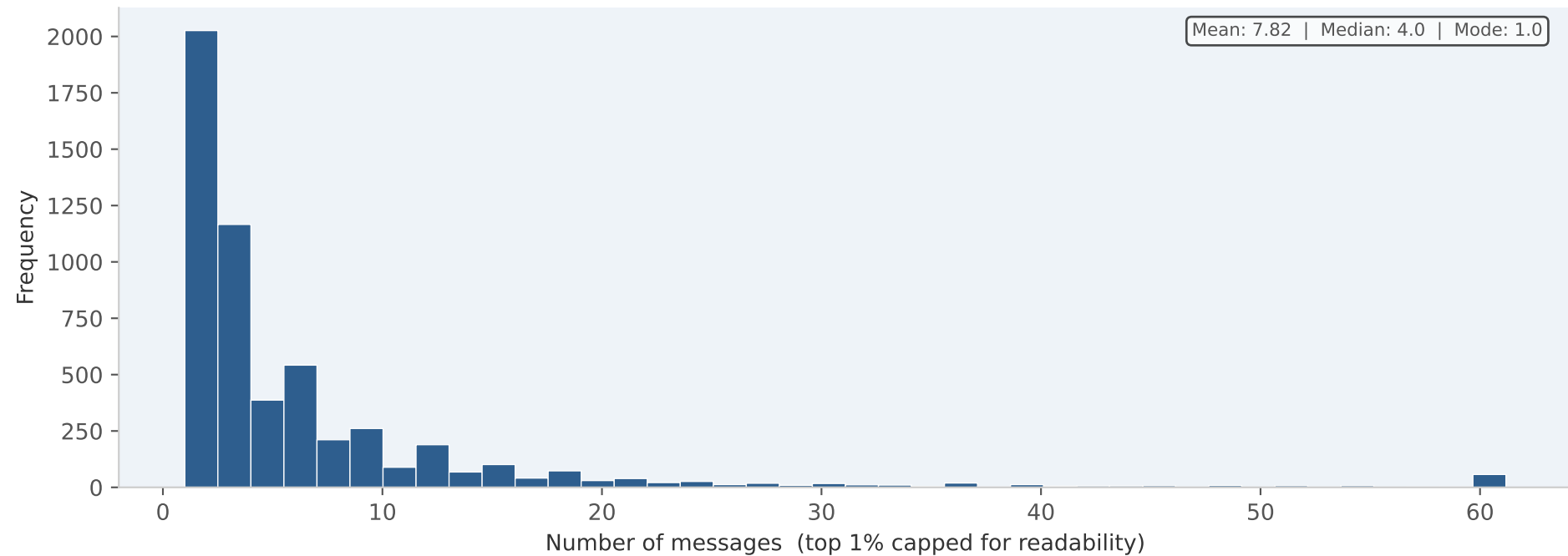
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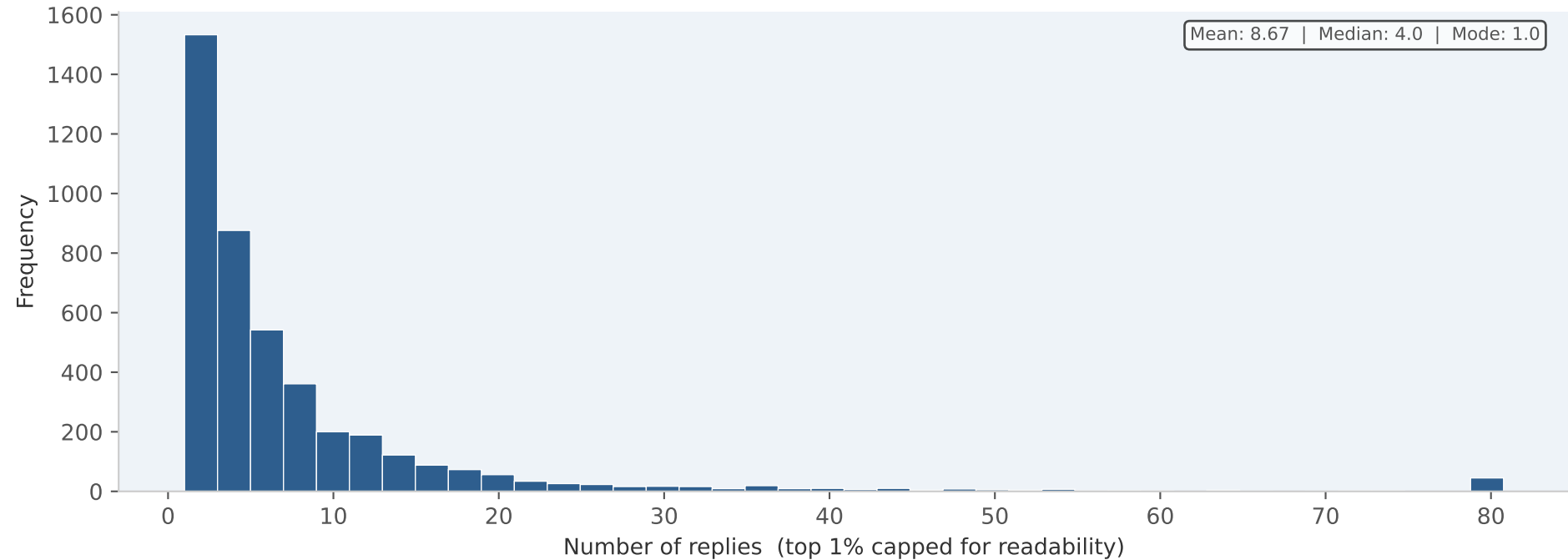
Distribution: Posts per User



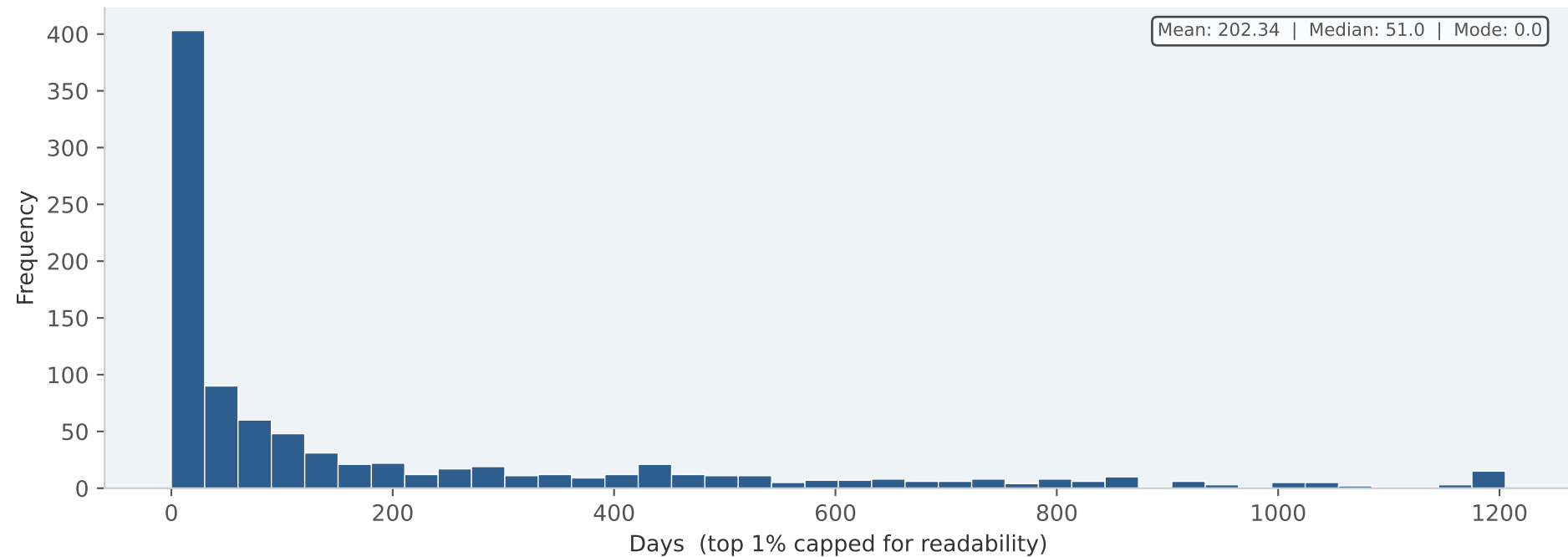
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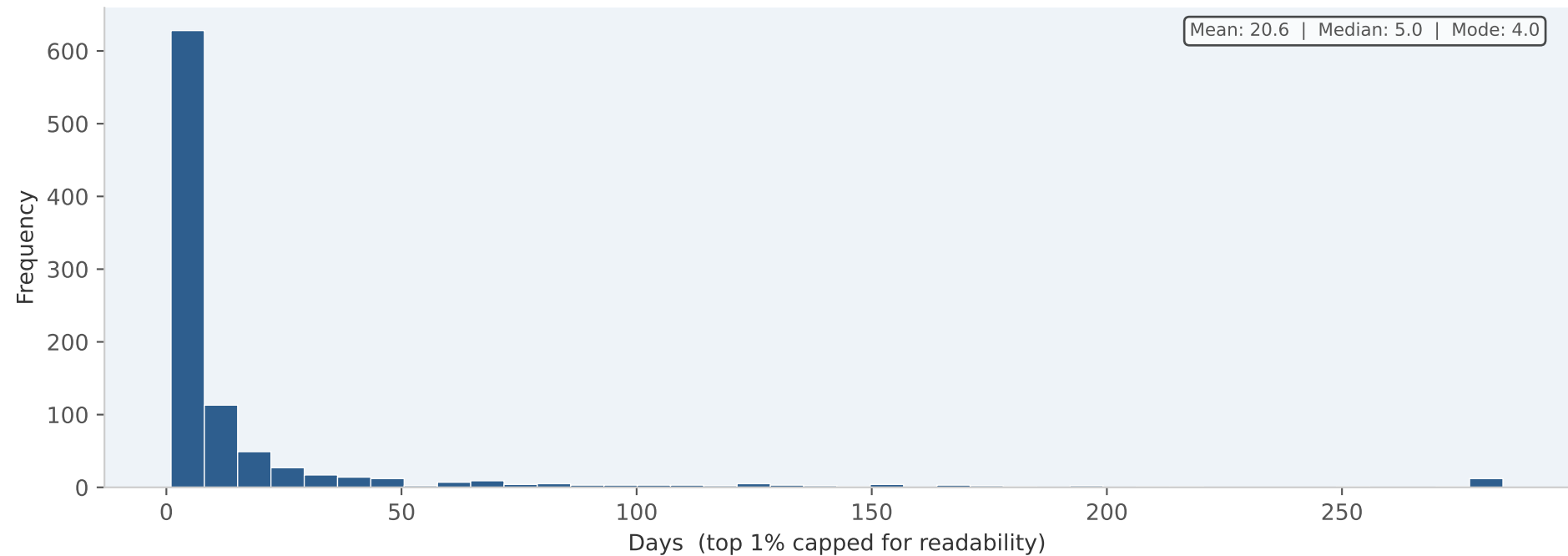
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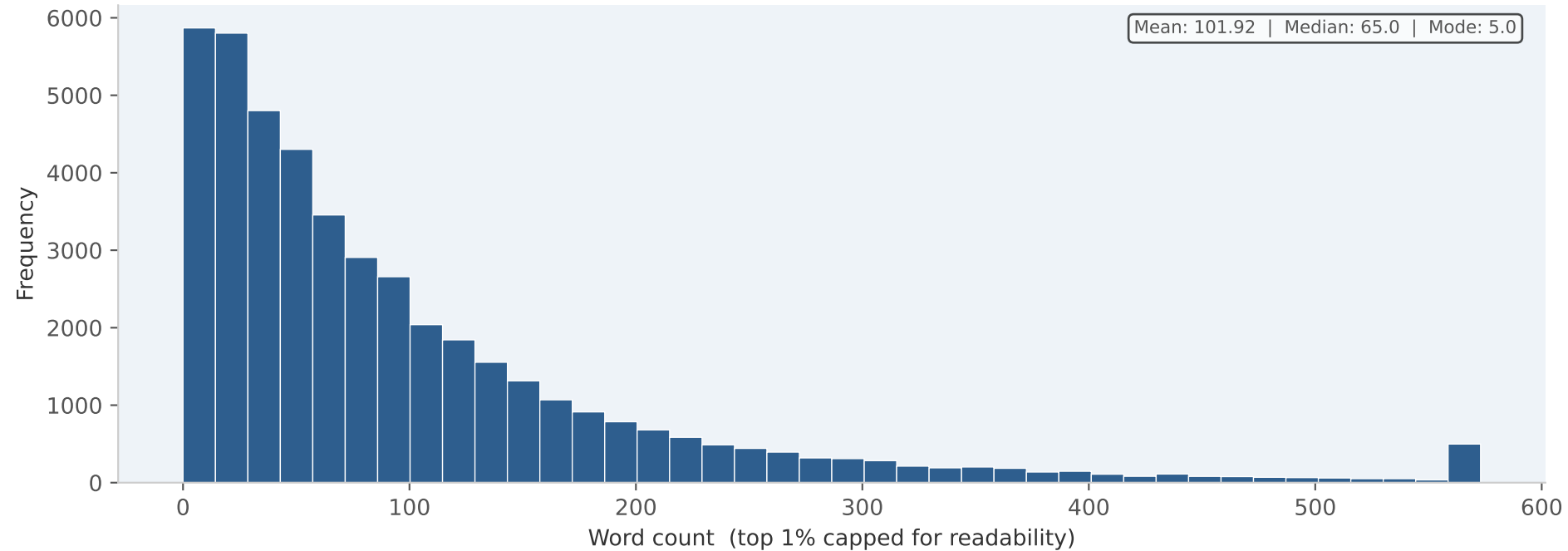
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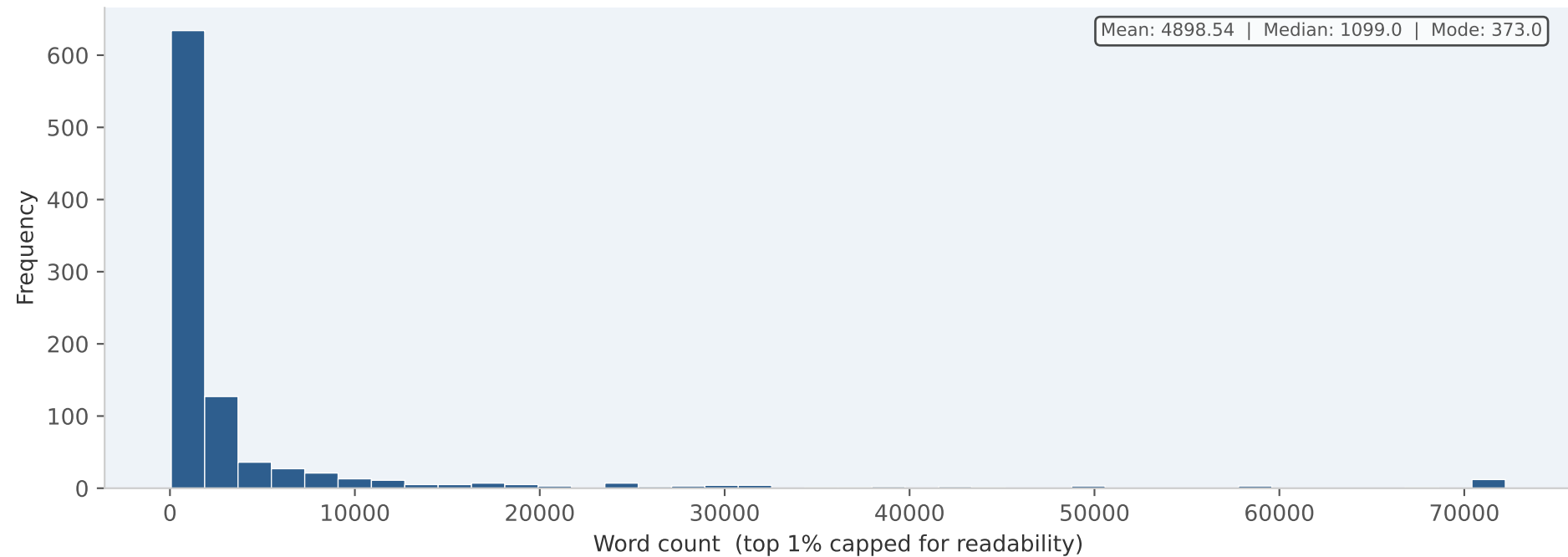
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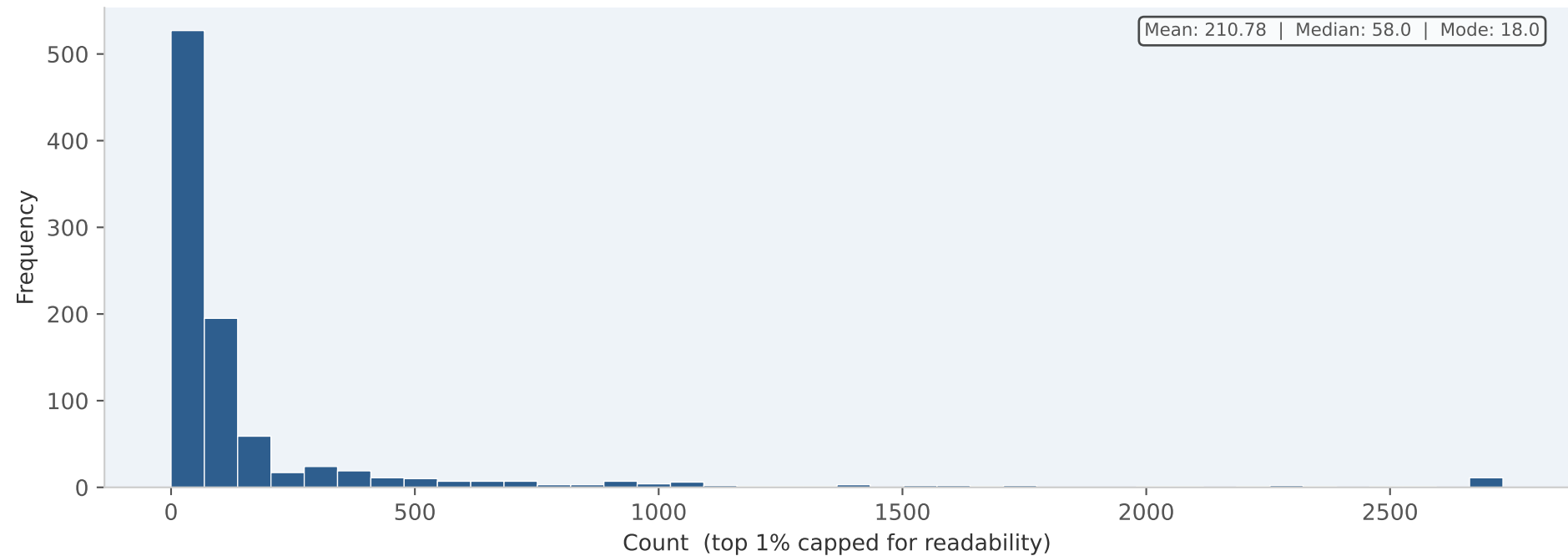
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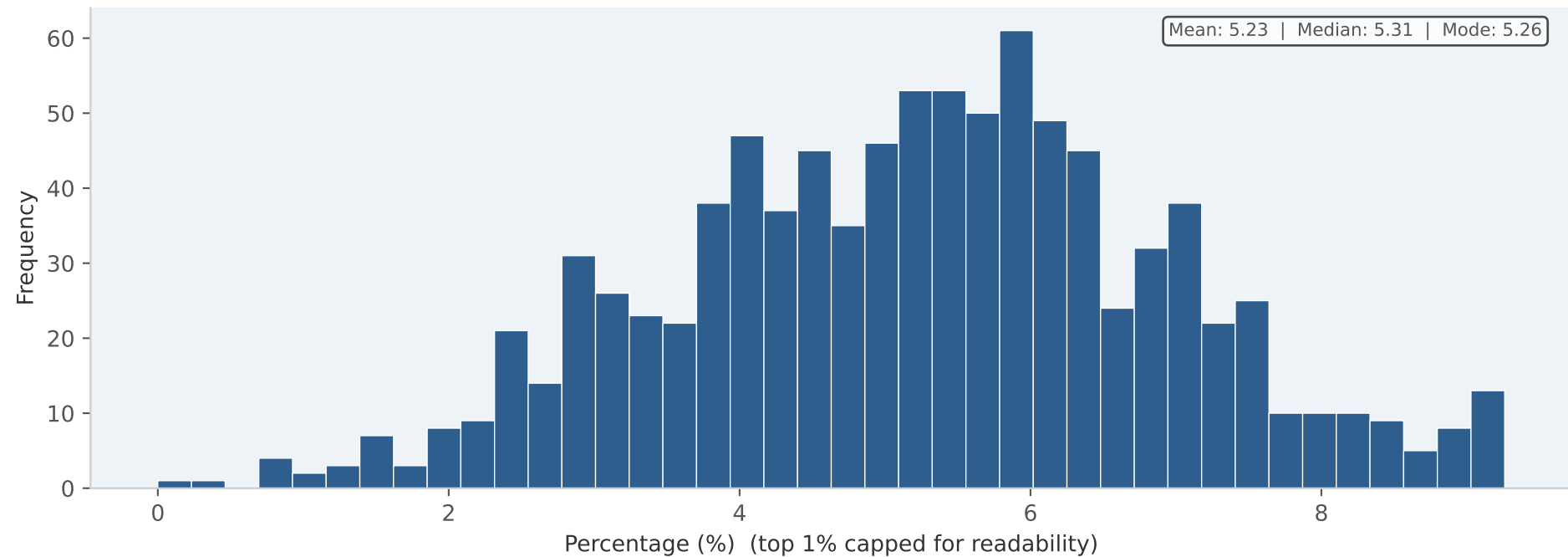
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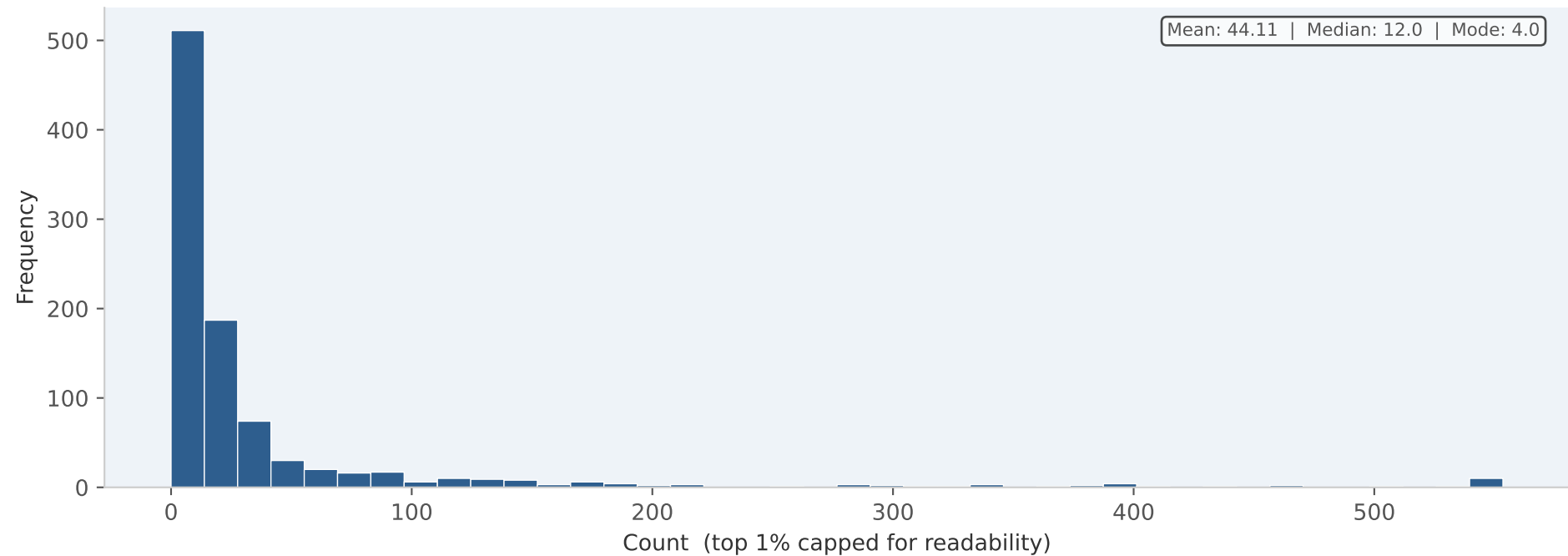
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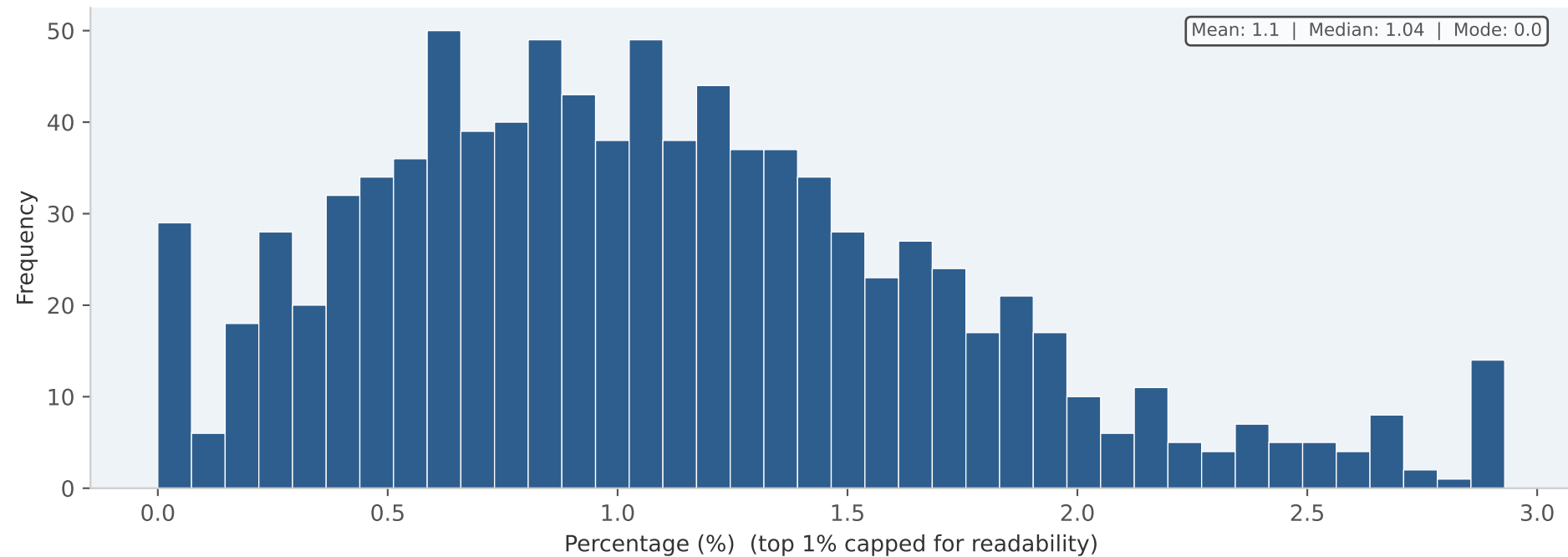
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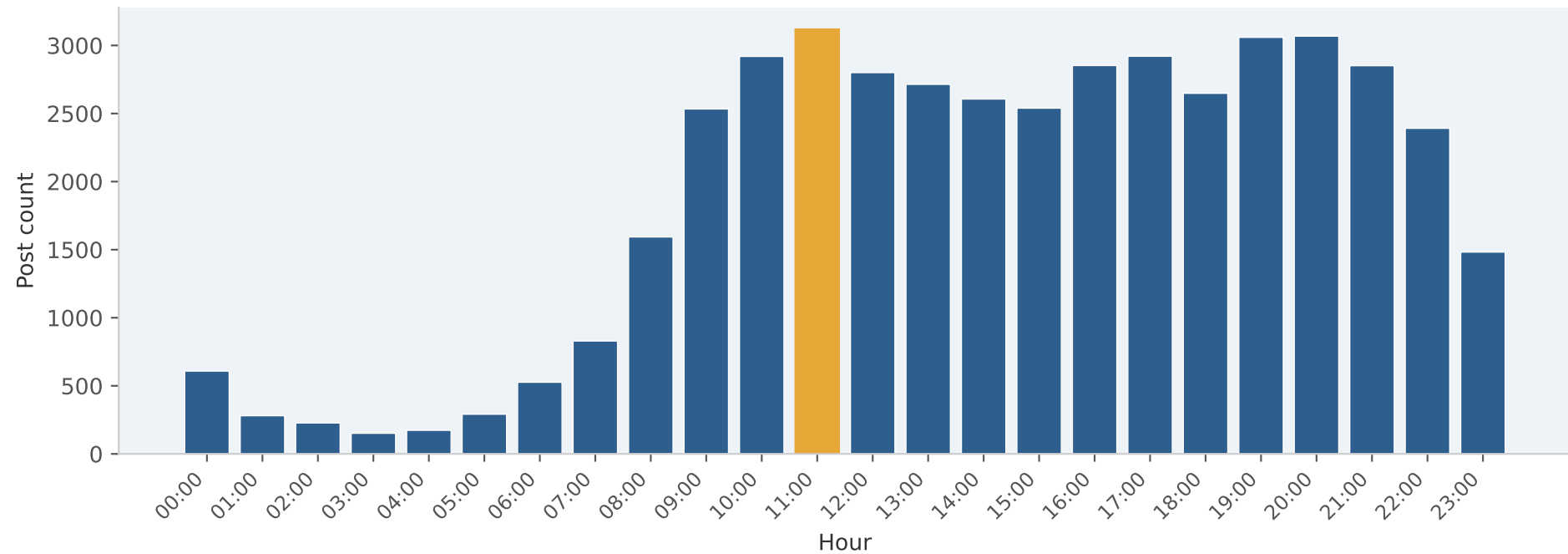
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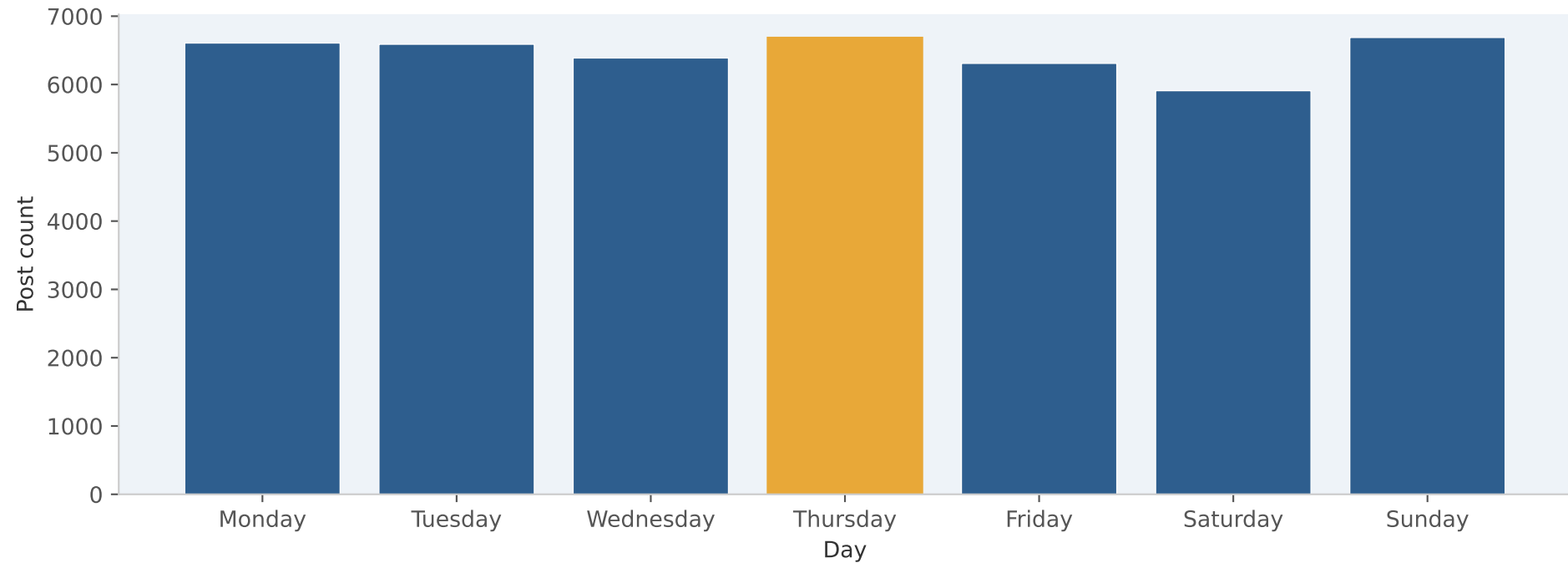
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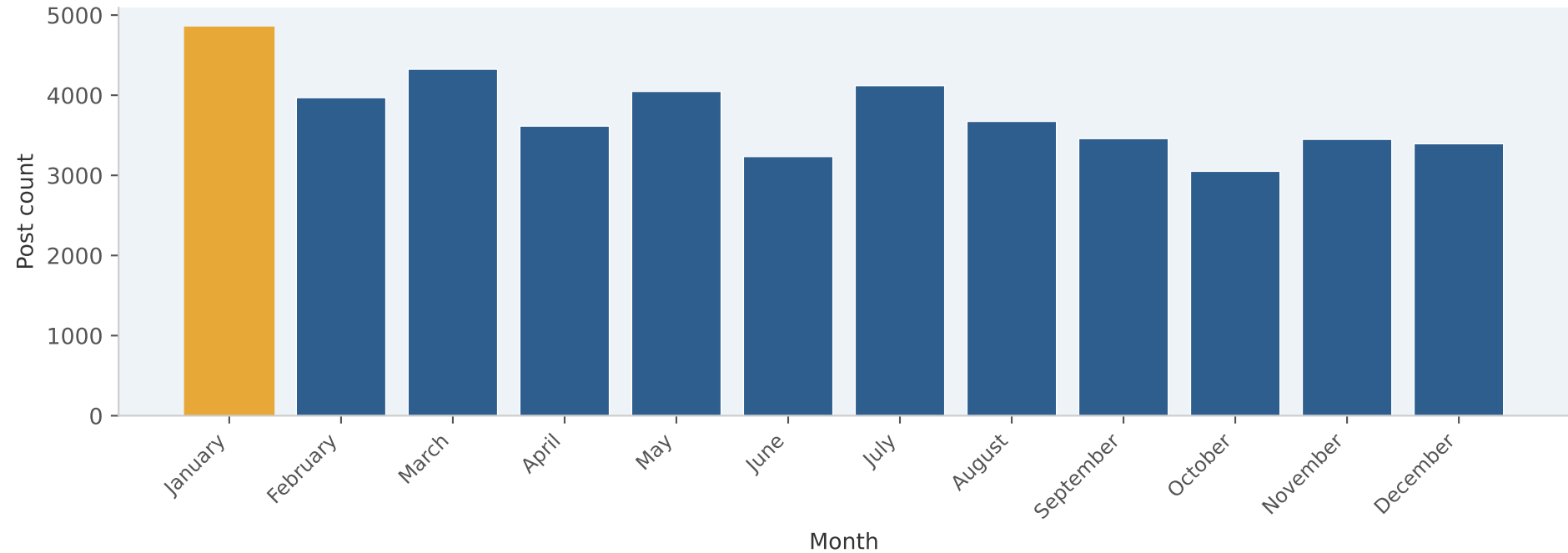
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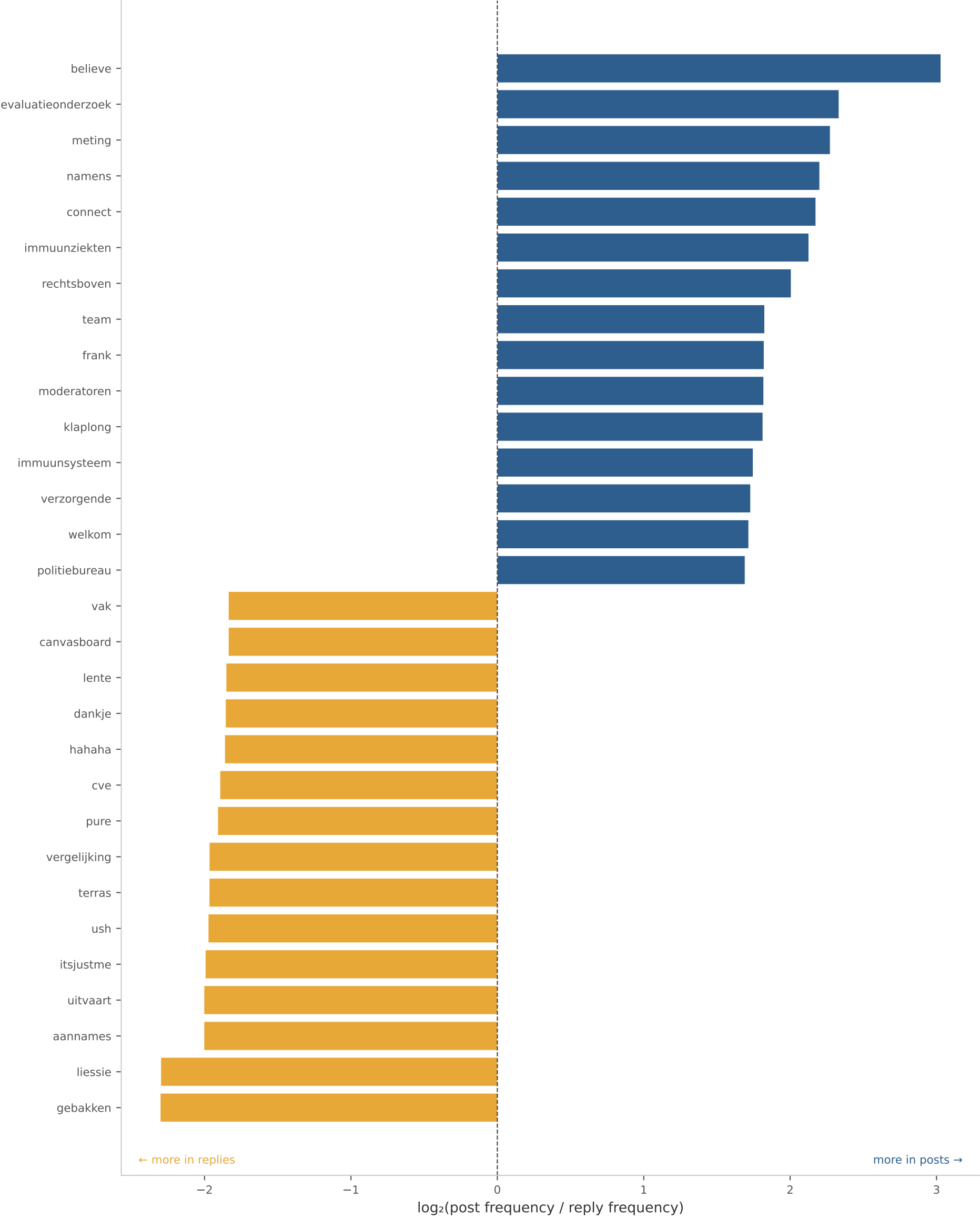


Popular Months



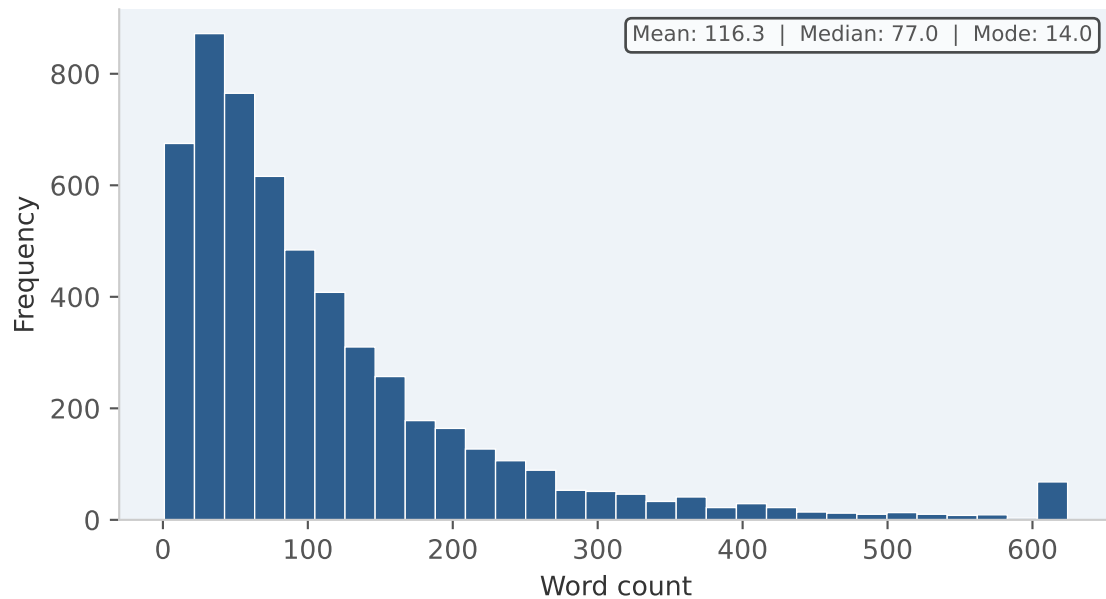
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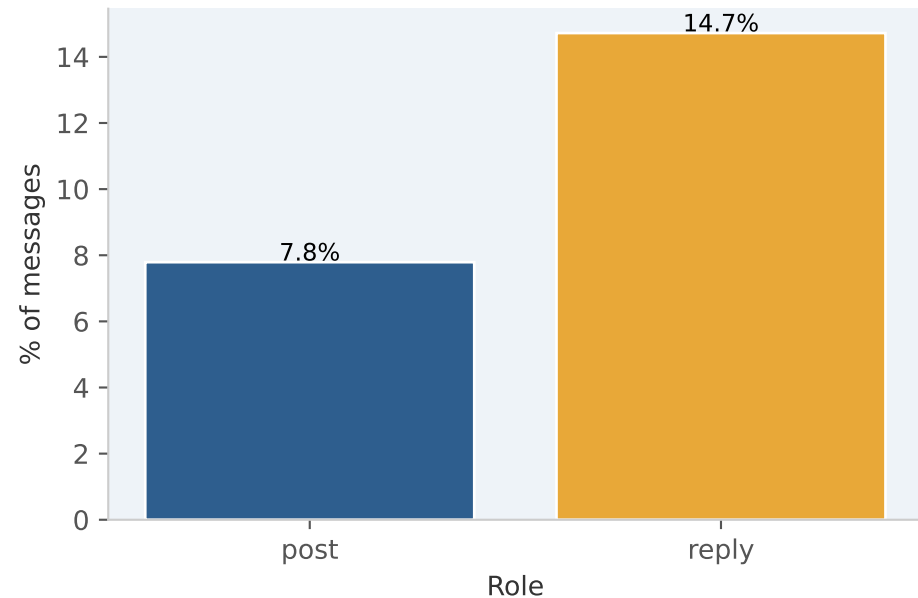
Words per Message: Initial Posts vs Replies

Initial Posts

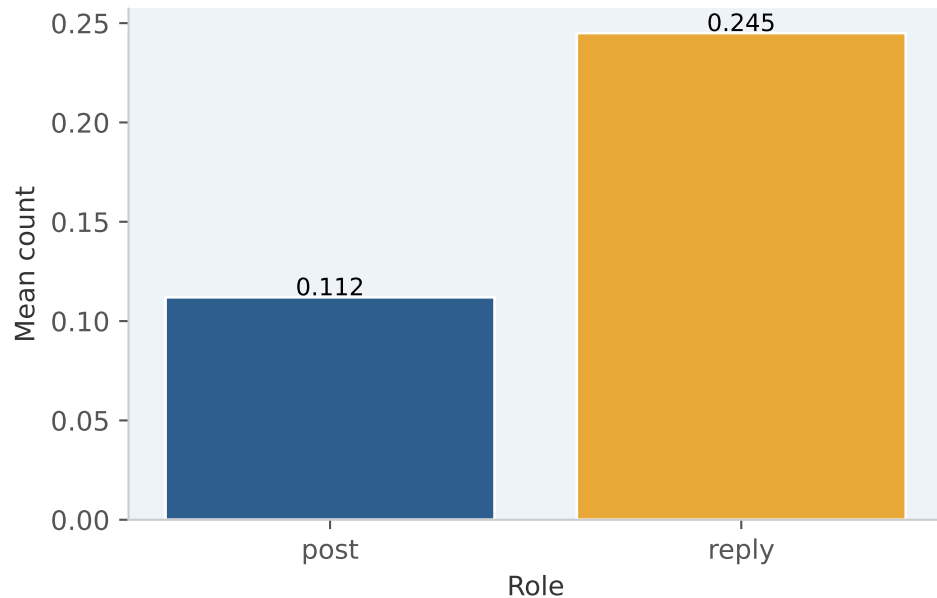


Emoji Use by Message Role

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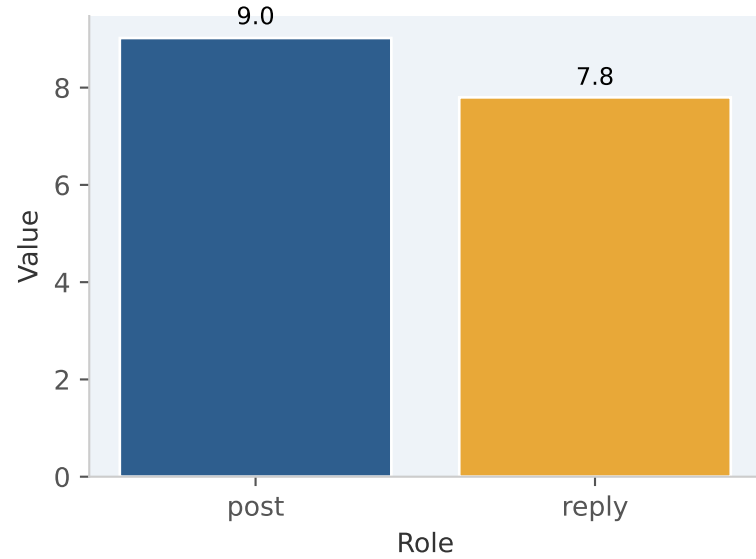
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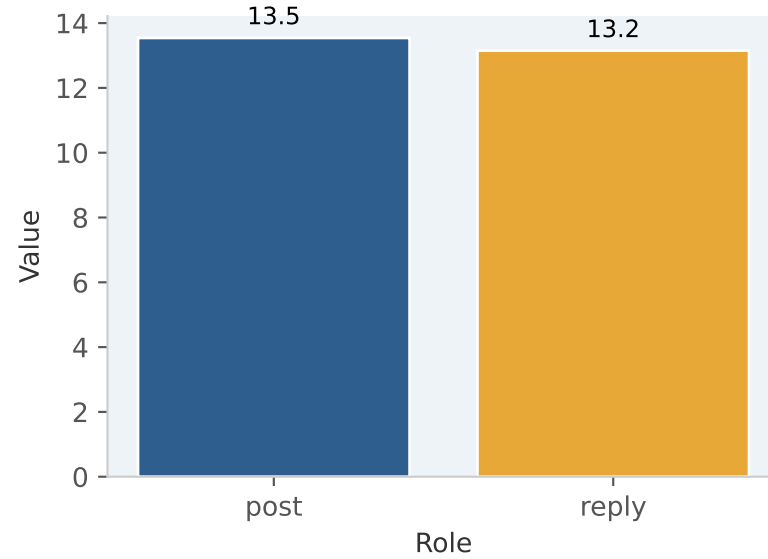
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Sentence Structure — Initial Posts vs Replies

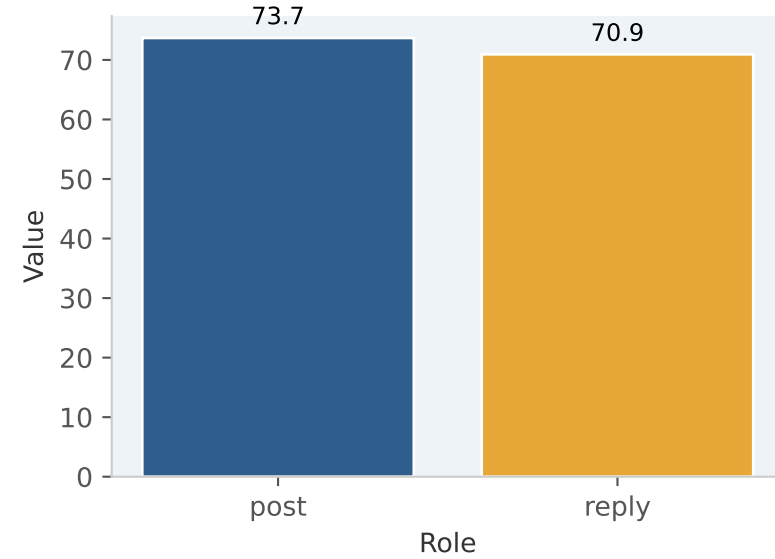
Avg Sentences per Message



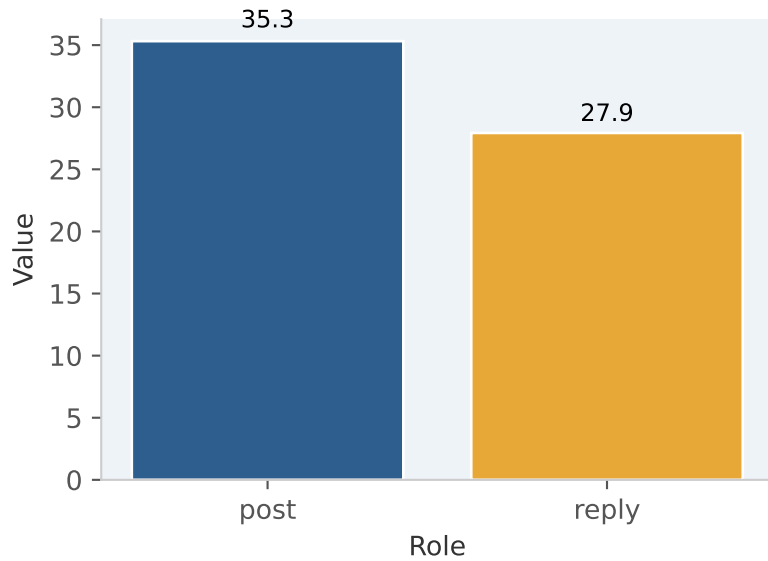
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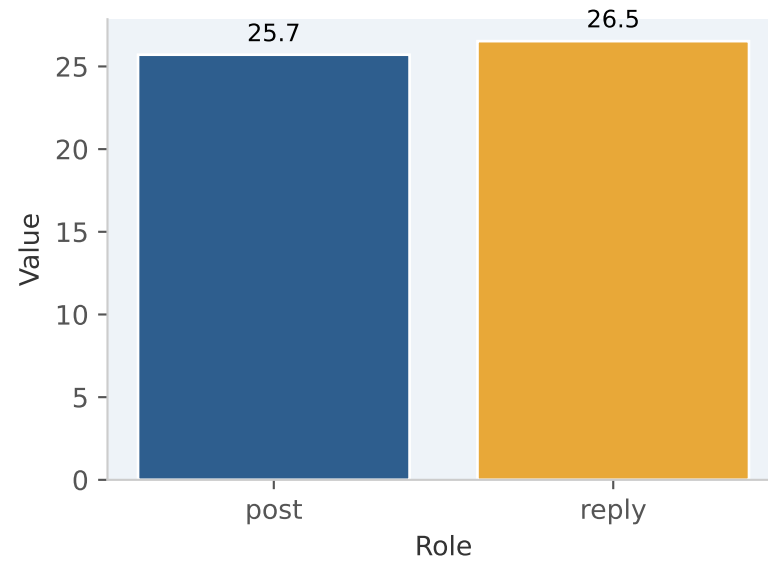
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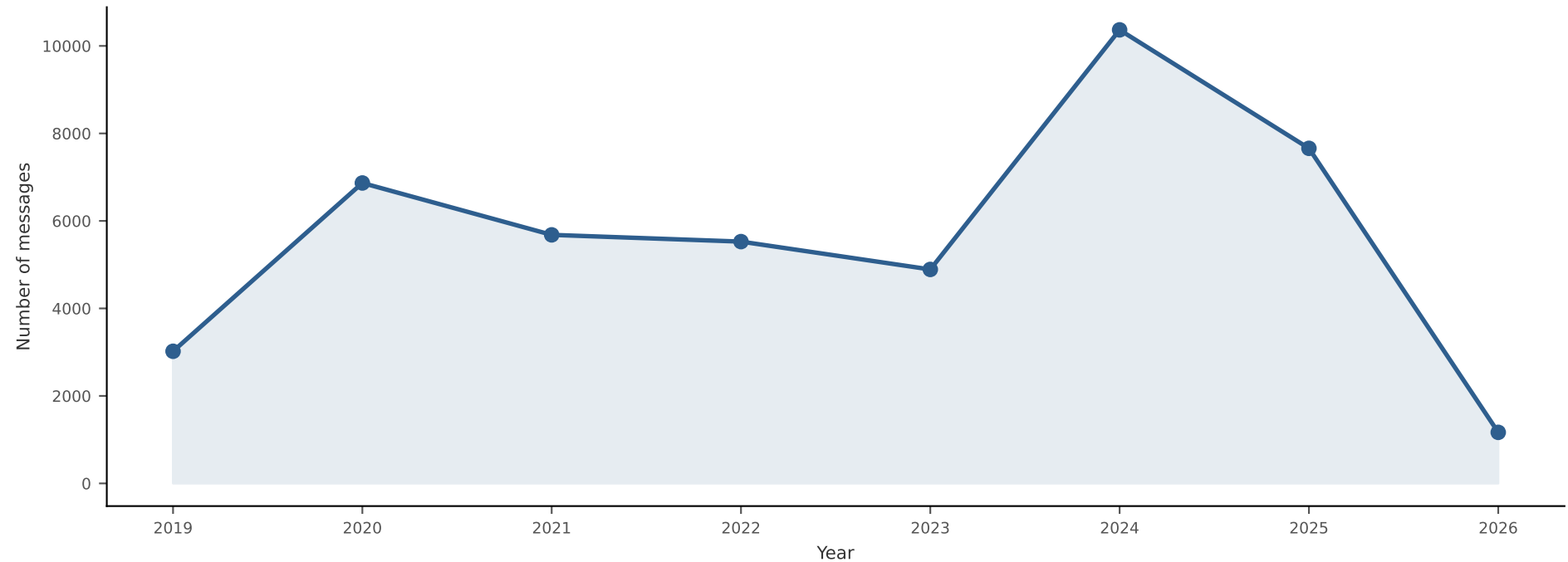
3. Time Series & CDS Prevalence

Depression Connect Forum

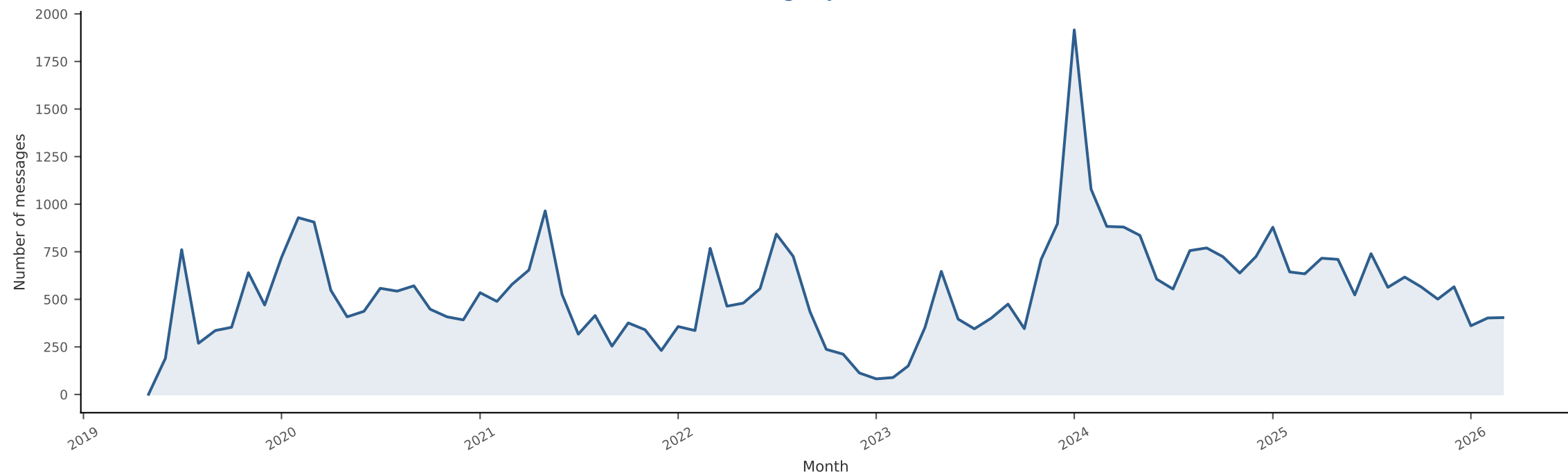
Exploratory Analysis — Time Series & CDS Prevalence

Section 1 — Forum Activity Over Time

Messages per Year

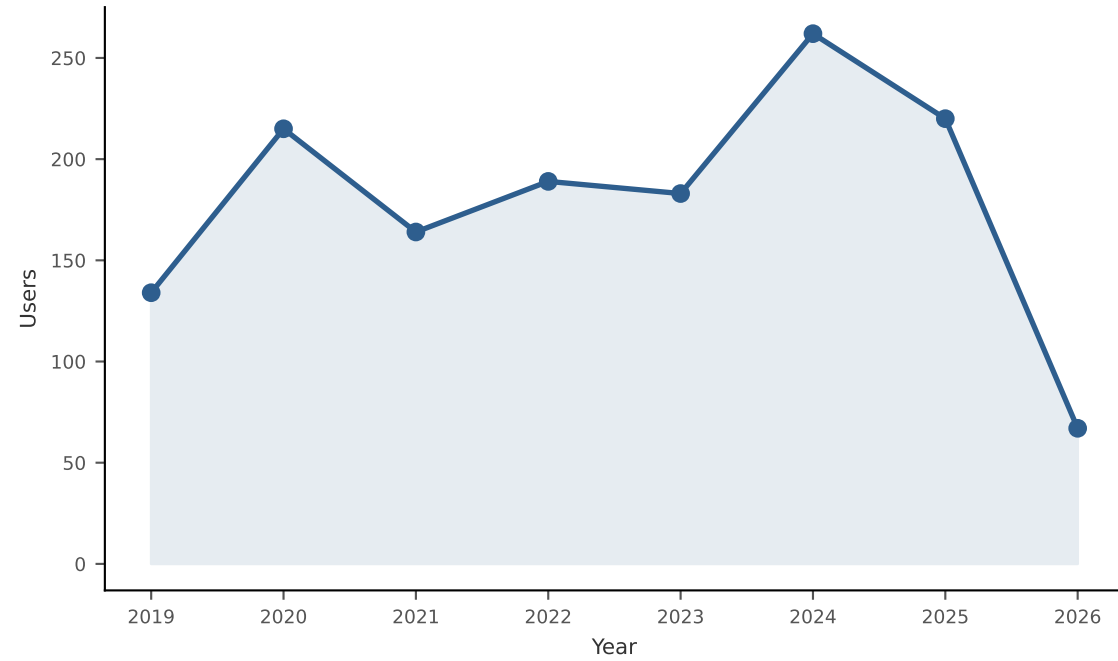


Messages per Month

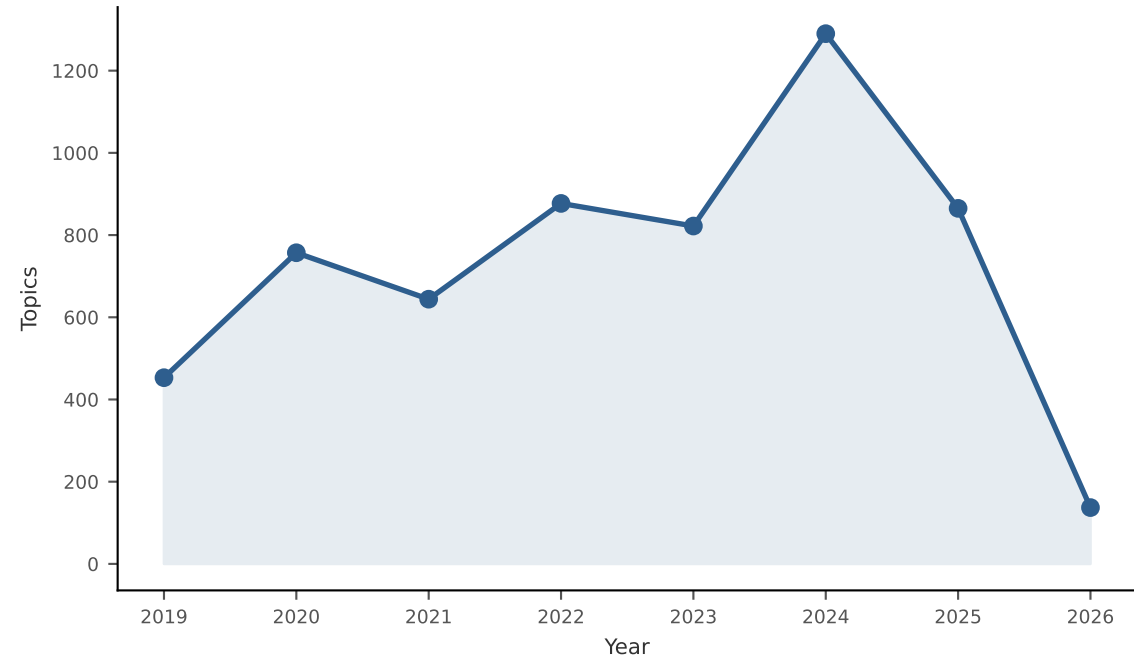


Forum Activity — Year-level Metrics

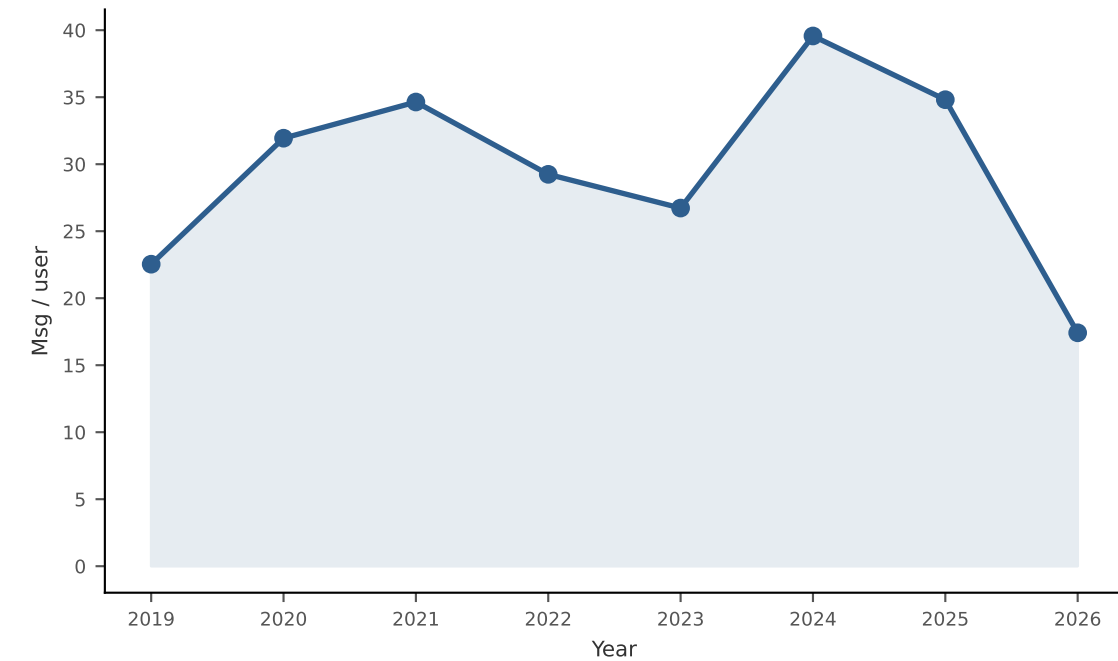
Unique Users per Year



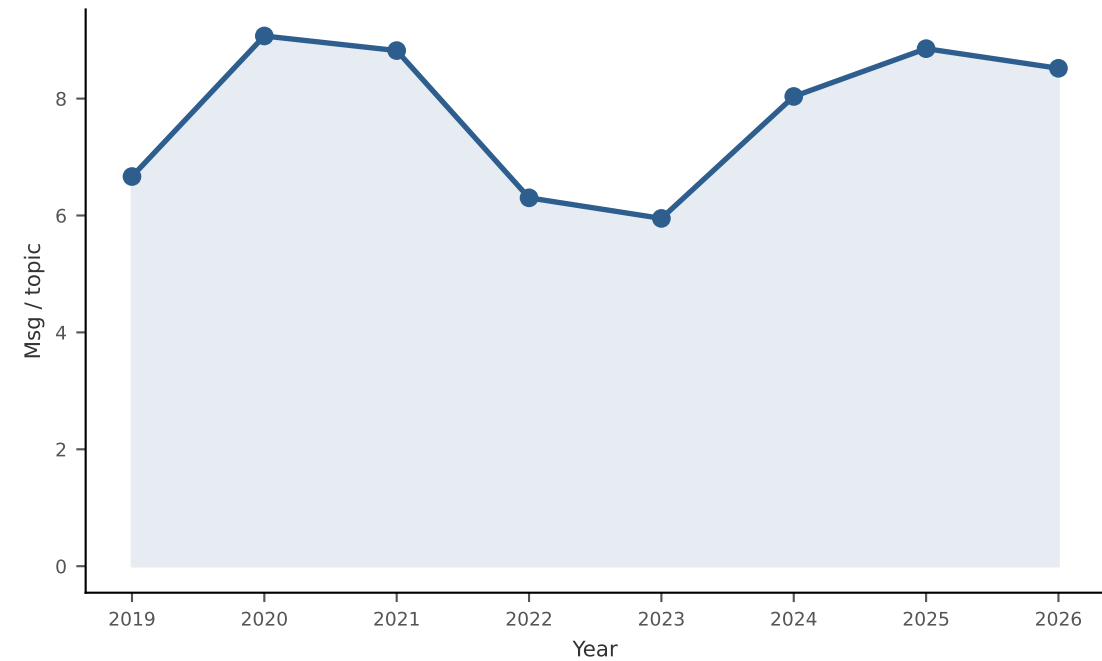
Unique Topics per Year



Messages per User per Year

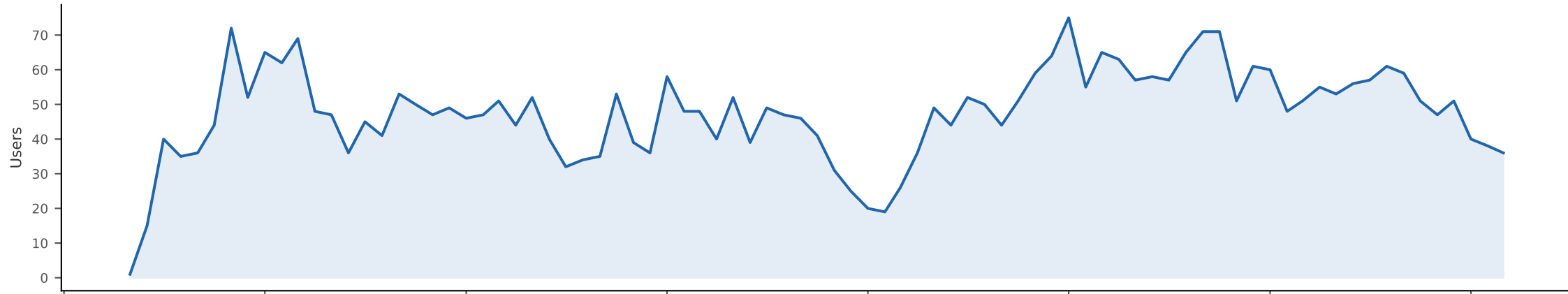


Messages per Topic per Year

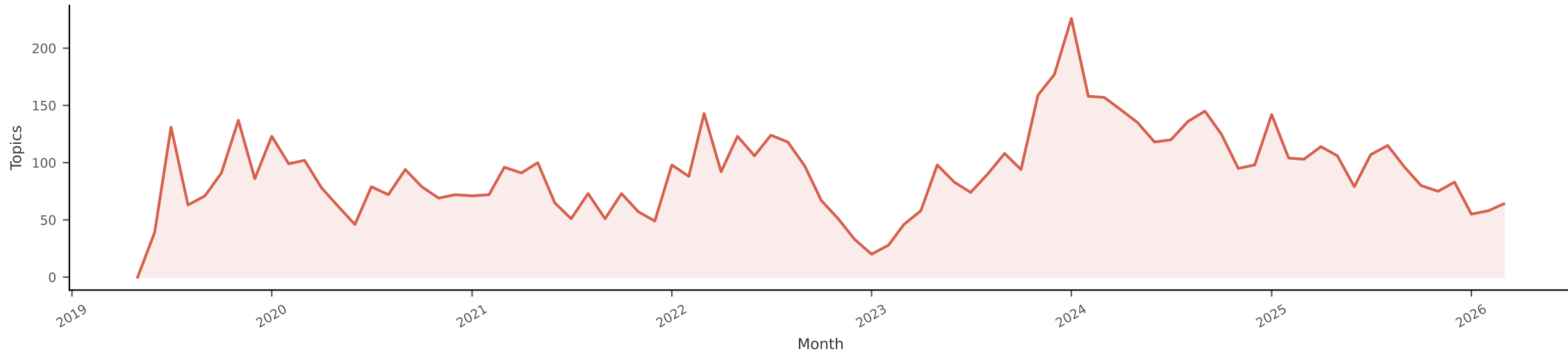


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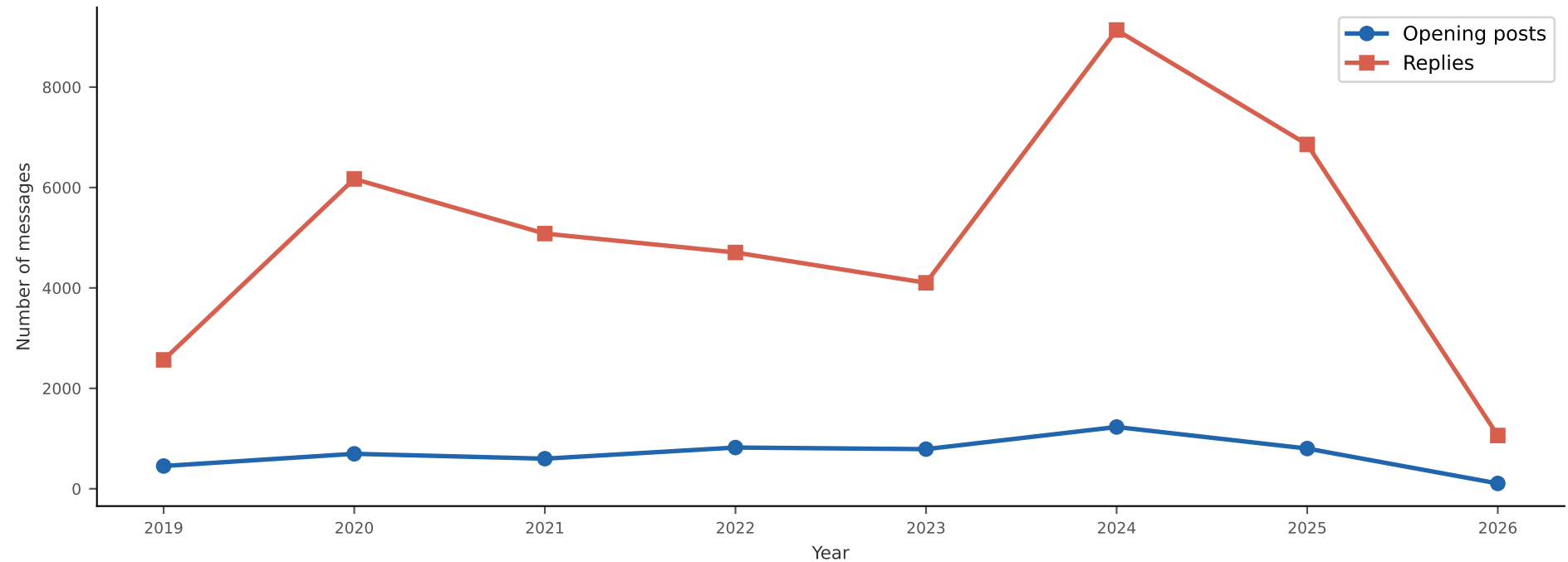
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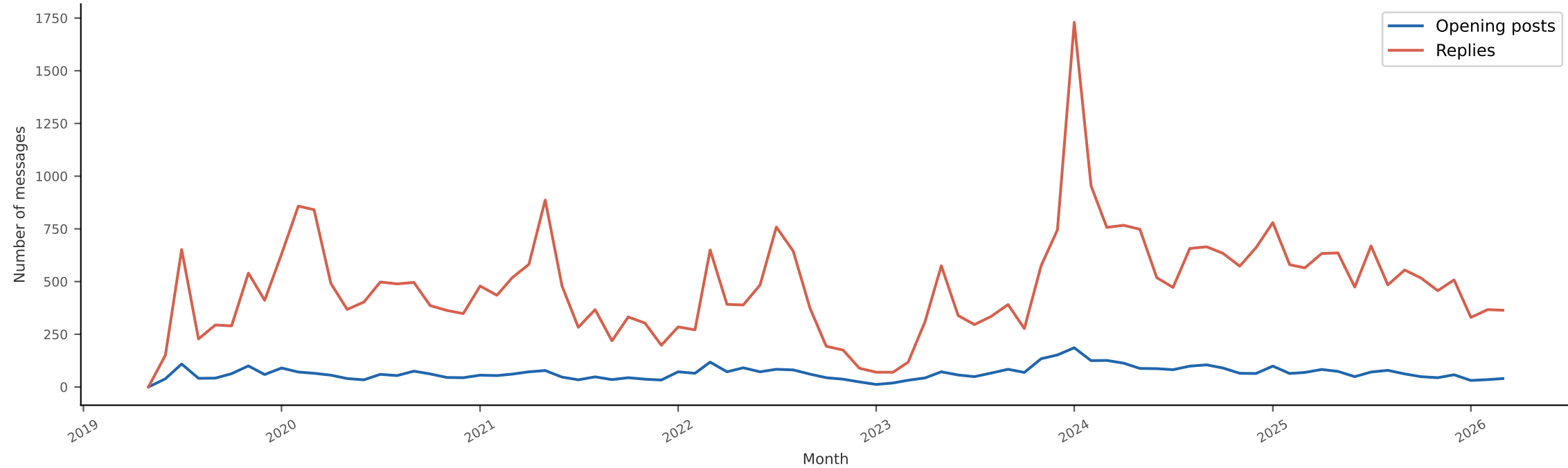
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Message Volume by Role — Year Level



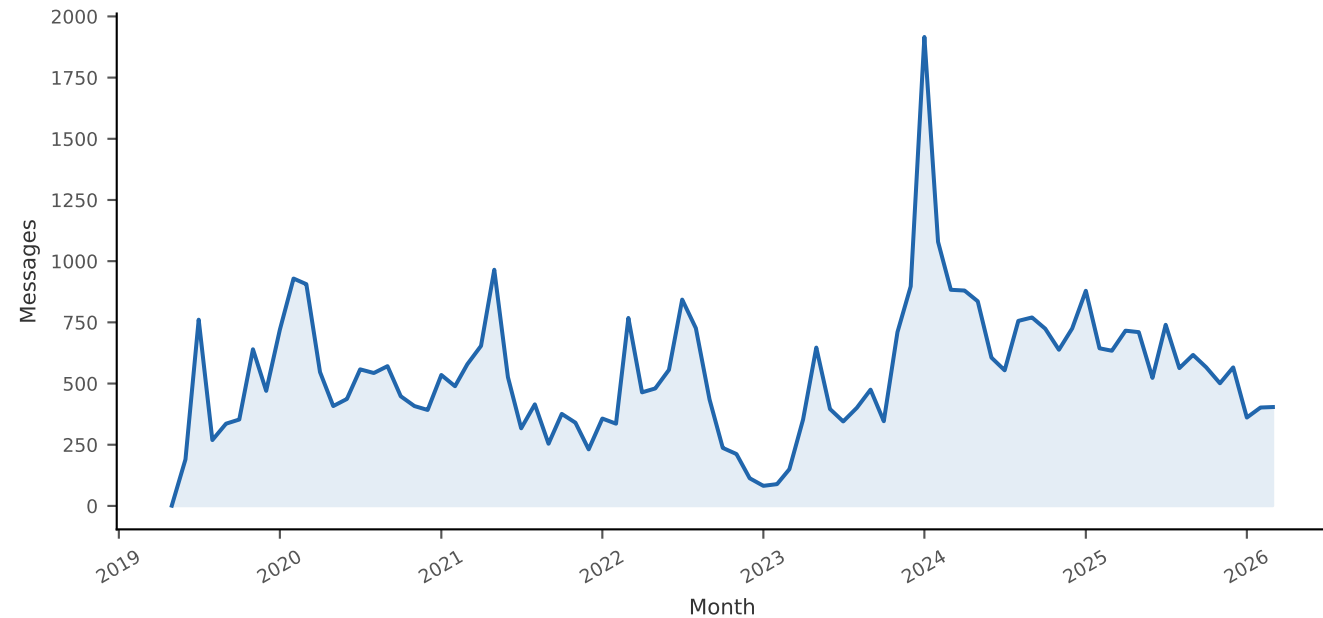
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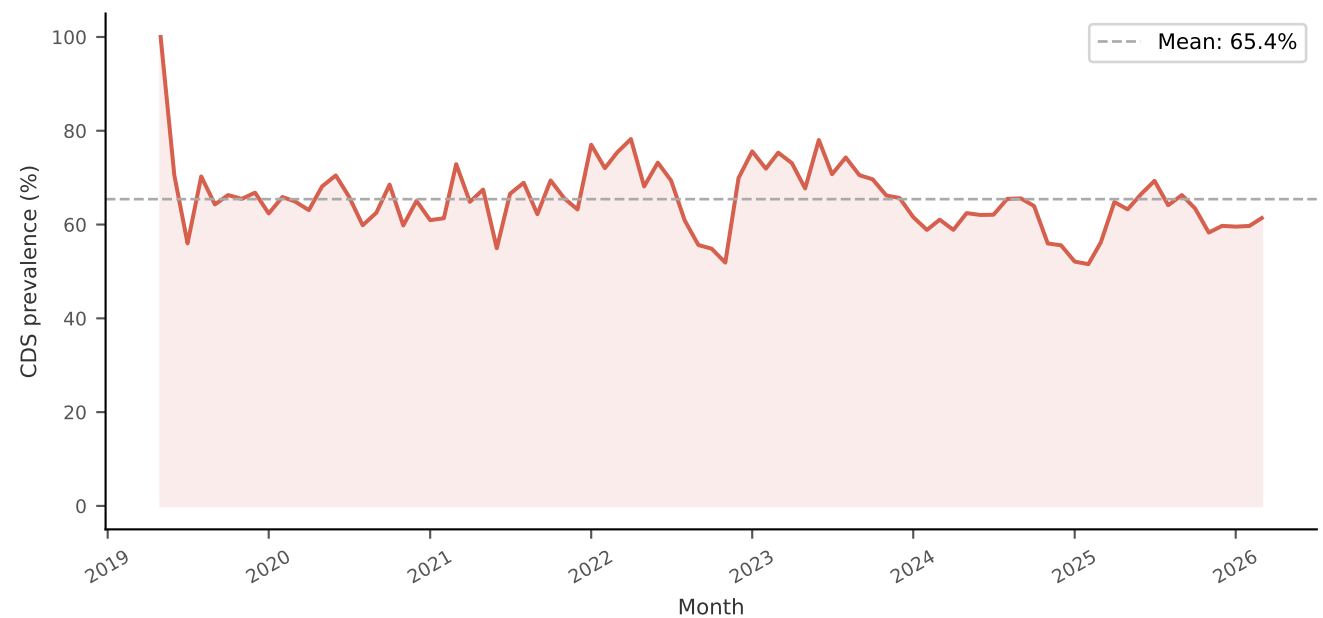
Section 2 — CDS Prevalence Over Time

Message Volume and CDS Prevalence Over Time (Monthly)

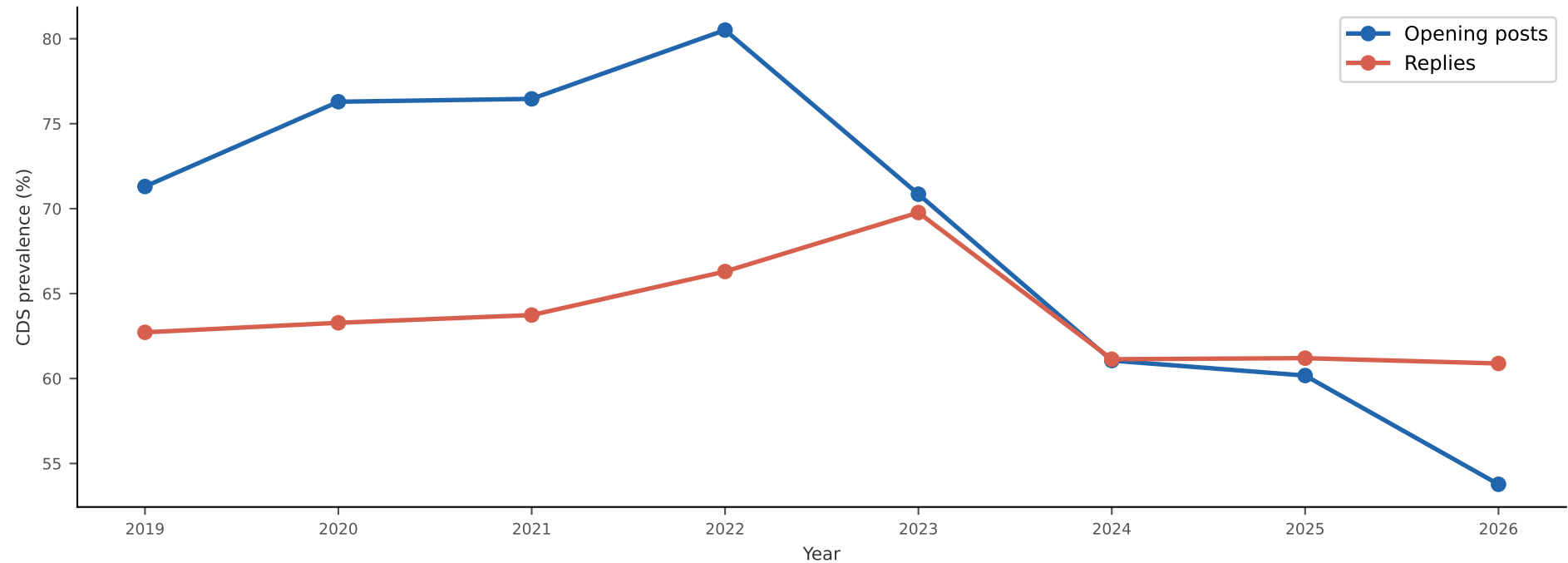
Monthly Message Volume



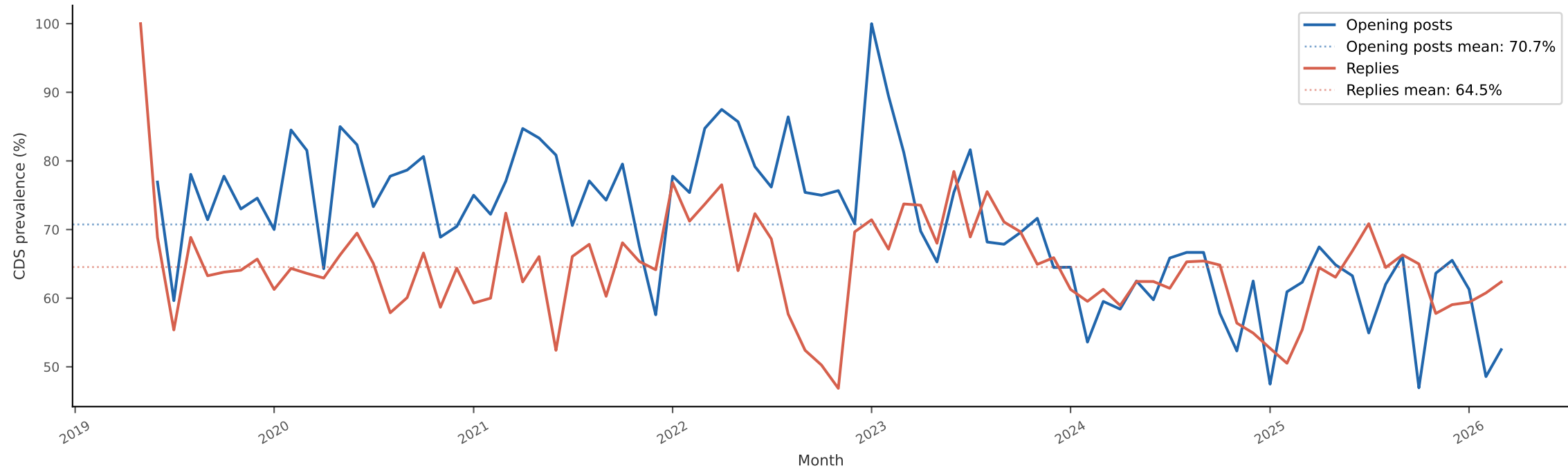
Monthly CDS Prevalence (%)



CDS Prevalence by Role — Year Level

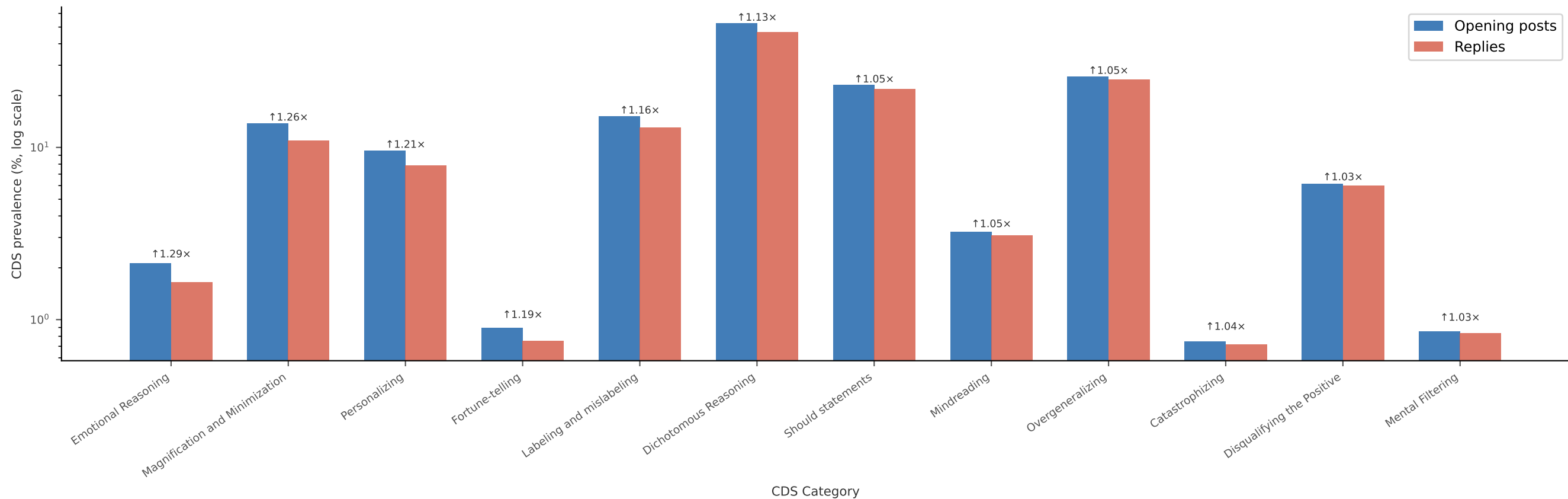


CDS Prevalence by Role — Month Level



Section 3 — CDS Prevalence by Category

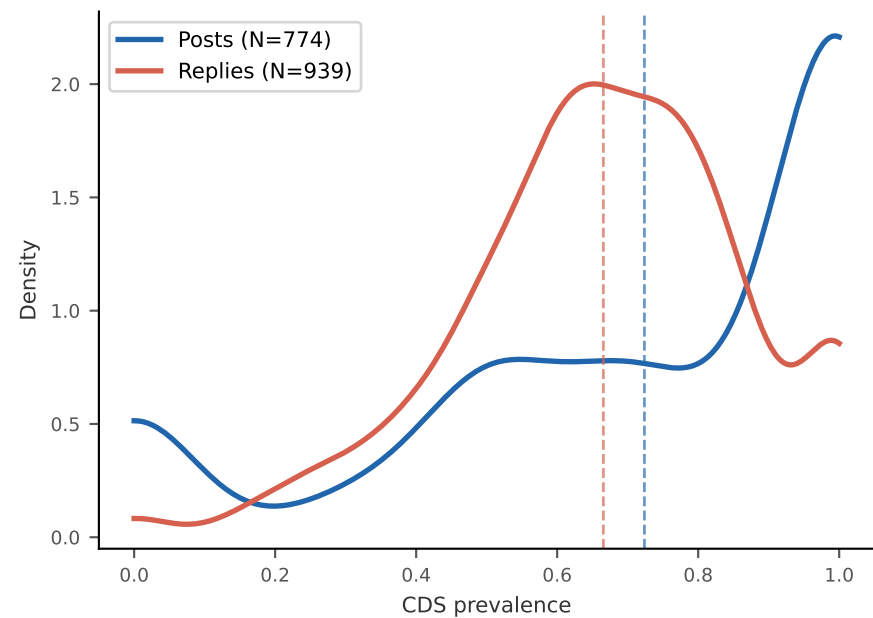
CDS Prevalence by Category — Posts vs Replies
(sorted by prevalence ratio, log scale)



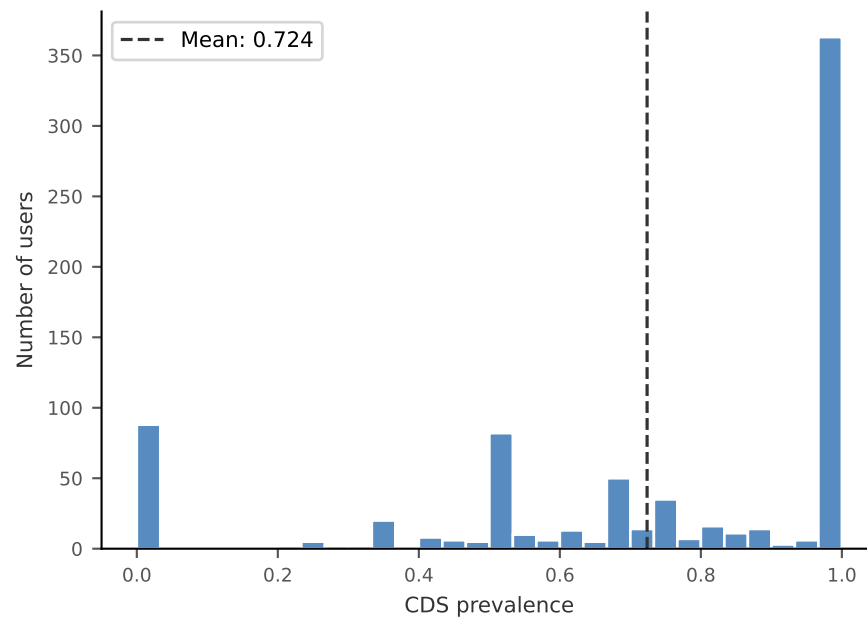
Section 4 — Within-User CDS Distribution

Within-User CDS Prevalence Distribution

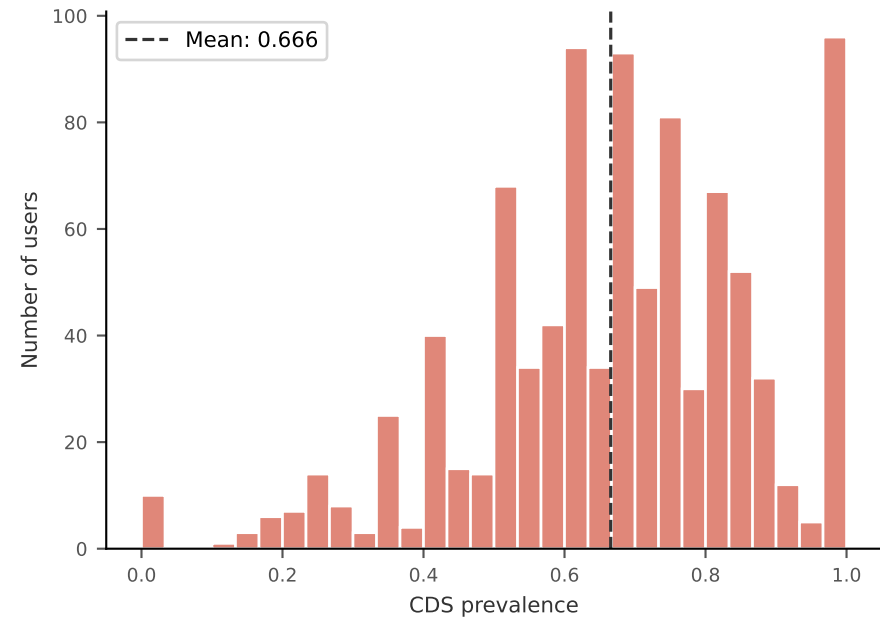
KDE: Within-user CDS Prevalence



Distribution: Posts



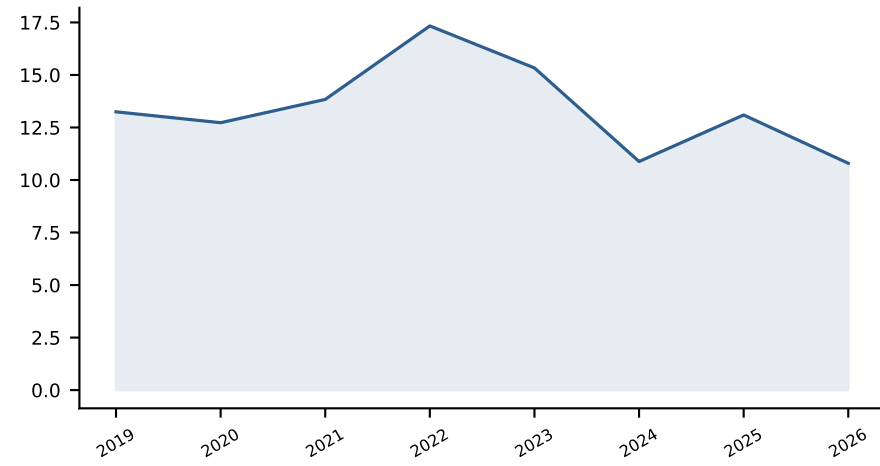
Distribution: Replies



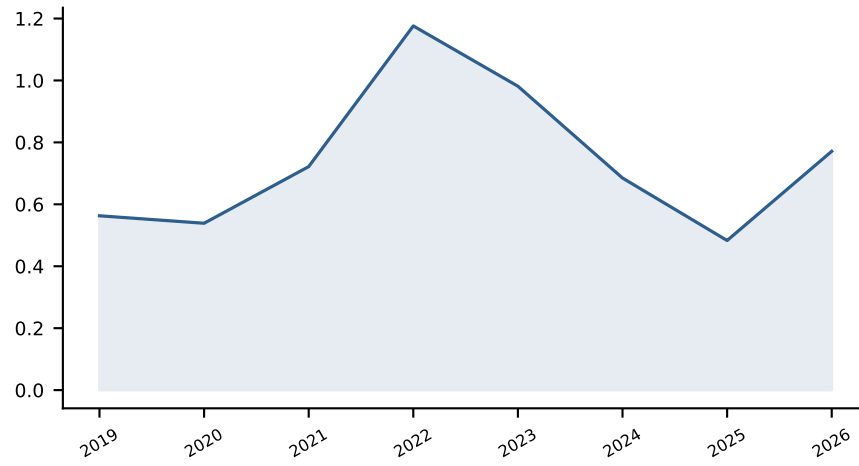
Section 5 — CDS per Category Over Time

CDS Prevalence per Category Over Time (Year Level)

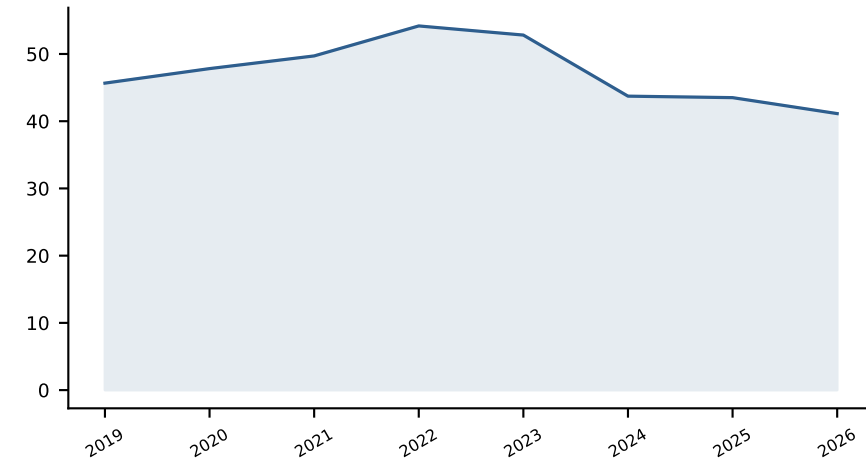
Labeling And Mislabeling



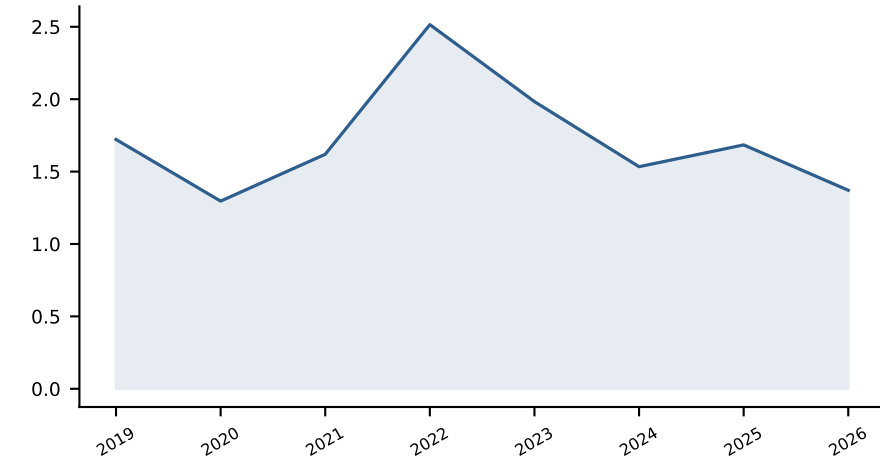
Catastrophizing



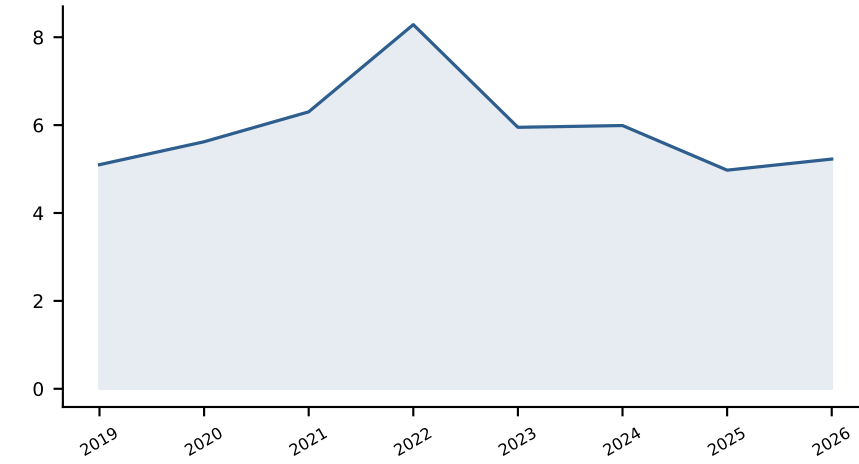
Dichotomous Reasoning



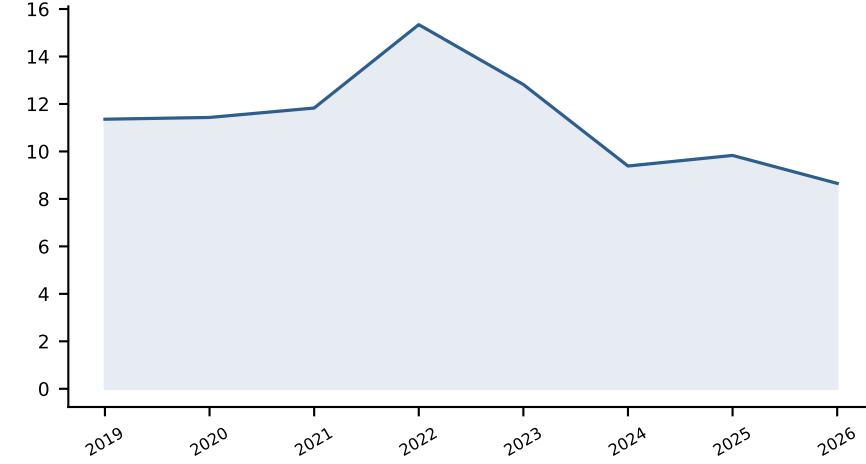
Emotional Reasoning



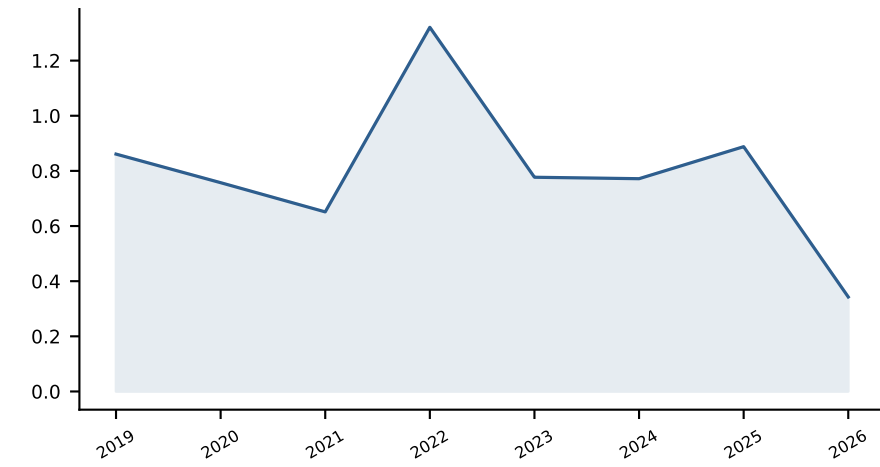
Disqualifying The Positive



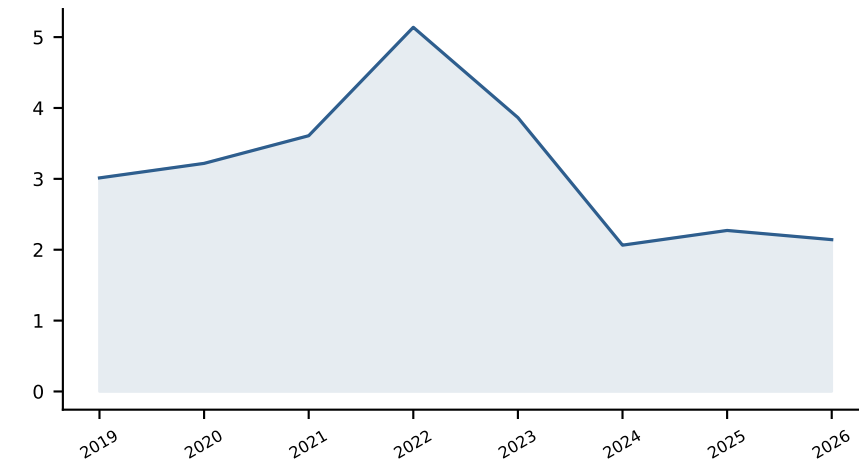
Magnification And Minimization



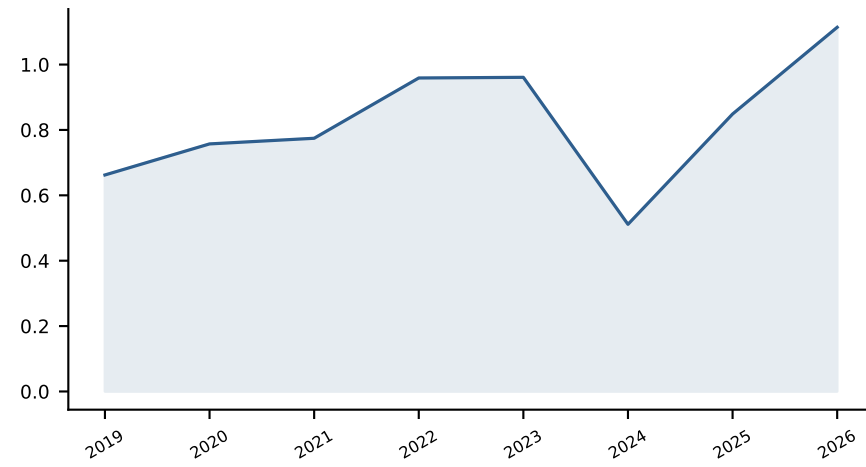
Mental Filtering



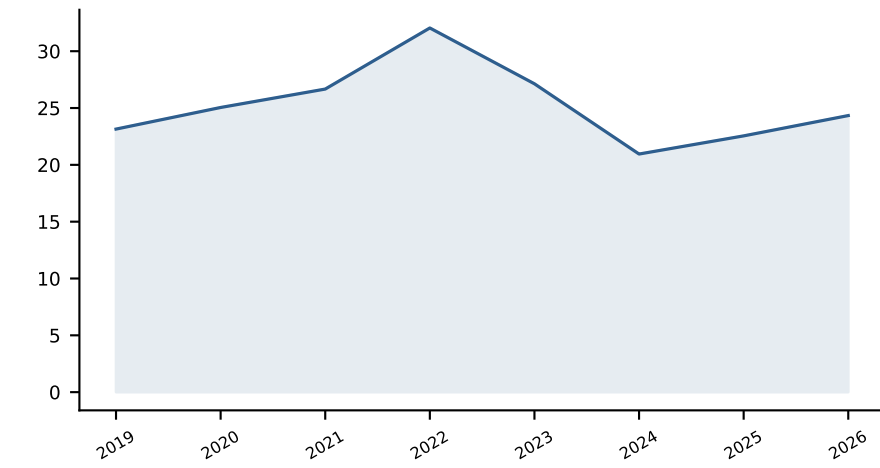
Mindreading



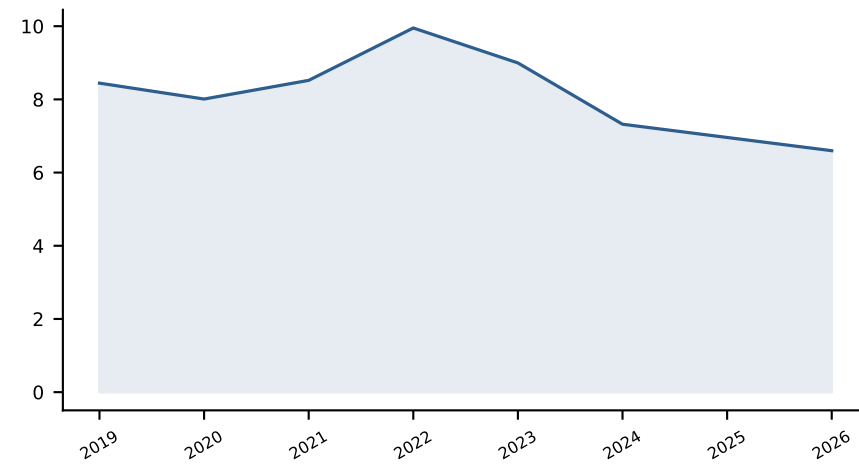
Fortune-Telling



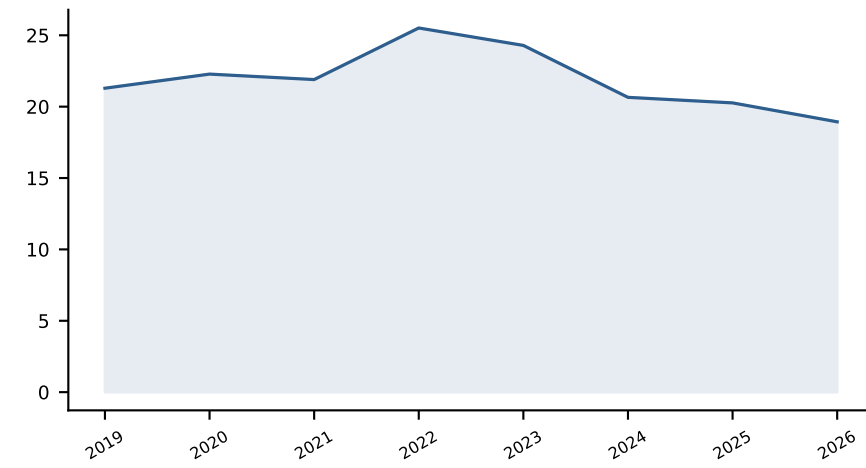
Overgeneralizing



Personalizing

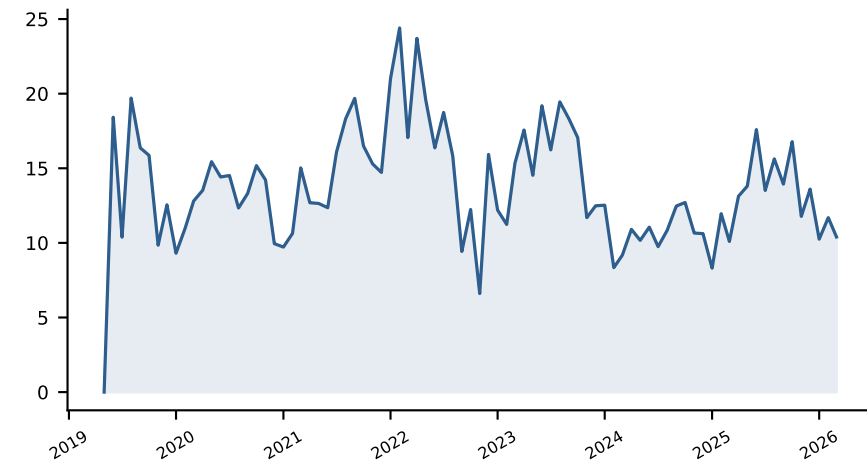


Should Statements

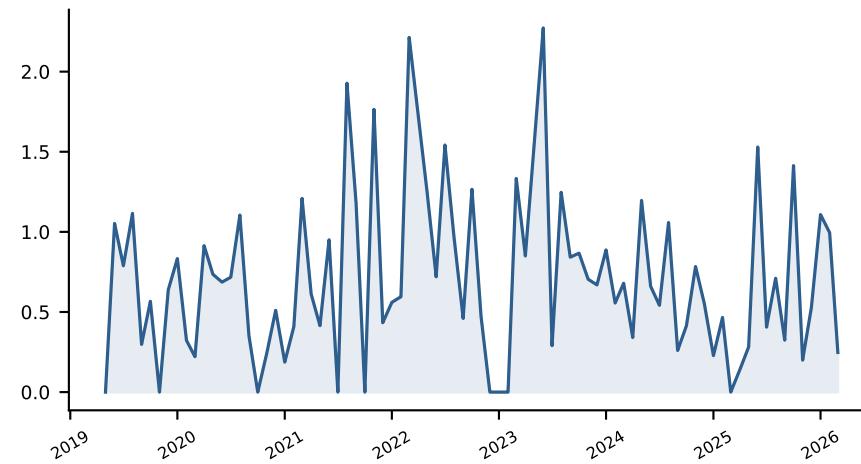


CDS Prevalence per Category Over Time (Month Level)

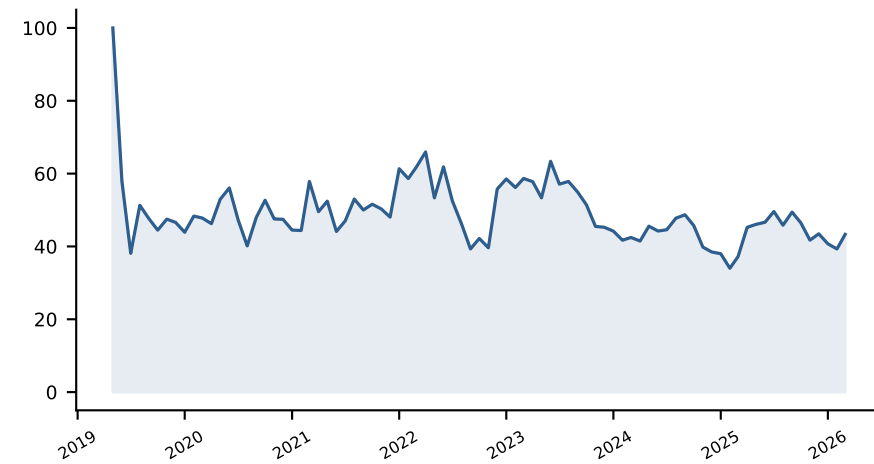
Labeling And Mislabeling



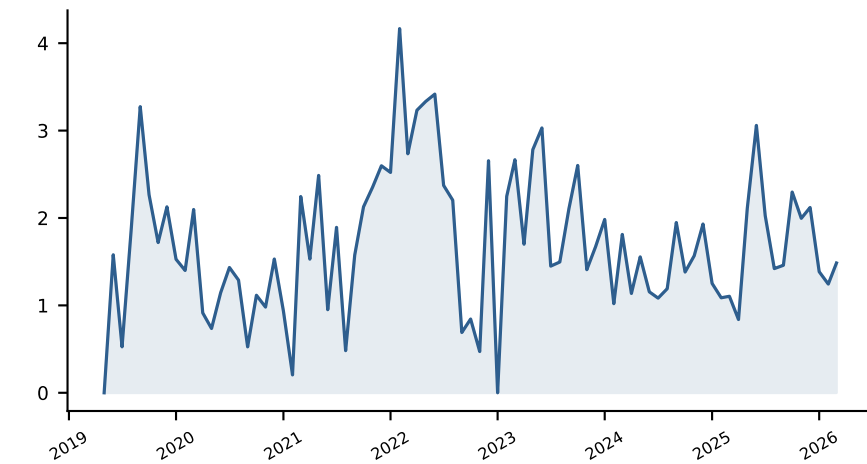
Catastrophizing



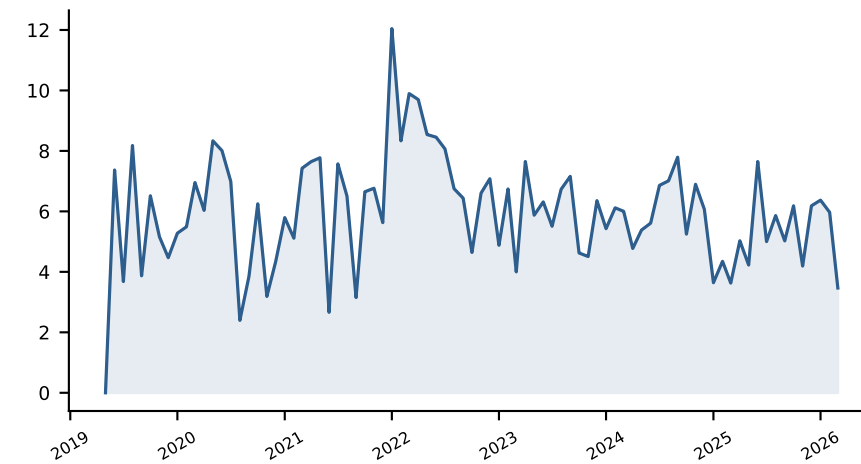
Dichotomous Reasoning



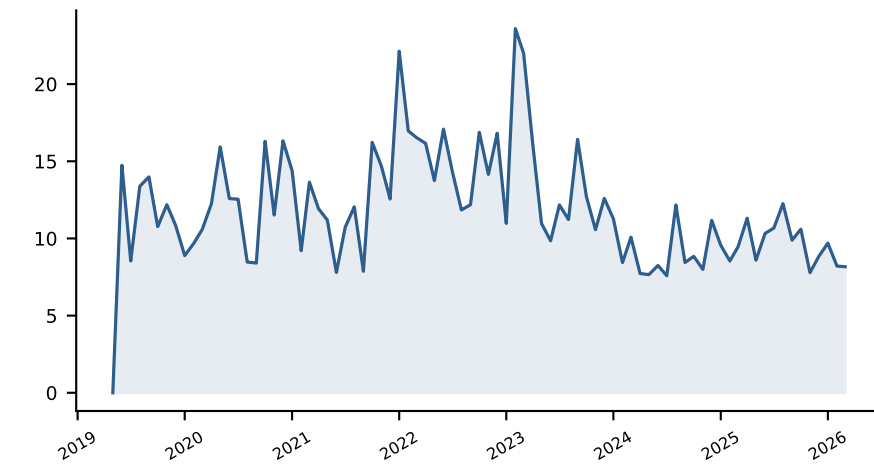
Emotional Reasoning



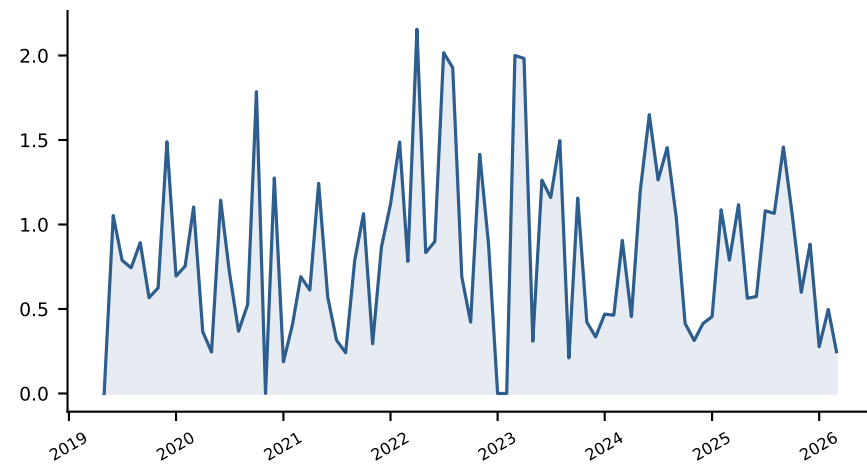
Disqualifying The Positive



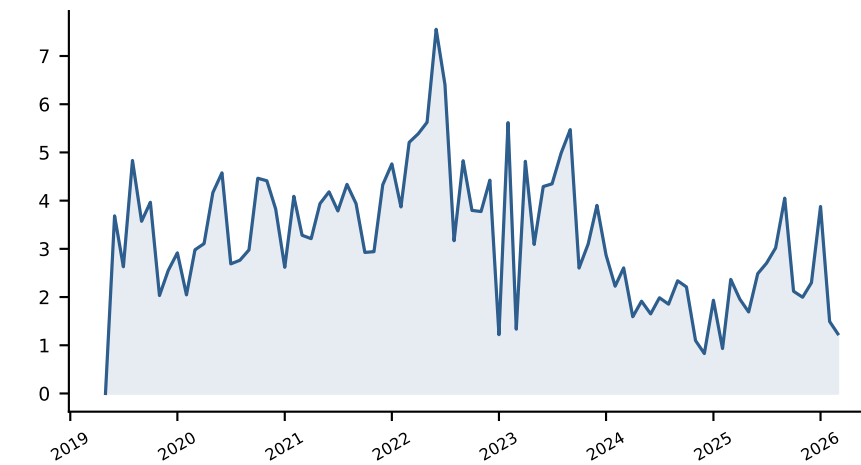
Magnification And Minimization



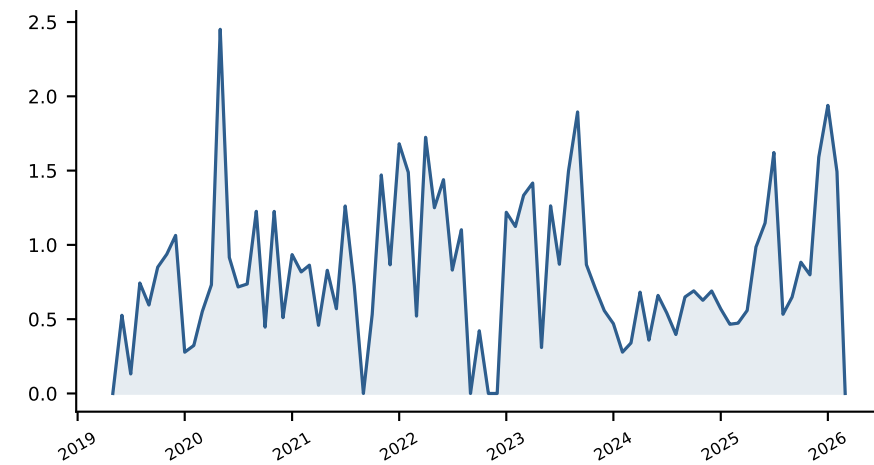
Mental Filtering



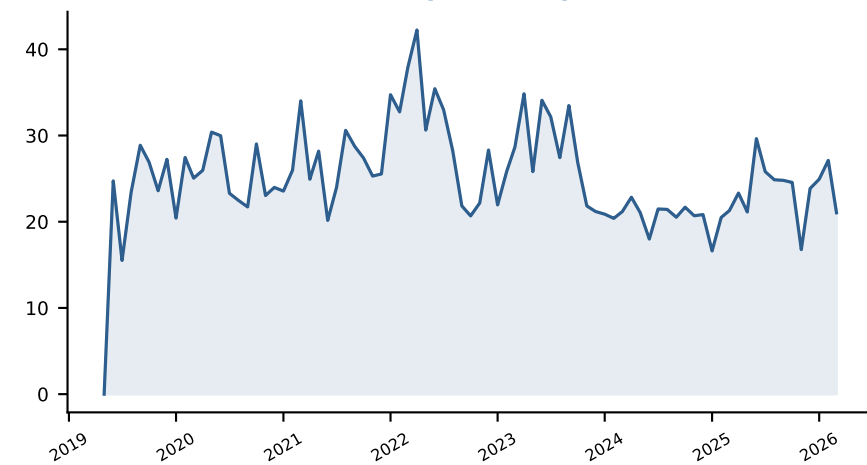
Mindreading



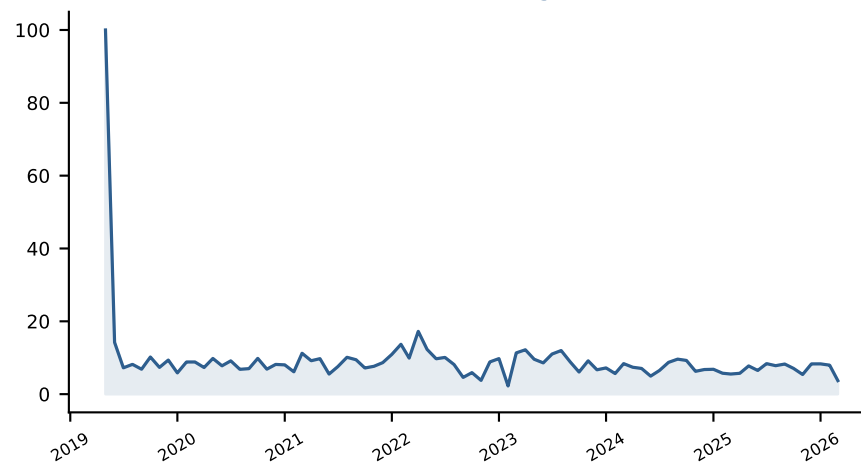
Fortune-Telling



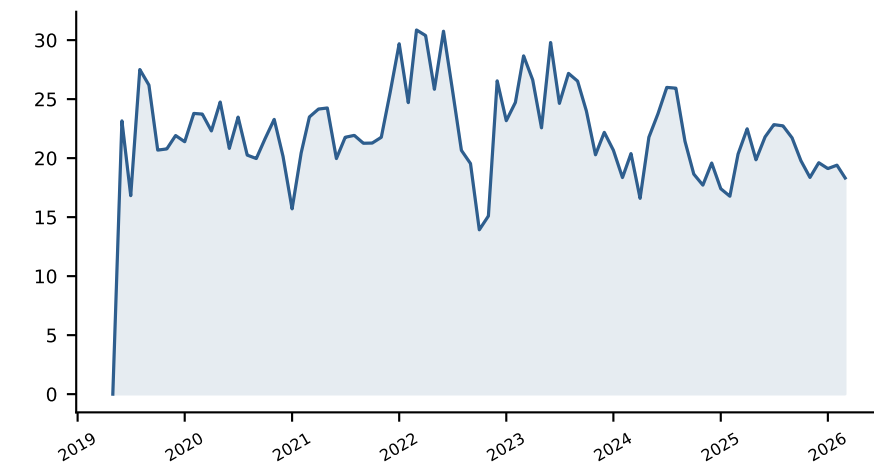
Overgeneralizing



Personalizing



Should Statements



4. Cognitive Distortion (CDS) Prevalence

Depression Connect Forum

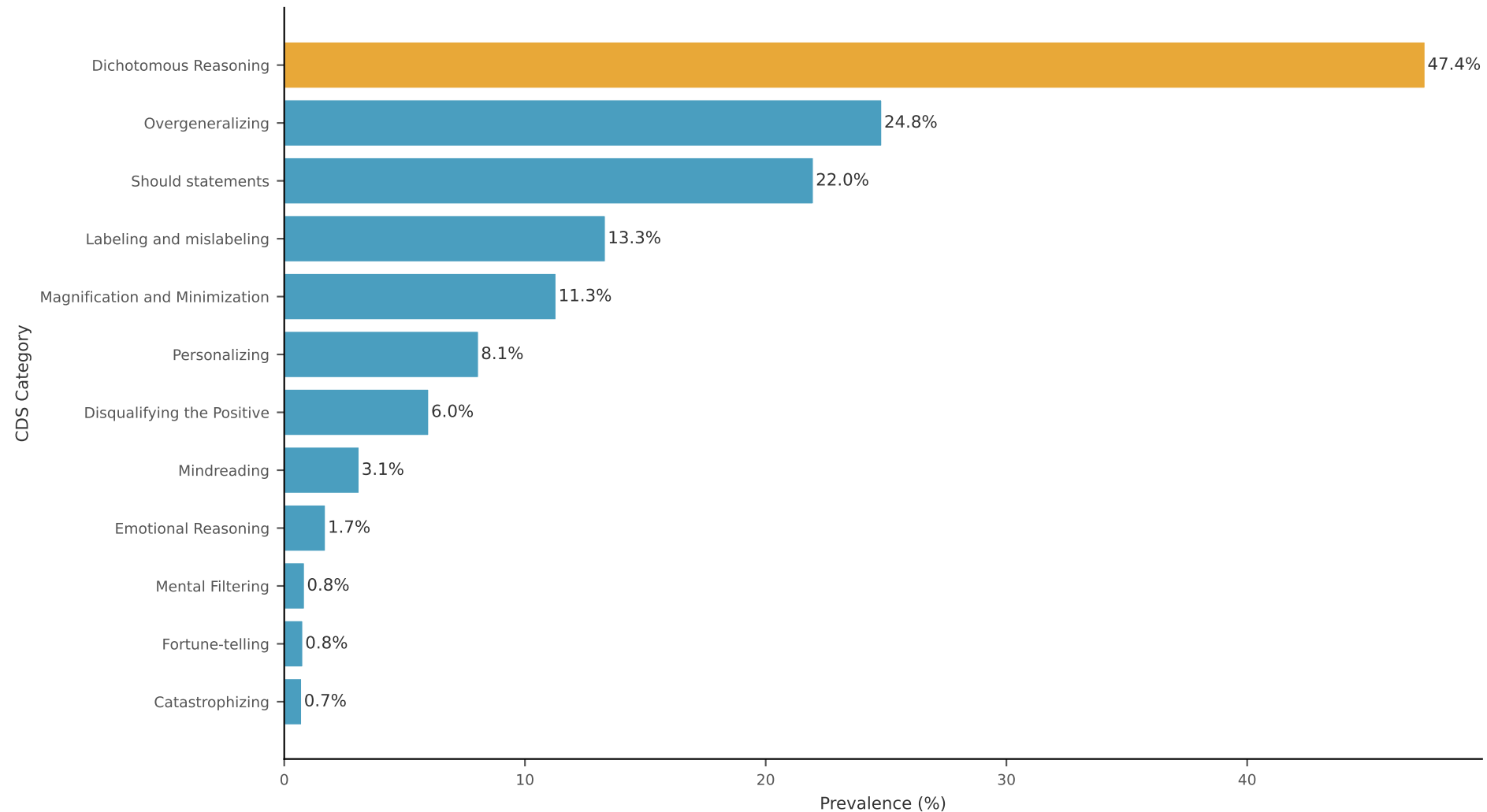
Most Common Cognitive Distortions (CDS Prevalence)

Section 1 — Category Ranking

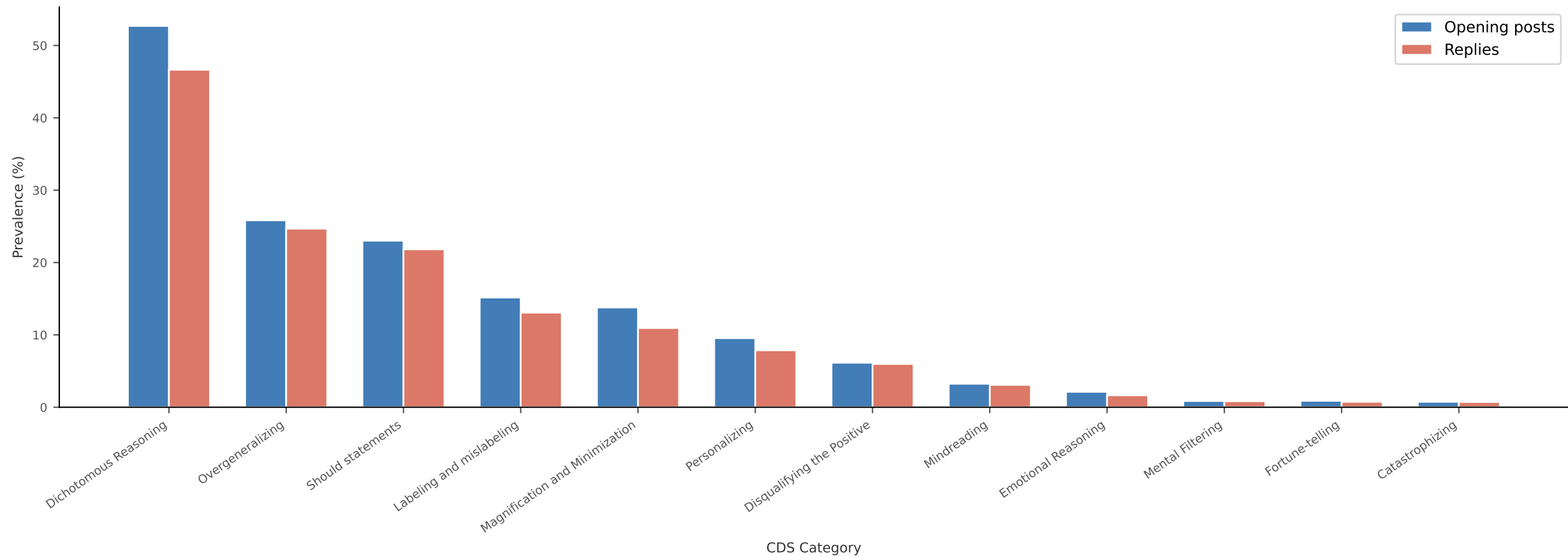
CDS Category Ranking — Mann-Whitney U (per-user rates) with BH correction
(p-BH = Benjamini-Hochberg corrected; yellow rows significant after correction)

Category	Overall %	Posts %	Replies %	Ratio P/R	N users (post)	N users (reply)	p-value (MWU)	p-BH	Rank-biserial r
Dichotomous Reasoning	47.39	52.7	46.65	1.13	774	939	0.0	0.0	-0.1446
Overgeneralizing	24.82	25.84	24.68	1.047	774	939	0.0147	0.0147	0.0675
Should statements	21.98	23.02	21.83	1.054	774	939	0.0002	0.0002	0.1042
Labeling and mislabeling	13.34	15.16	13.08	1.159	774	939	0.0	0.0	0.1282
Magnification and Minimization	11.29	13.79	10.95	1.26	774	939	0.0	0.0	0.1741
Personalizing	8.07	9.55	7.87	1.214	774	939	0.0	0.0	0.1506
Disqualifying the Positive	6.0	6.15	5.98	1.029	774	939	0.0	0.0	0.1801
Mindreading	3.11	3.24	3.09	1.049	774	939	0.0	0.0	0.142
Emotional Reasoning	1.71	2.13	1.65	1.288	774	939	0.0	0.0	0.1073
Mental Filtering	0.84	0.86	0.83	1.025	774	939	0.0	0.0	0.0848
Fortune-telling	0.77	0.89	0.75	1.187	774	939	0.0	0.0	0.0747
Catastrophizing	0.72	0.75	0.72	1.043	774	939	0.0	0.0	0.0901

CDS Category Prevalence Ranking (% of messages containing at least one phrase from this category)



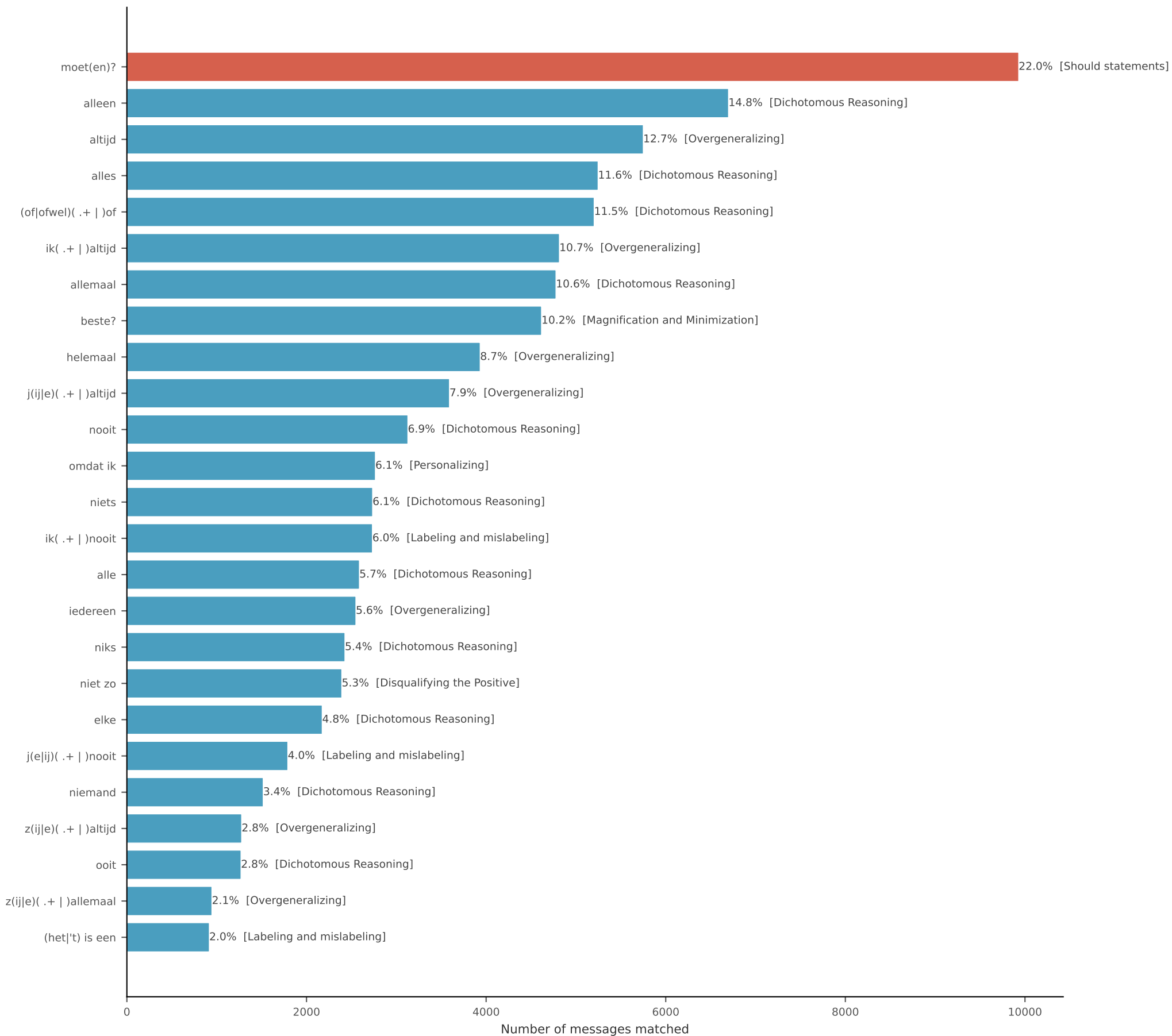
CDS Prevalence by Category — Opening Posts vs Replies



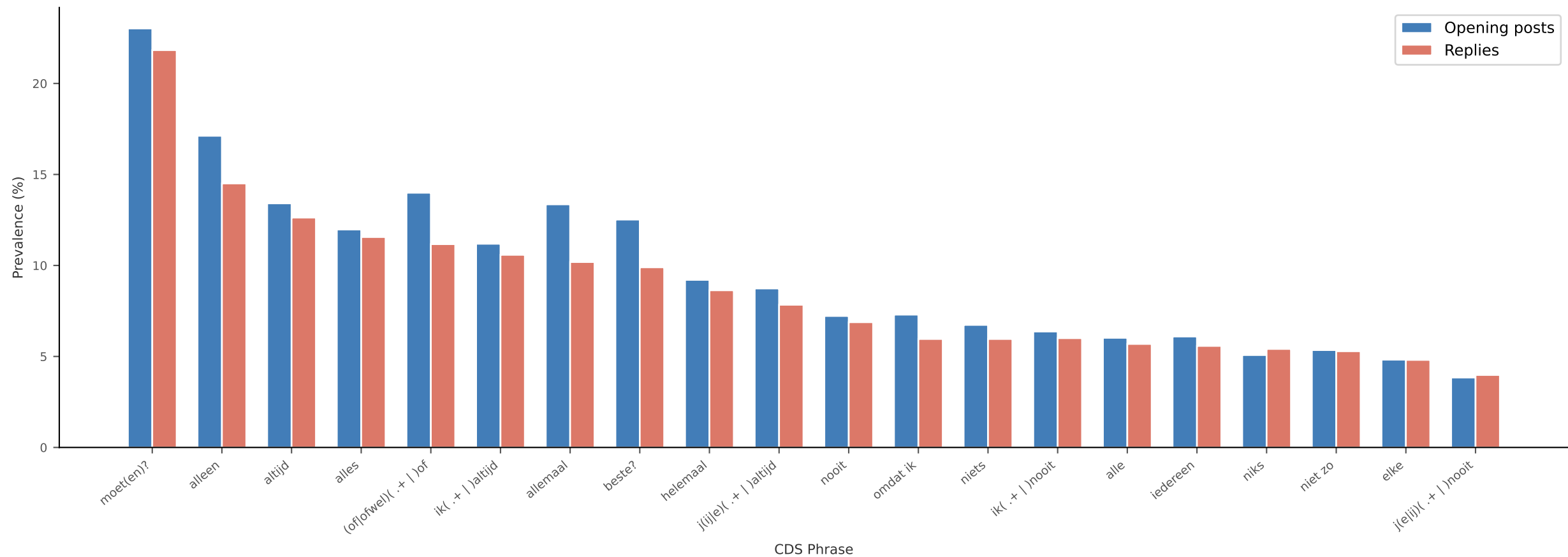
Section 2 — Individual CDS Phrases

Top 25 Most Frequent CDS Phrases (label = prevalence % | category)

CDS Phrase



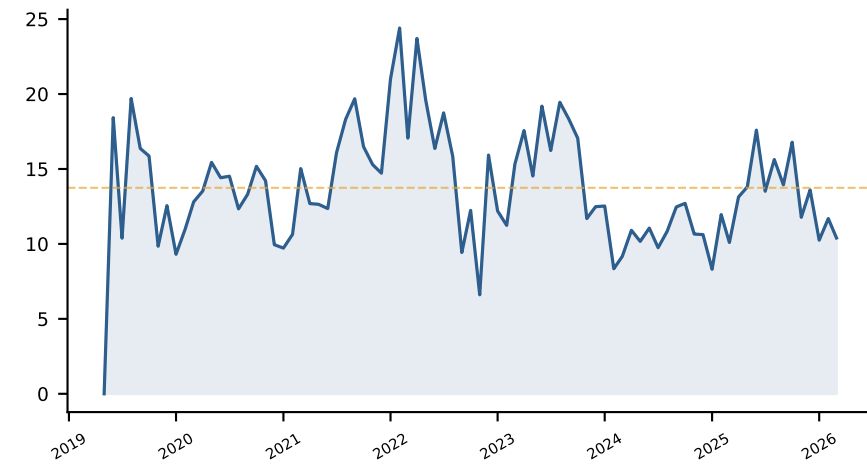
Top 20 CDS Phrases — Opening Posts vs Replies (prevalence %)



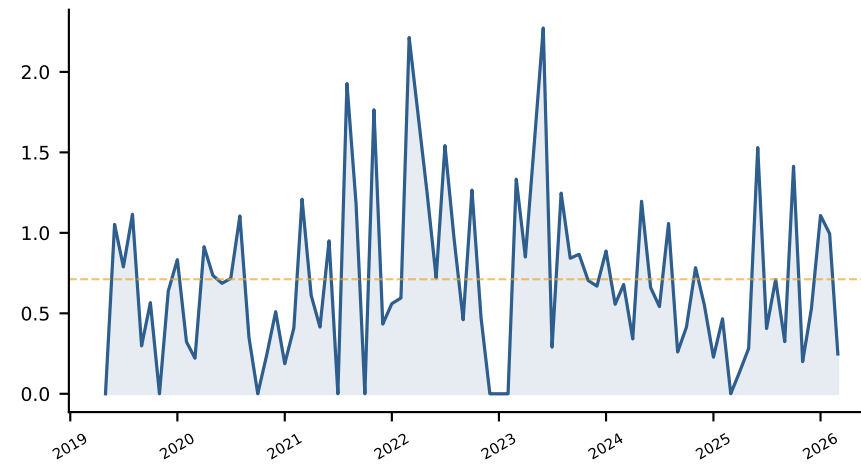
Section 3 — Category Trends Over Time

CDS Category Prevalence Over Time (Monthly, dashed = mean)

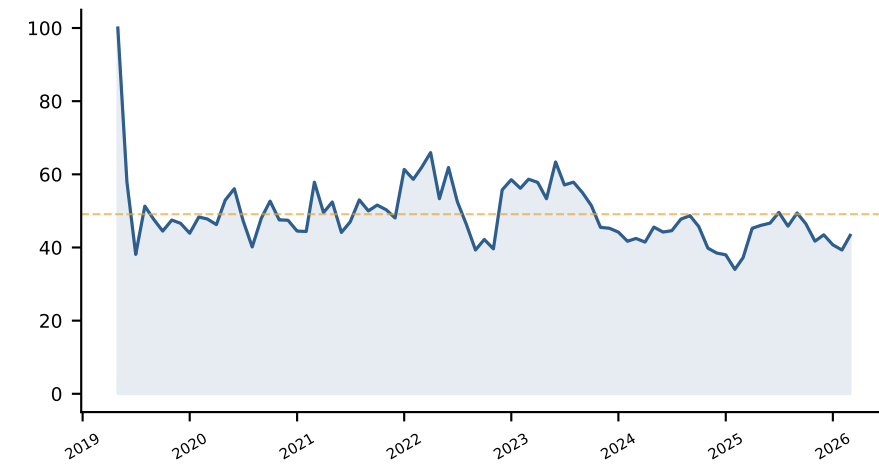
Labeling and mislabeling



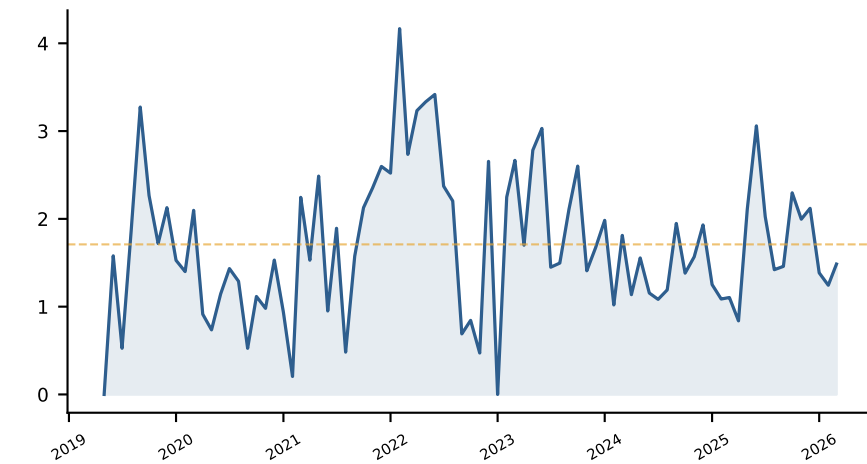
Catastrophizing



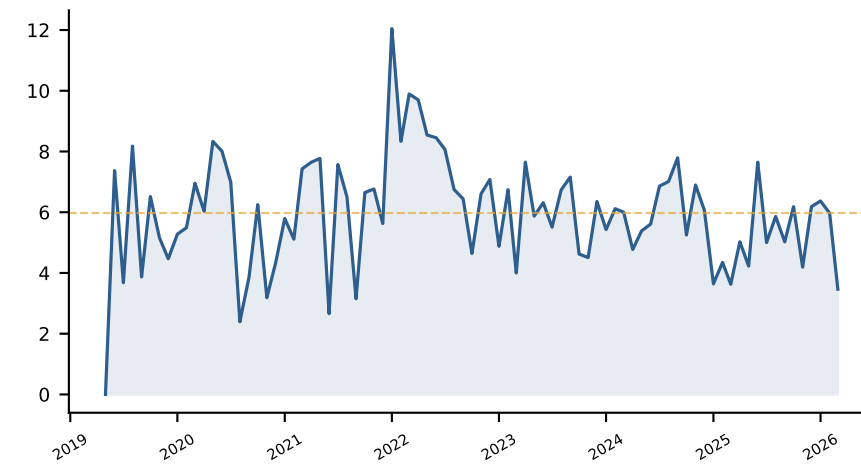
Dichotomous Reasoning



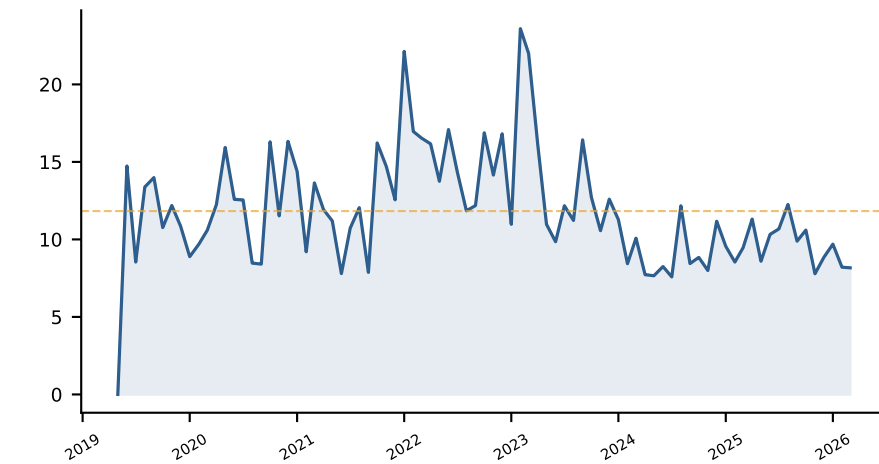
Emotional Reasoning



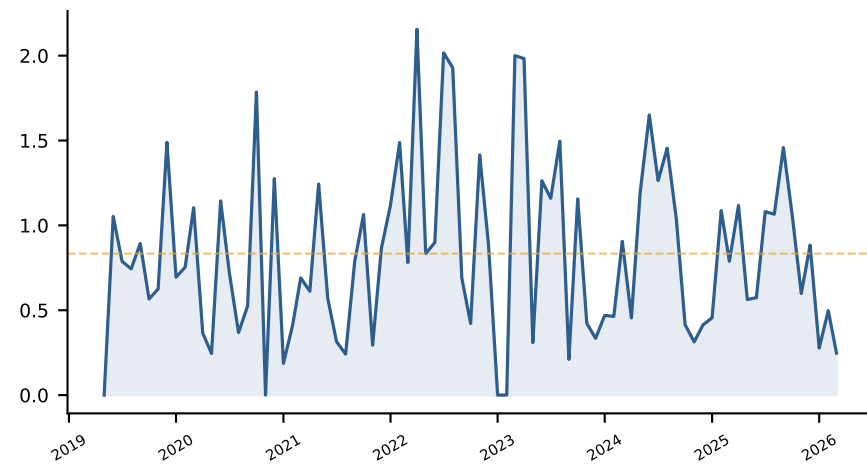
Disqualifying the Positive



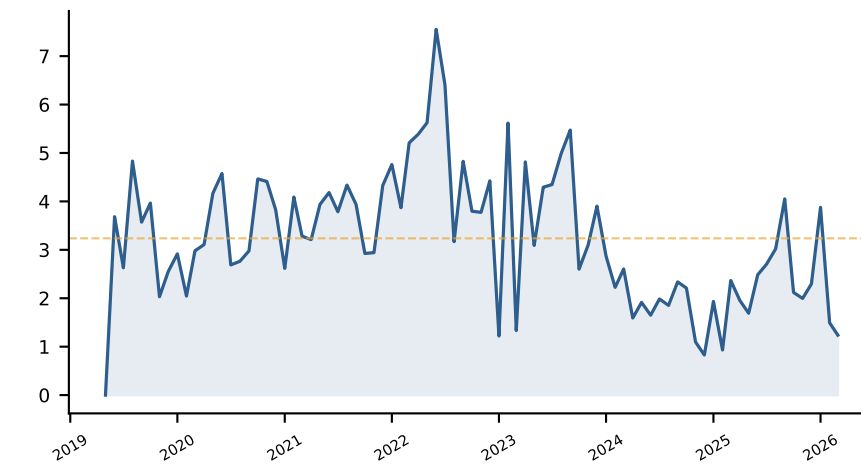
Magnification and Minimization



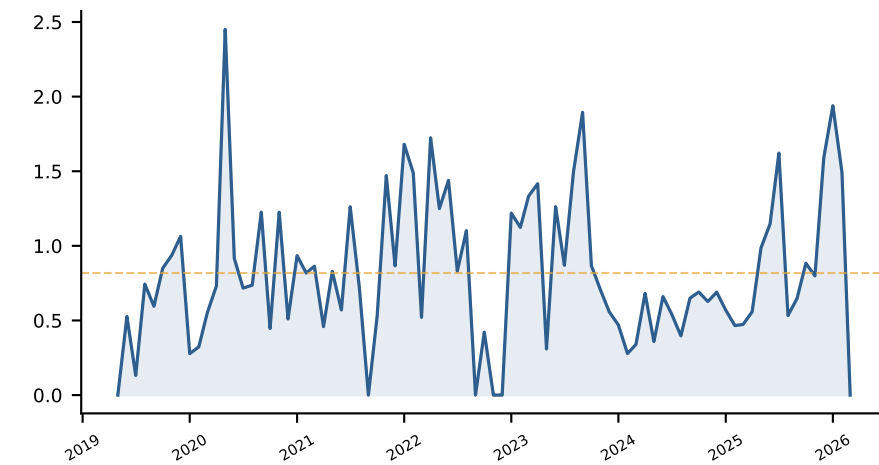
Mental Filtering



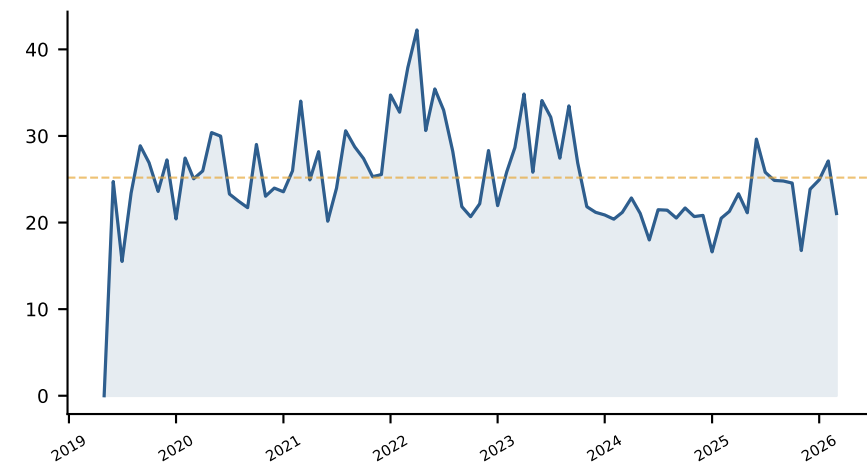
Mindreading



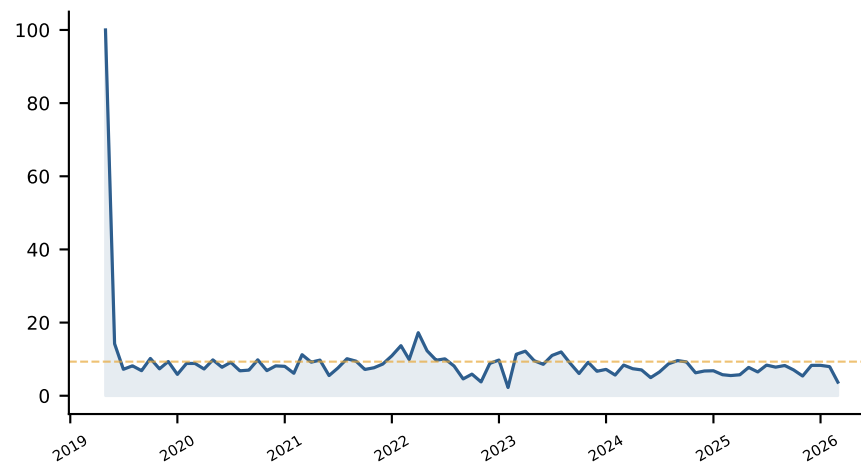
Fortune-telling



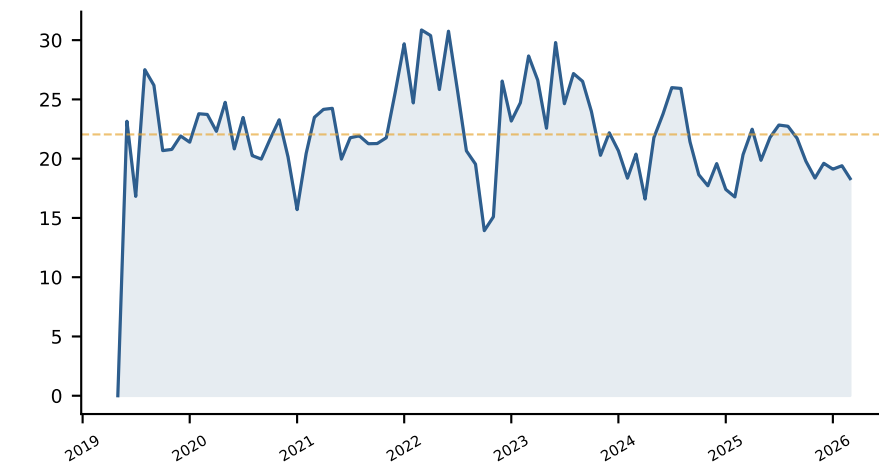
Overgeneralizing



Personalizing



Should statements



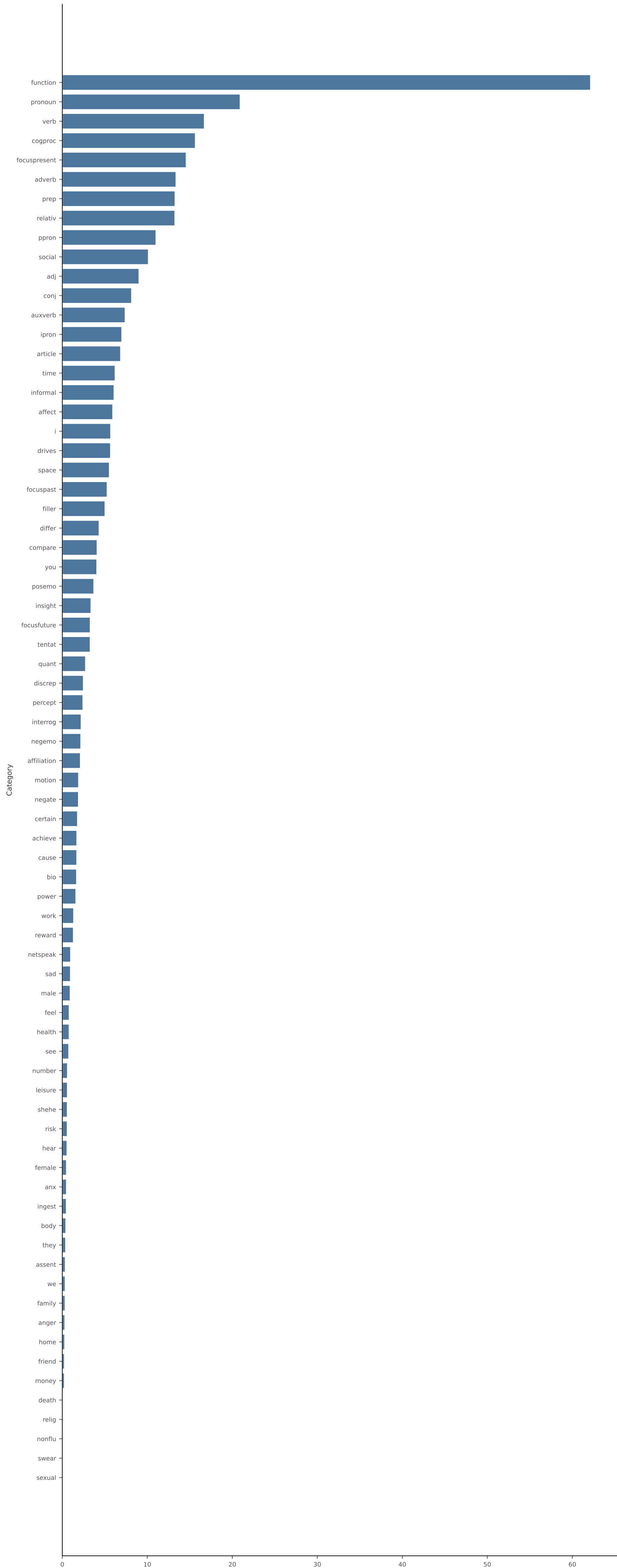
5. LIWC Psycholinguistic Analysis

Depression Connect Forum

LIWC Psycholinguistic Feature Analysis

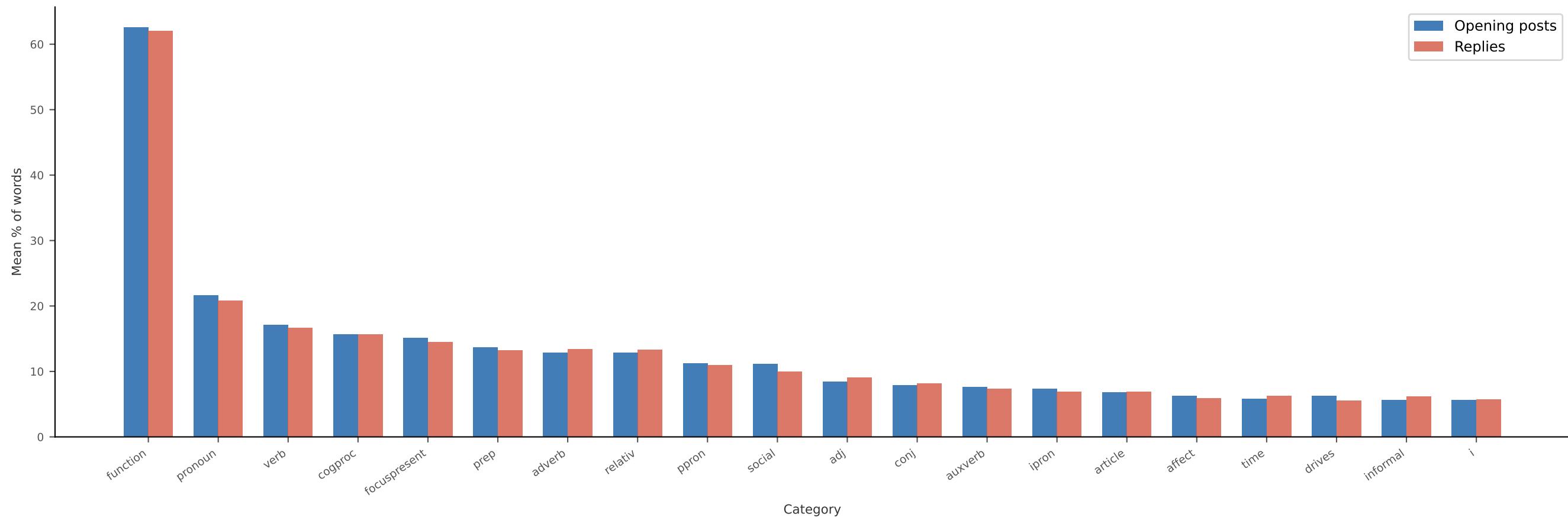
Section 1 — Overall Category Prevalence

Overall LIWC Category Prevalence (mean % of words)



Section 2 — Posts vs Replies

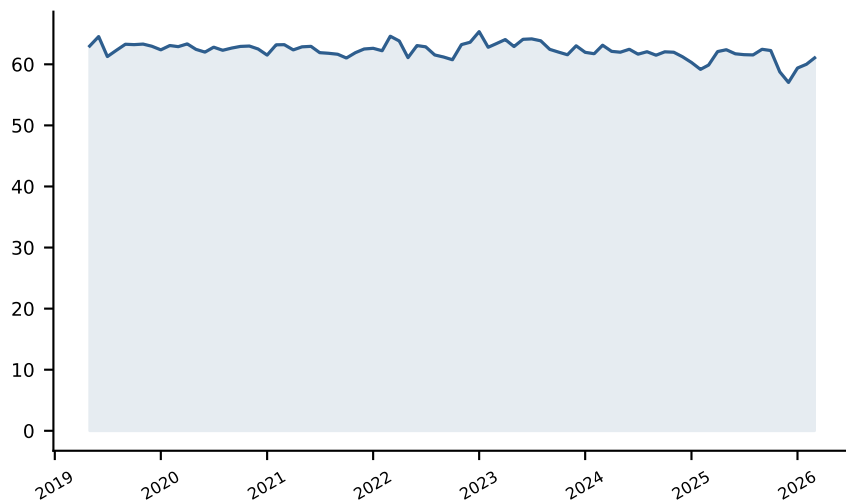
Top 20 LIWC Categories — Posts vs Replies (mean % of words)



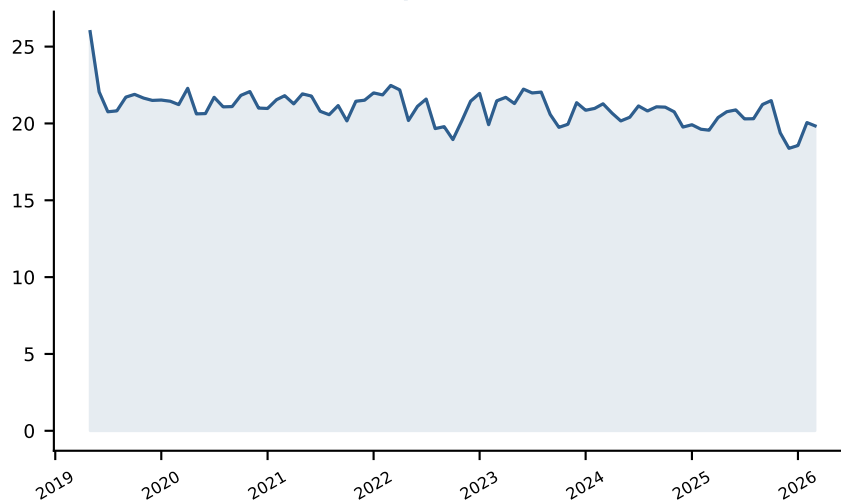
Section 3 — Top Categories Over Time

Top 12 LIWC Categories — Monthly Prevalence Over Time

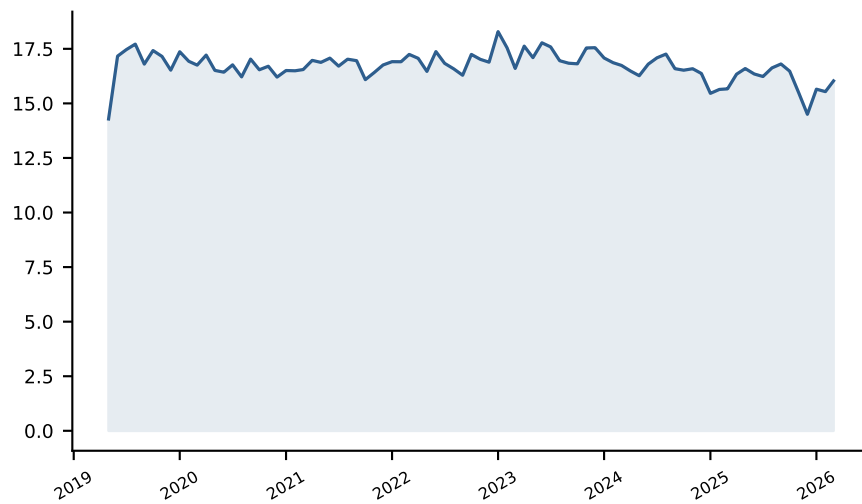
function



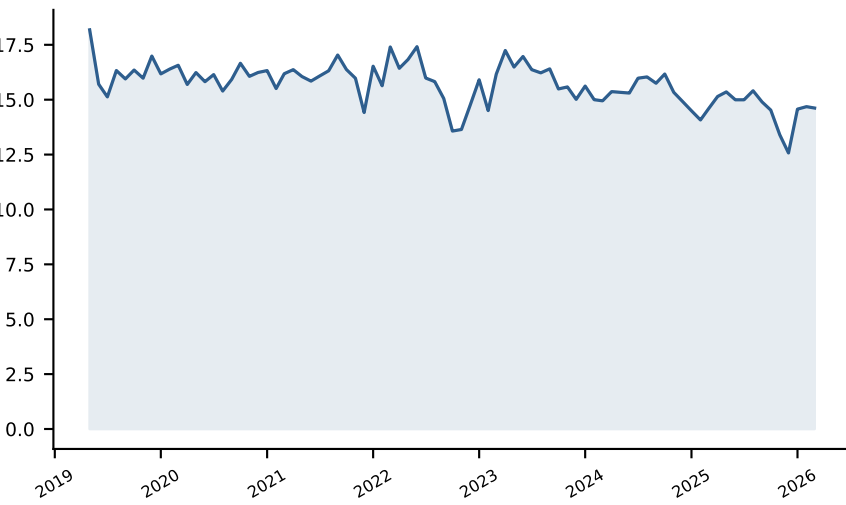
pronoun



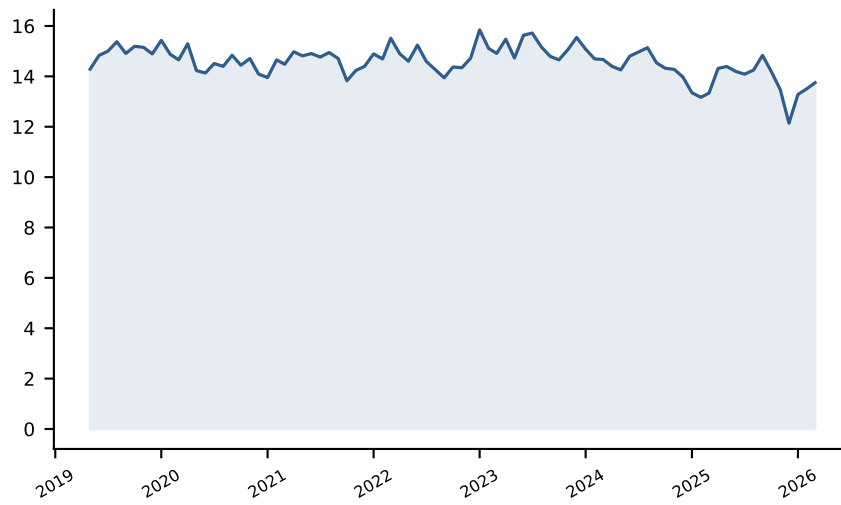
verb



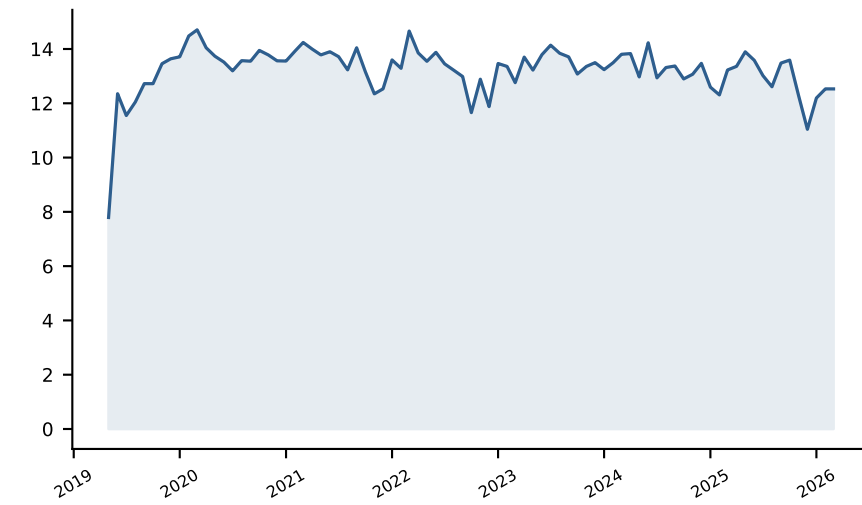
cogproc



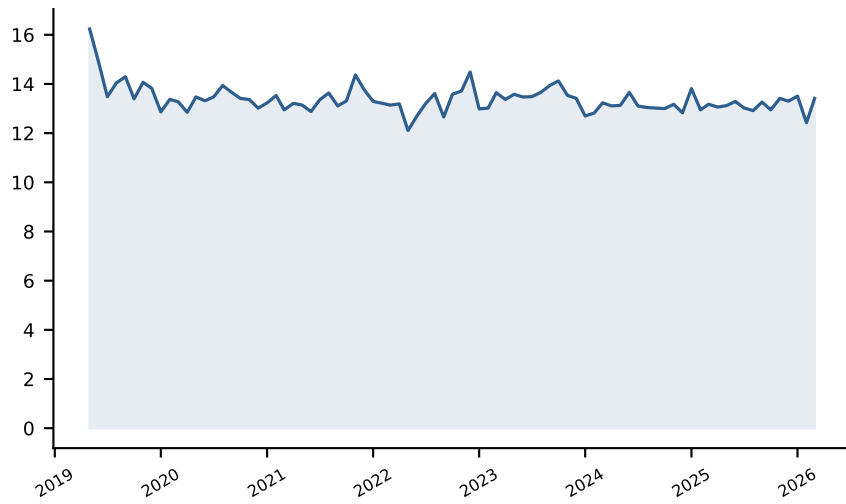
focuspresent



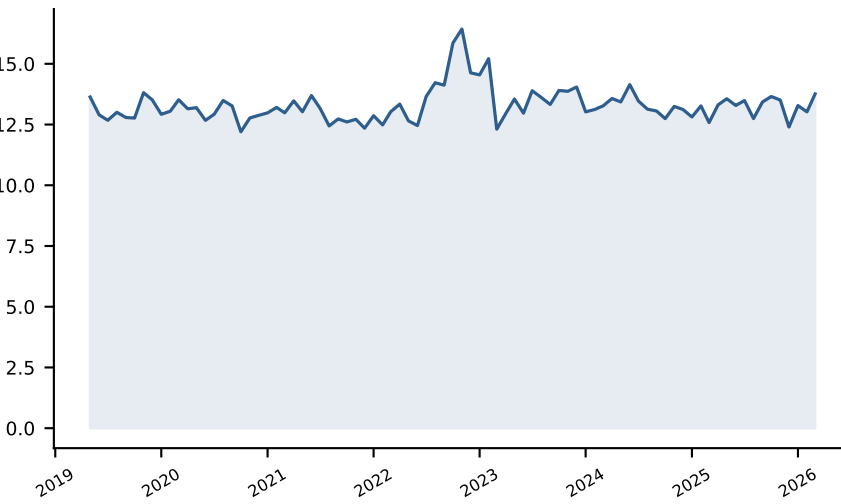
adverb



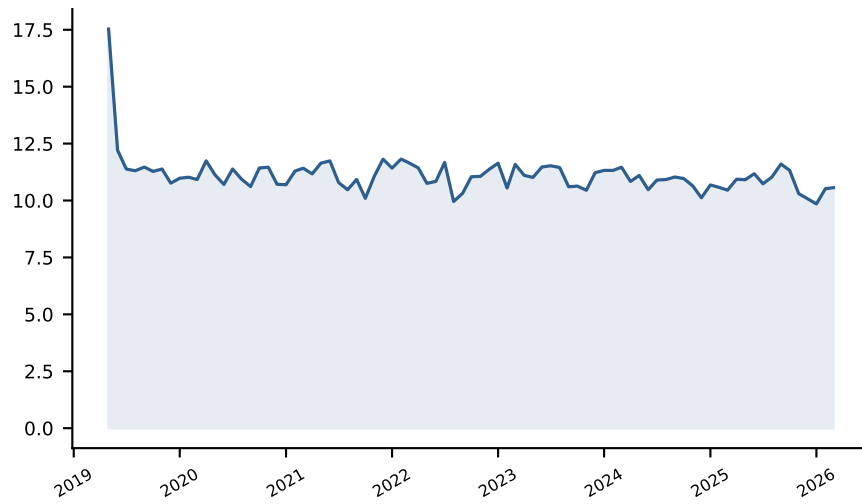
prep



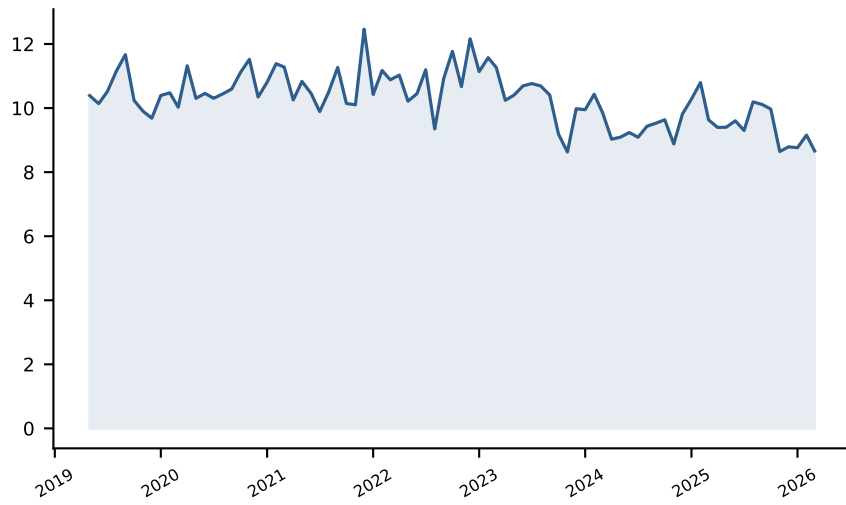
relativ



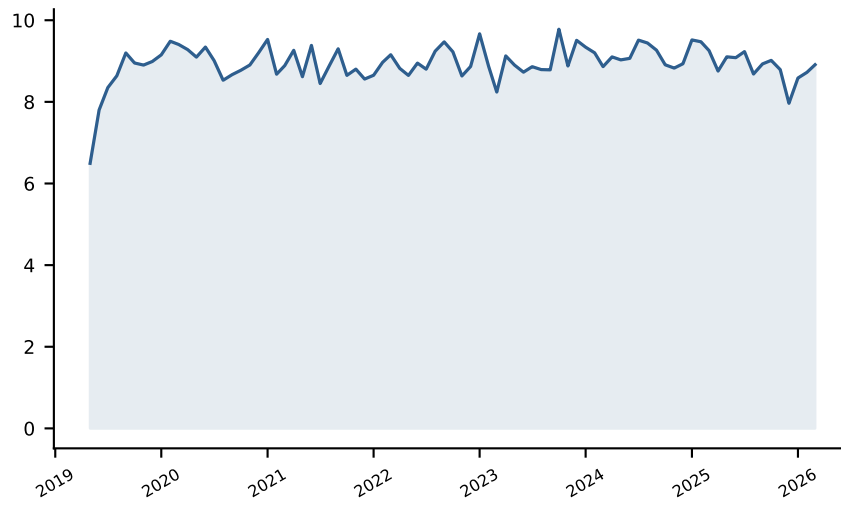
ppron



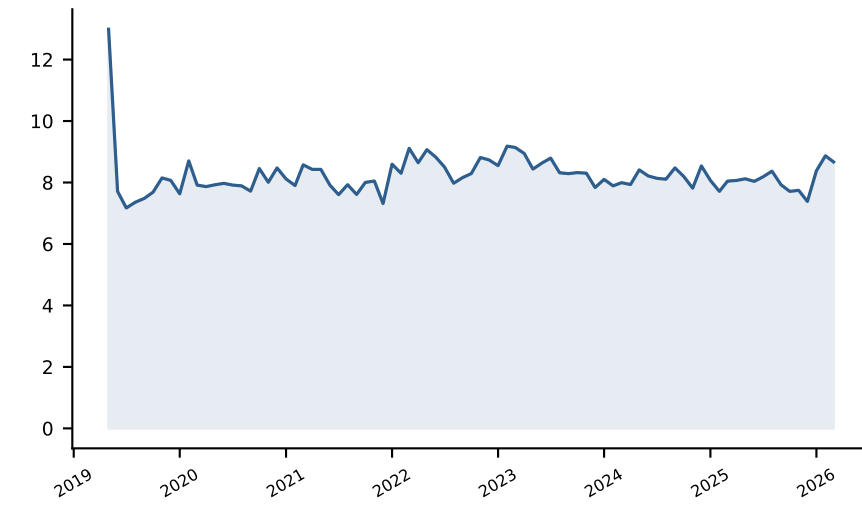
social



adj

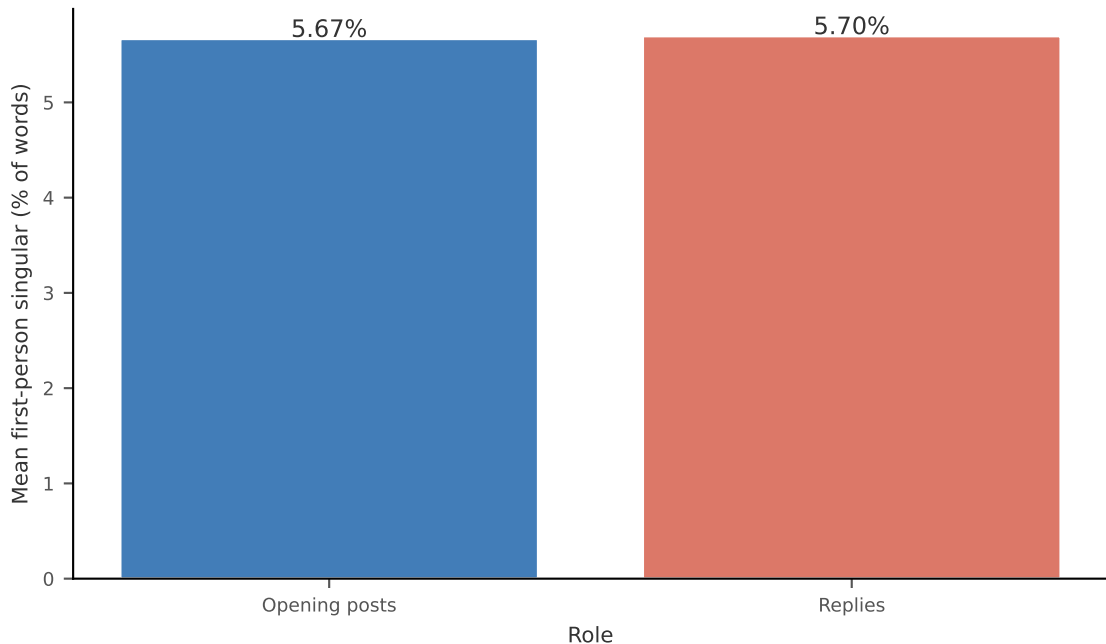


conj



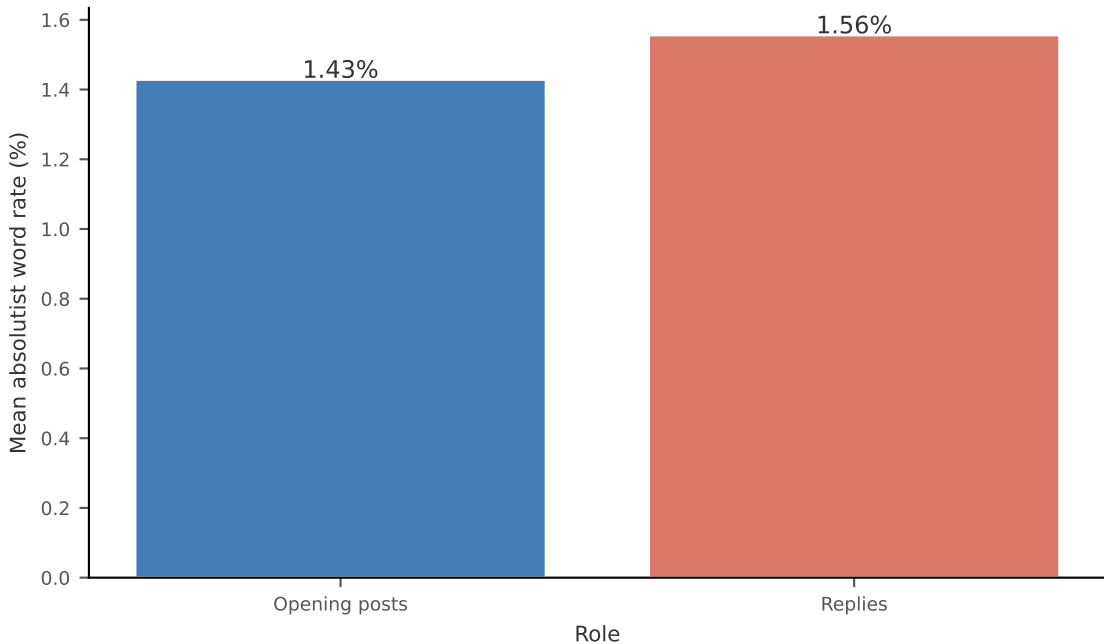
Section 4 — First-Person Singular by Role

First-Person Singular Rate by Role (self-focus marker; Smirnova 2018, Eichstaedt et al. 2018)



Section 5 — Absolutist Words

Absolutist Word Rate by Role (Dutch function-word list; Al-Mosaiwi & Johnstone 2018)



6. Per-User Longitudinal Trends

Depression Connect Forum

Longitudinal Analysis — Top 5 Most Active Posters

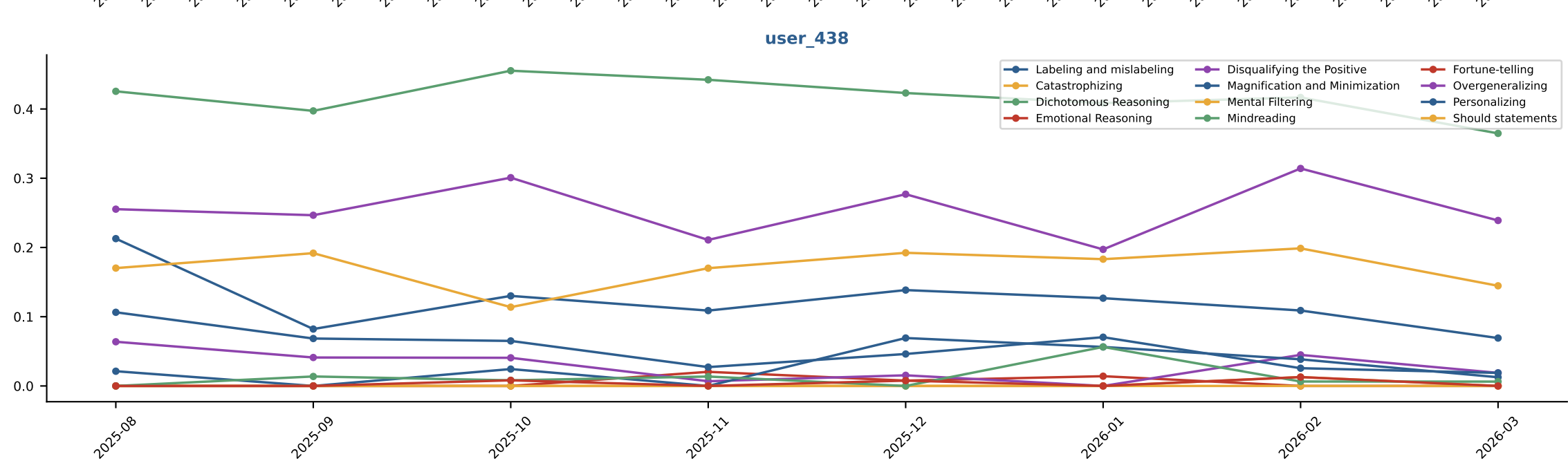
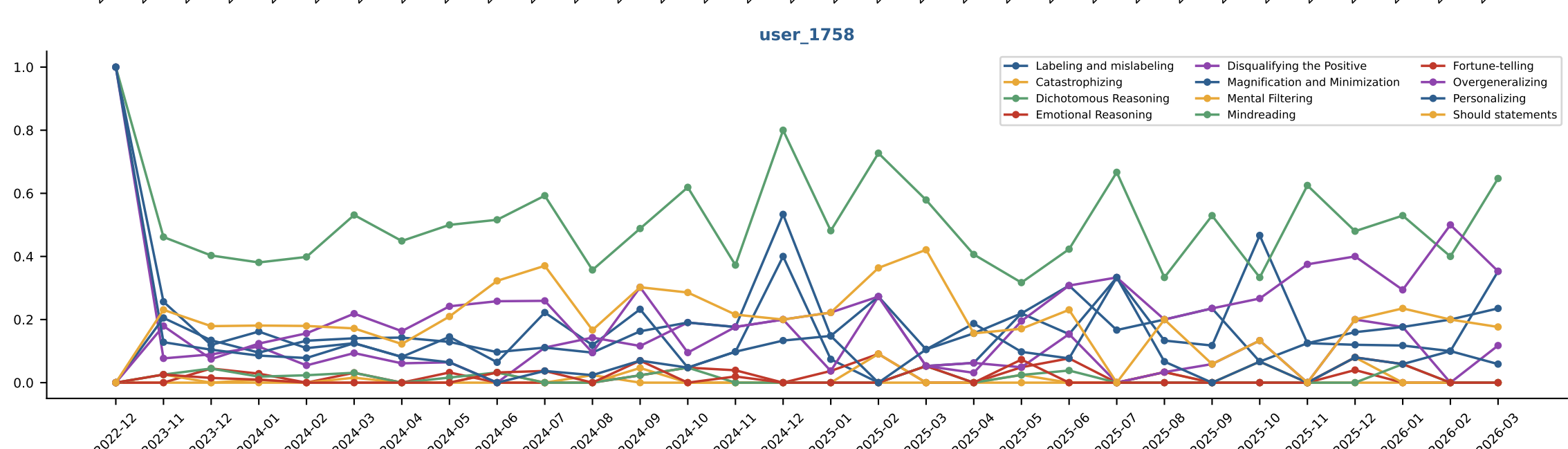
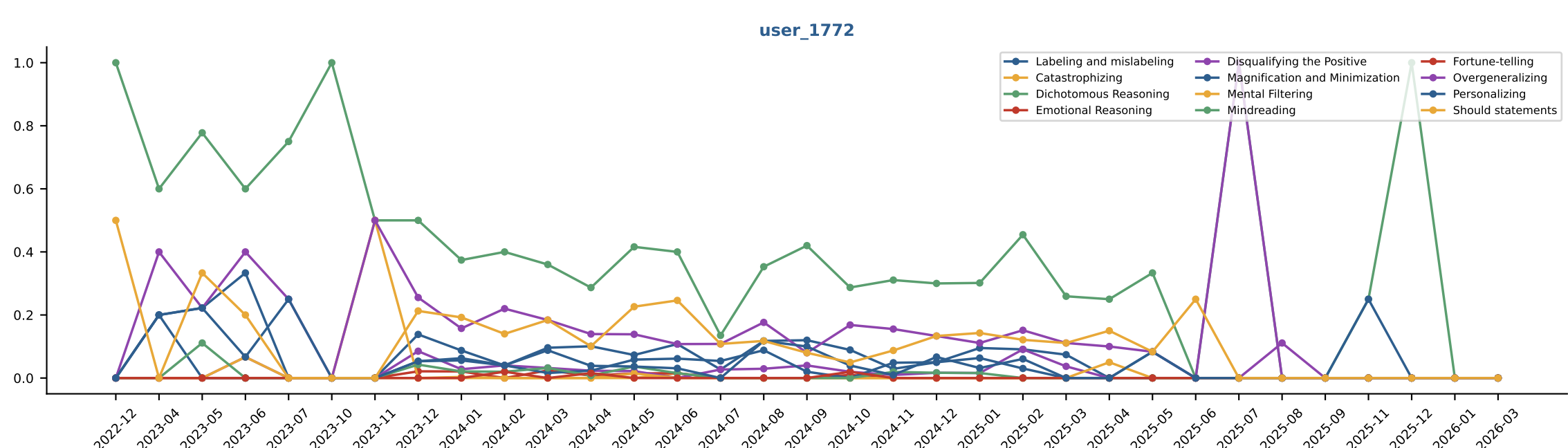
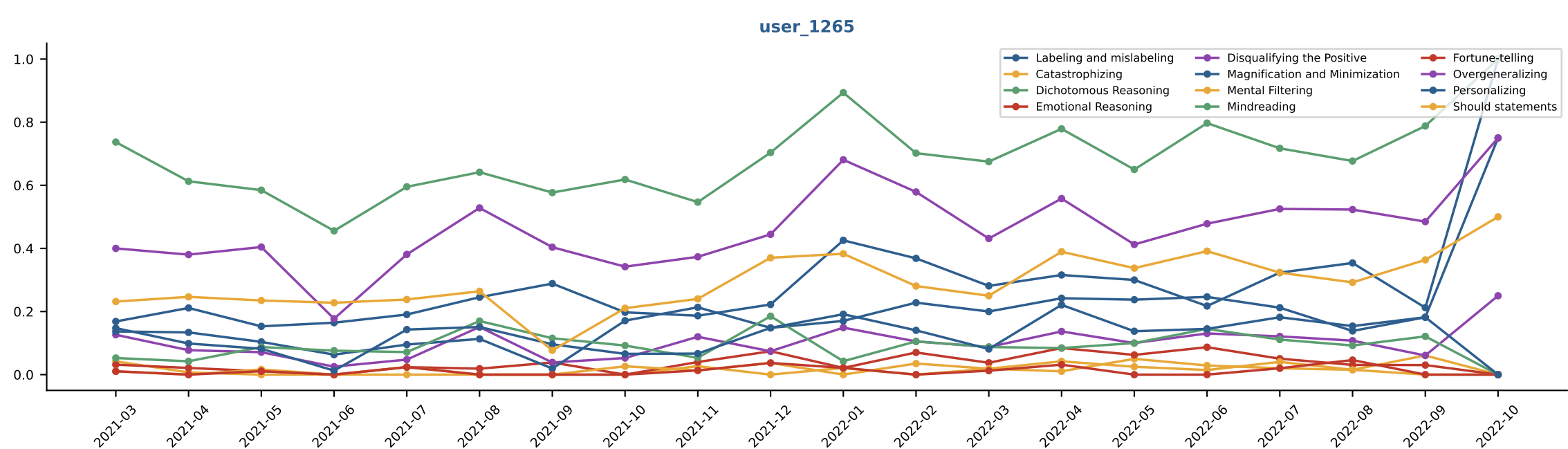
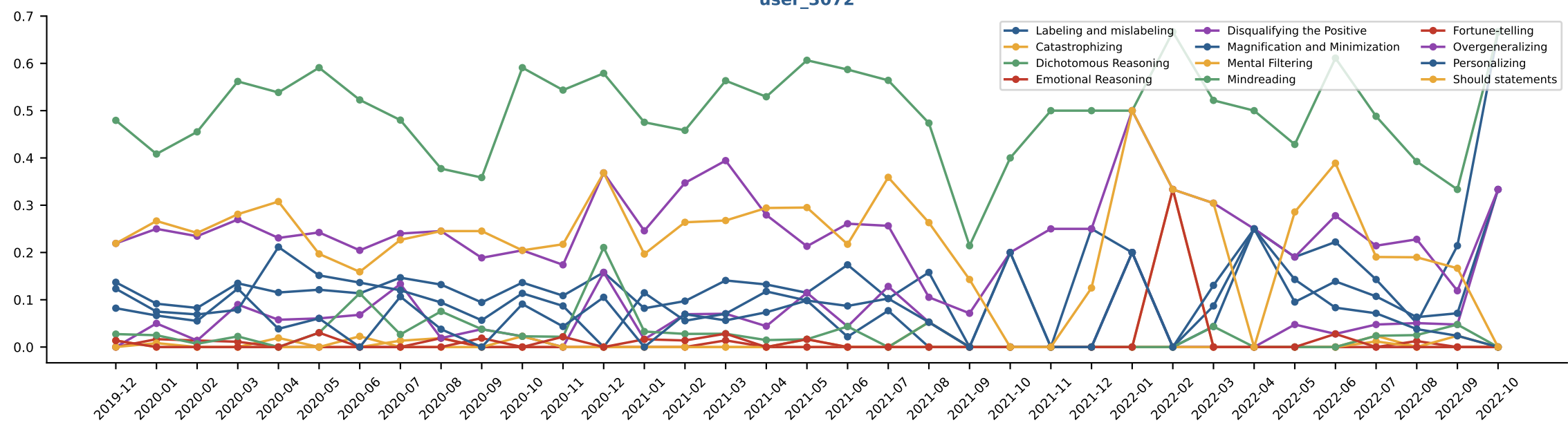
Selected Users — Top by Raw Post Volume

User	Posts	Active days	Posts / active month
user_3072	1656	1030	48.94
user_1265	1533	579	80.6
user_1772	1486	1160	38.99
user_1758	1046	1181	26.96
user_438	906	229	120.43

Selection is by total post count only. Note this over-weights high-intensity users and can miss long-active, moderate-volume users (see --select sustained).

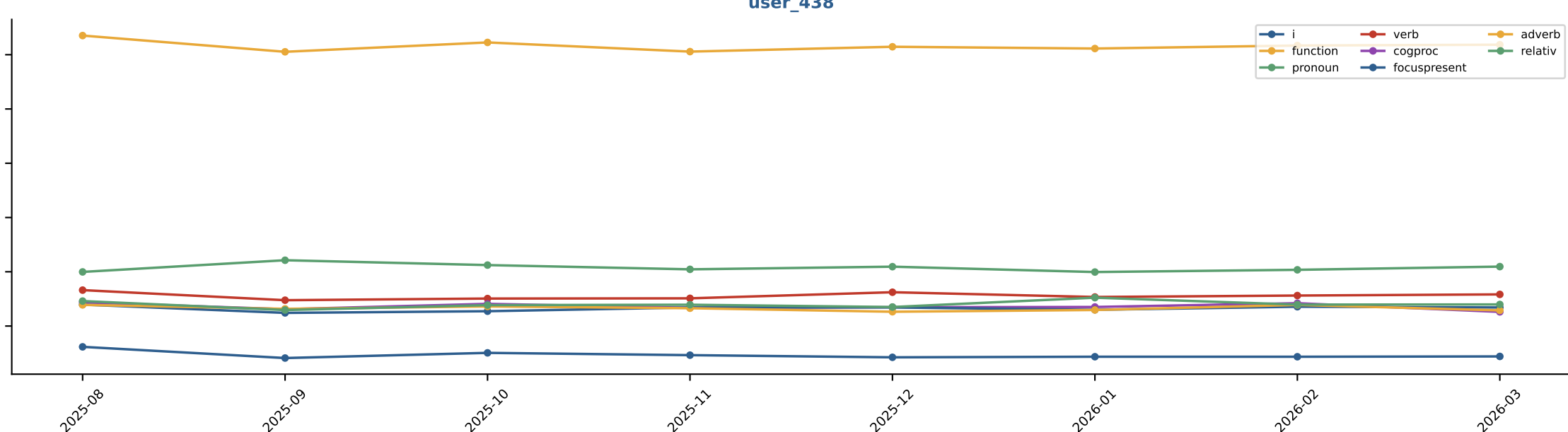
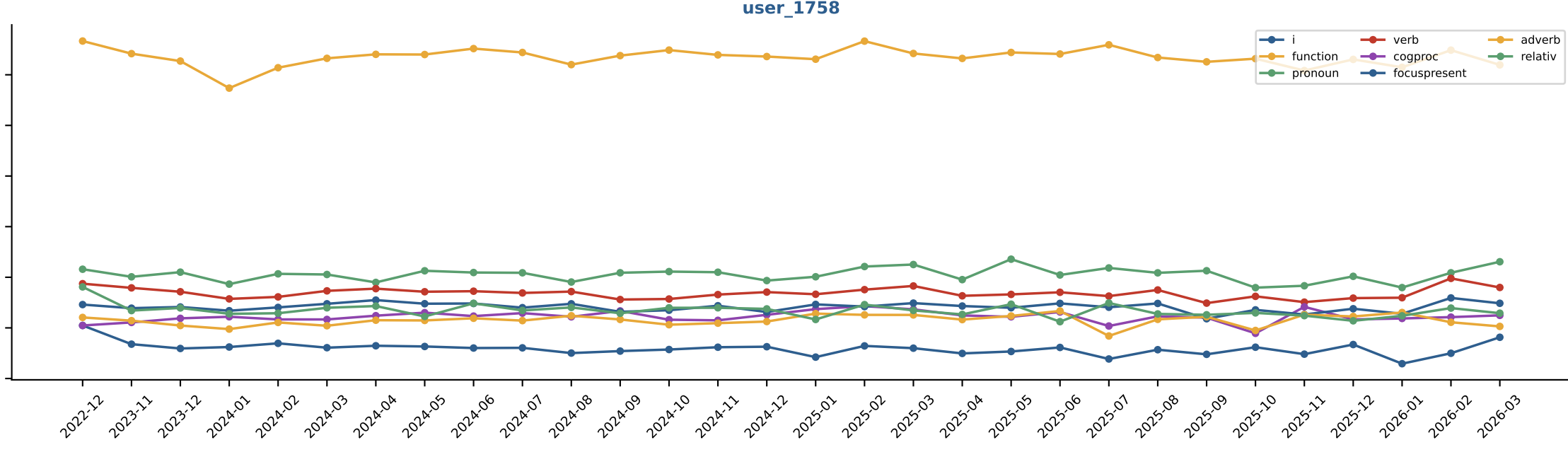
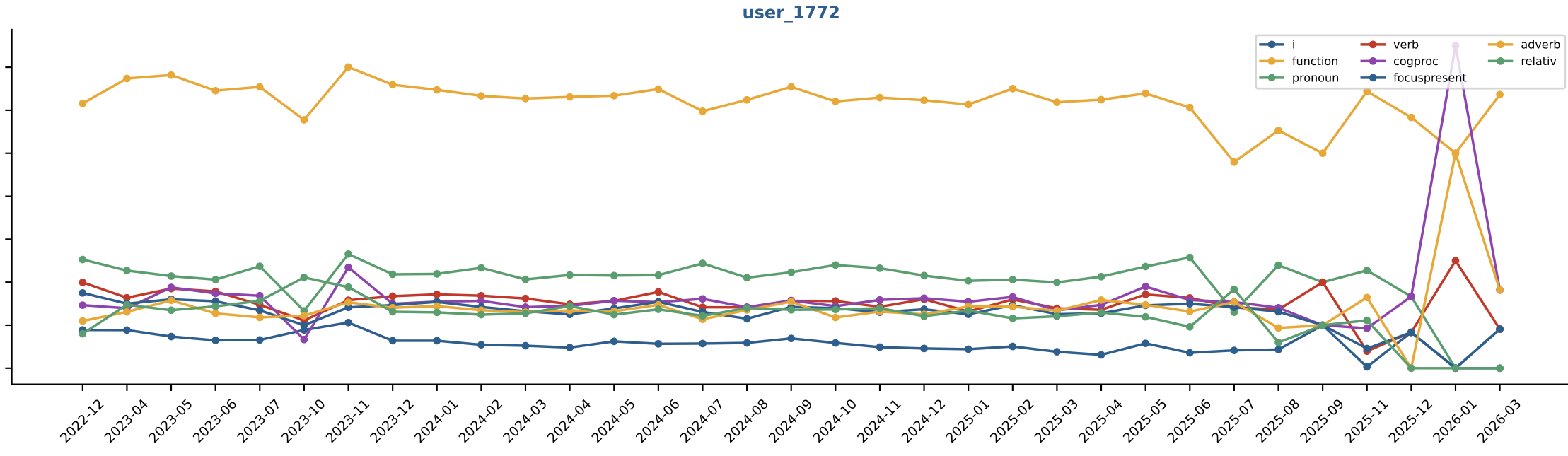
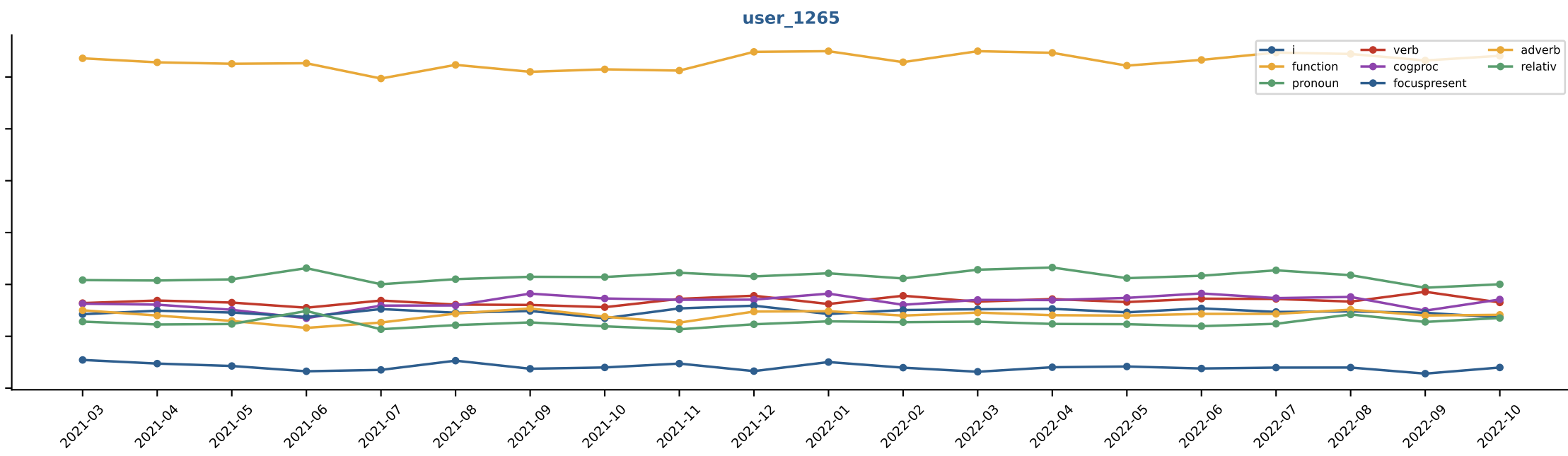
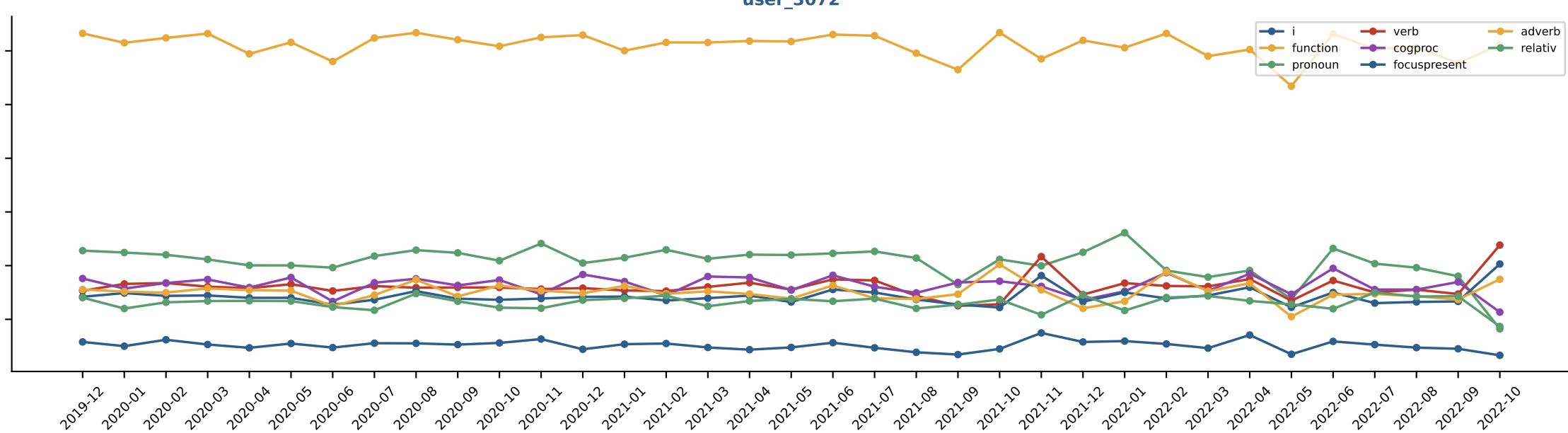
CDS Category Scores Per Month

CDS Categories Over Time (mean per month)



LIWC Category Scores Per Month

Top LIWC Categories Over Time (mean % of words per month)



7. LIWC-22 Validation

LIWC-22 Validation Report

Custom scorer vs LIWC-22 CLI (Dutch LIWC-2015 dictionary)

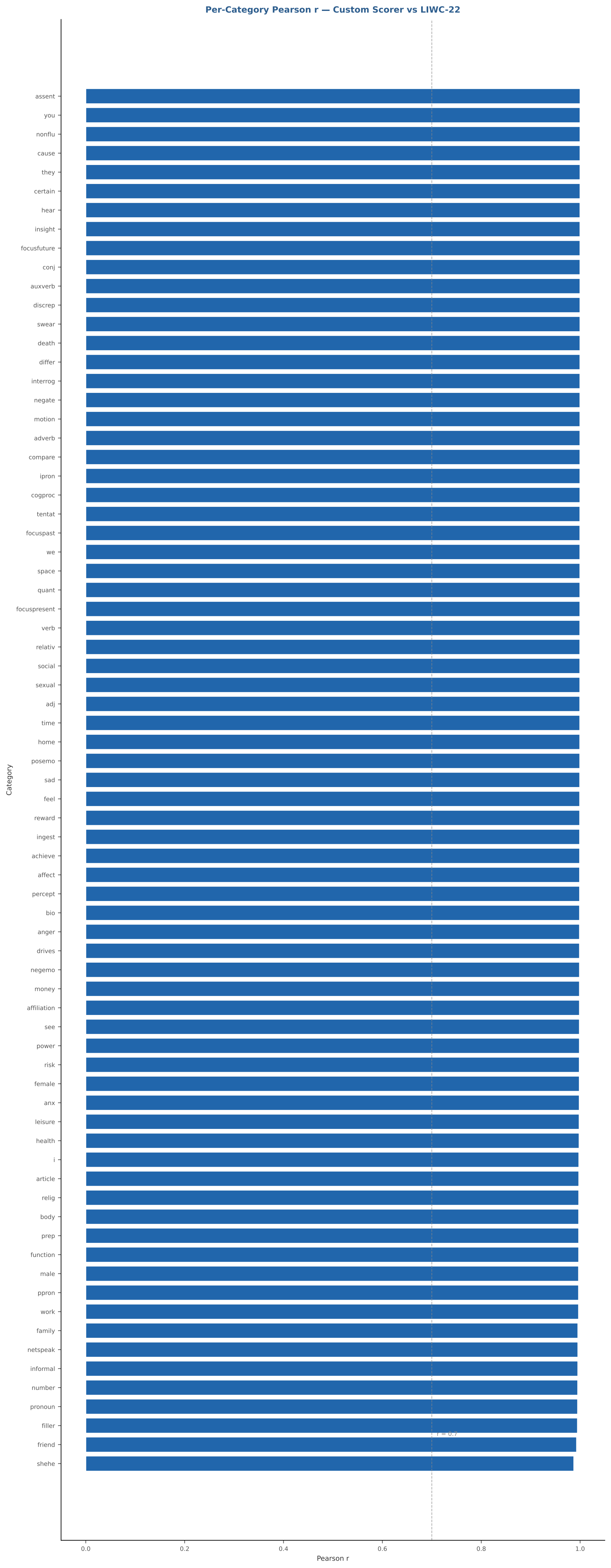
Dataset: Combined Dataset

Section 1 — Scoring Agreement

Per-Category Scoring Agreement (sorted by MAE descending)
MAE = mean absolute difference in percentage points; yellow = MAE > 2 pp

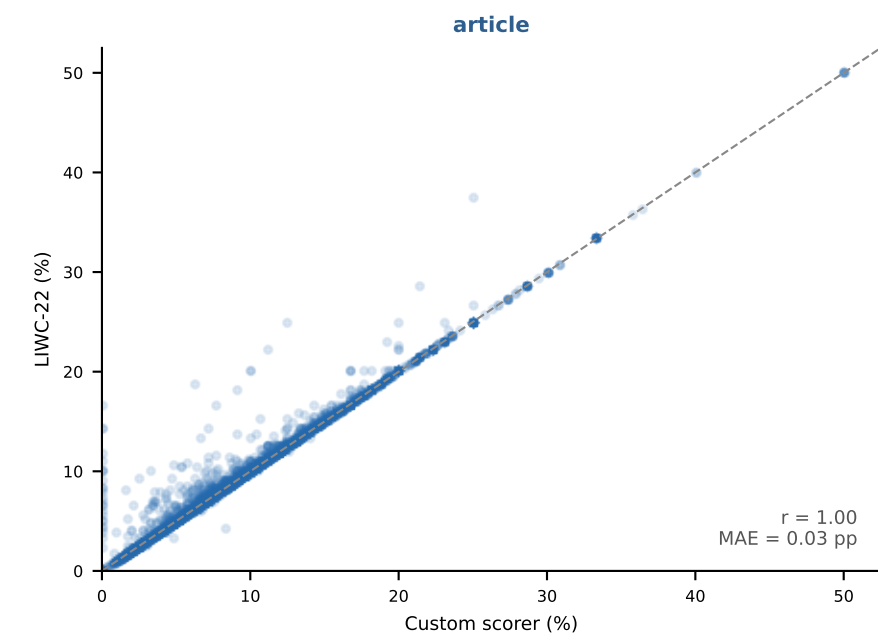
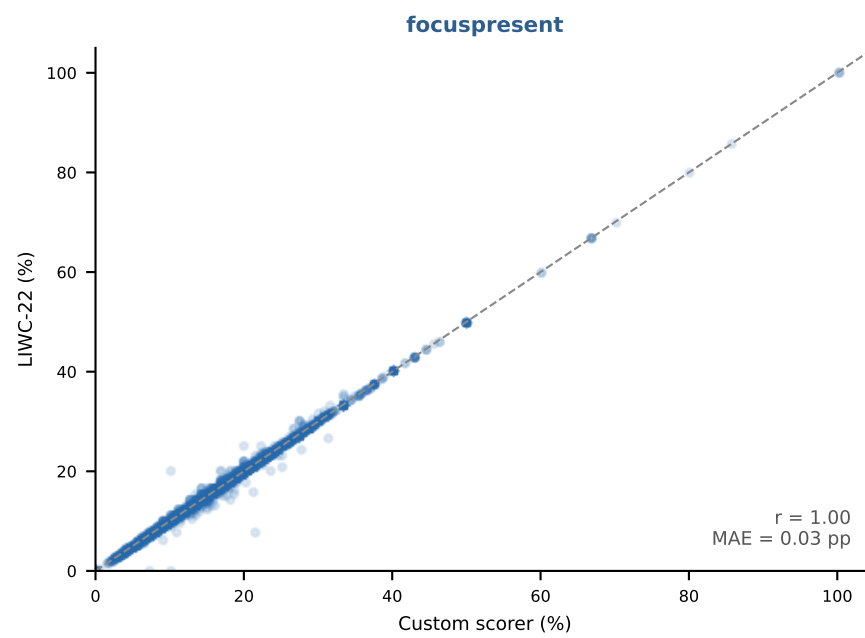
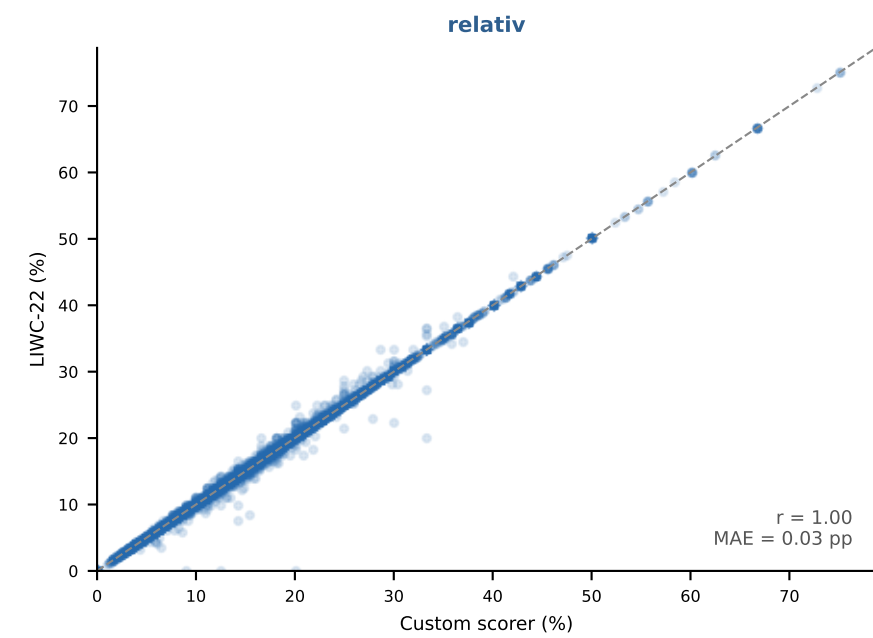
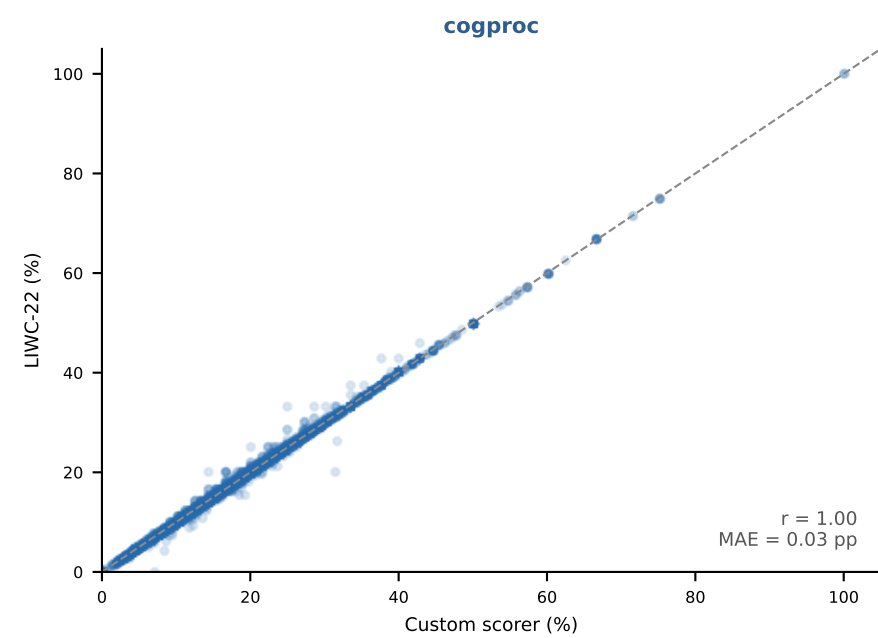
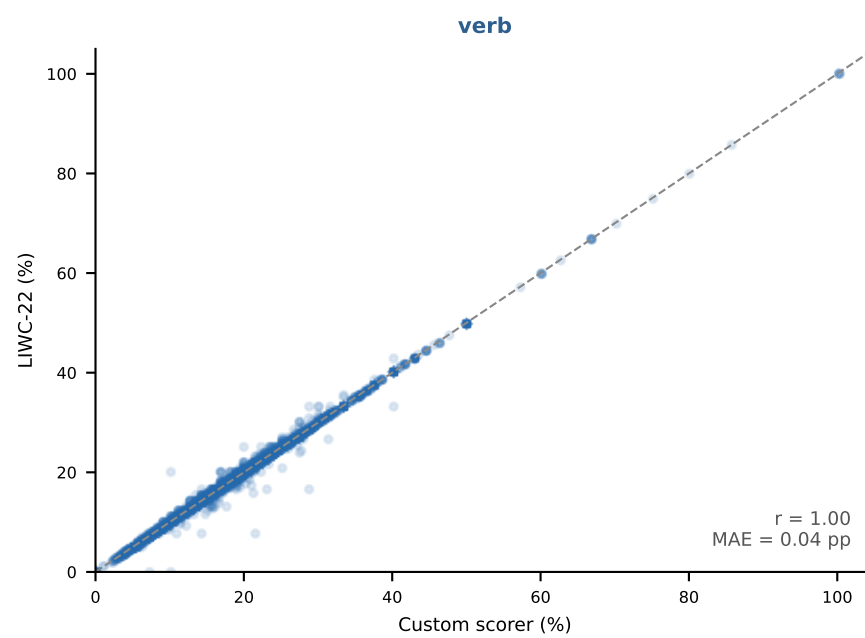
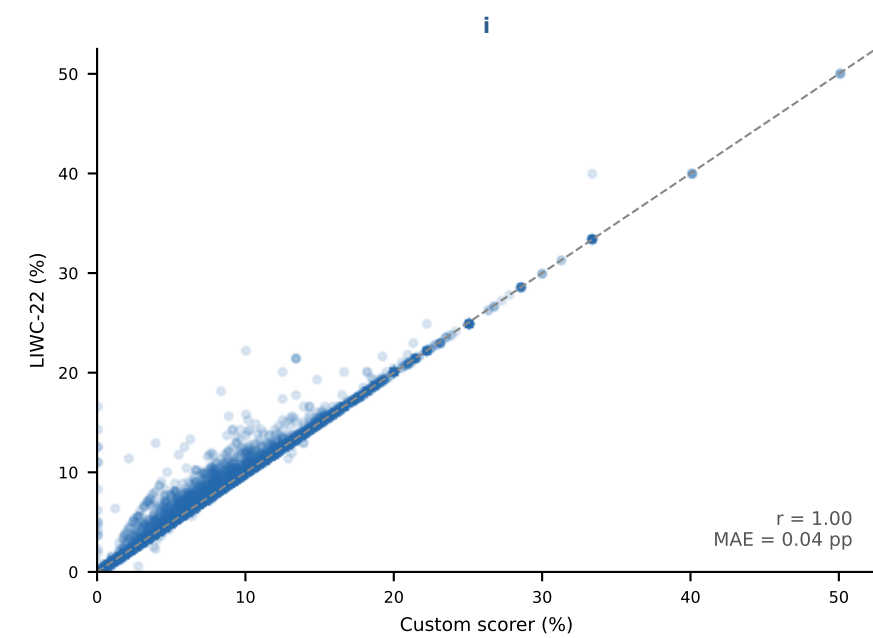
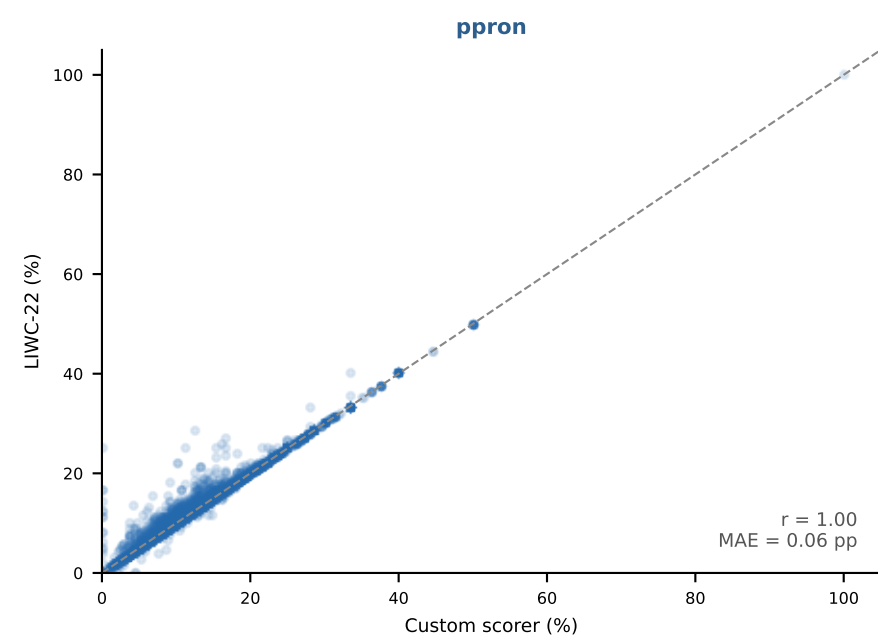
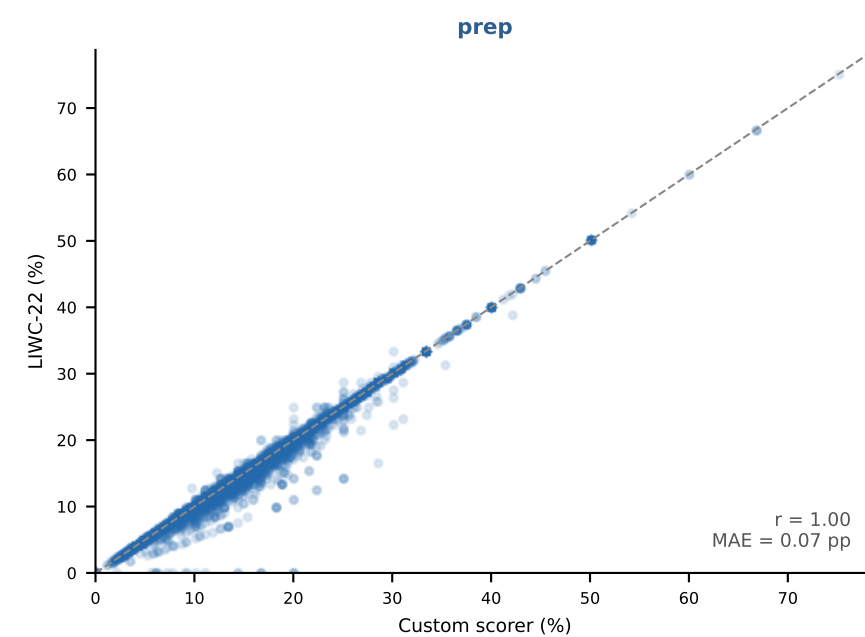
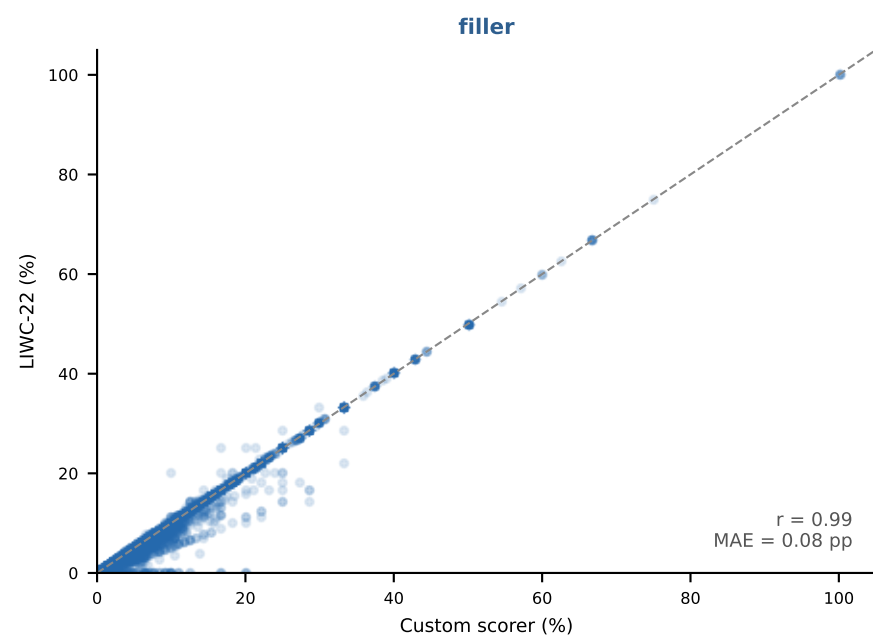
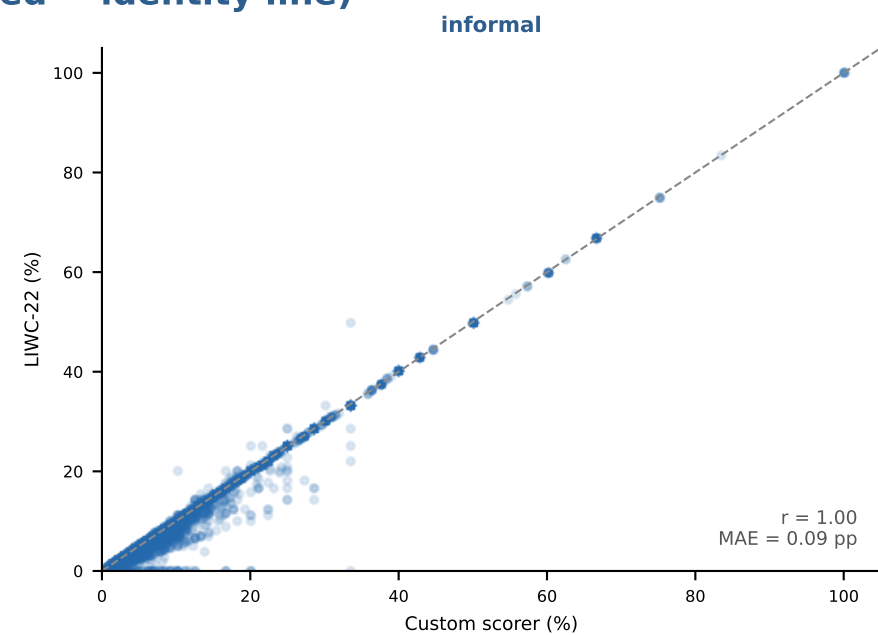
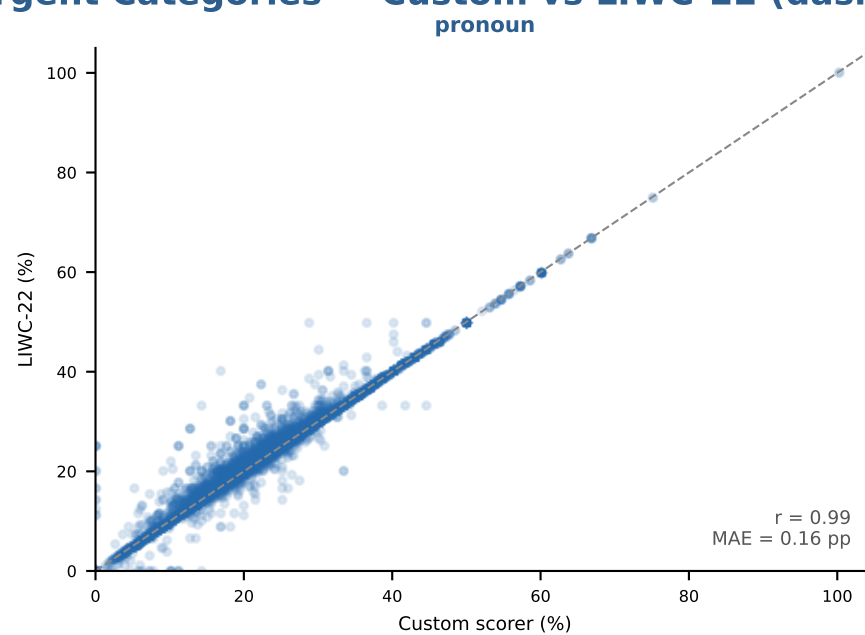
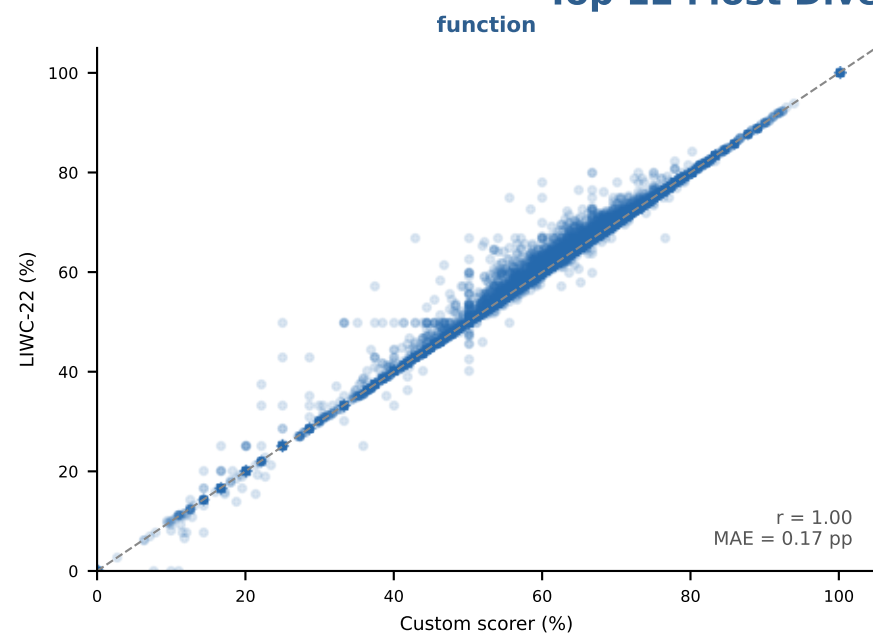
Category	Pearson r	p-value	MAE (pp)	Mean (custom)	Mean (LIWC-22)
function	0.9972	0.0	0.1709	62.1421	62.2935
pronoun	0.995	0.0	0.1552	20.9227	21.0426
informal	0.9954	0.0	0.0918	6.0867	6.0106
filler	0.9947	0.0	0.0843	5.026	4.9552
prep	0.9972	0.0	0.0738	13.2631	13.2271
ppron	0.9969	0.0	0.0649	11.0202	11.0802
i	0.9975	0.0	0.0448	5.6928	5.7352
verb	0.9995	0.0	0.0386	16.7093	16.7365
cogproc	0.9997	0.0	0.0346	15.6491	15.6758
relativ	0.9995	0.0	0.0346	13.2446	13.2639
focuspresent	0.9996	0.0	0.0332	14.5736	14.598
article	0.9974	0.0	0.0318	6.861	6.8904
adverb	0.9997	0.0	0.0313	13.3743	13.3997
social	0.9995	0.0	0.0275	10.1251	10.1345
drives	0.9988	0.0	0.0269	5.6738	5.6695
adj	0.9994	0.0	0.0236	9.0211	9.0333
affect	0.999	0.0	0.023	5.934	5.9343
conj	0.9998	0.0	0.0196	8.1537	8.1686
time	0.9994	0.0	0.0184	6.2095	6.2181
auxverb	0.9998	0.0	0.0182	7.3923	7.4062
ipron	0.9997	0.0	0.0161	7.0006	7.0133
space	0.9996	0.0	0.0157	5.5364	5.5457
affiliation	0.9987	0.0	0.0152	2.1266	2.1202
work	0.9969	0.0	0.0151	1.3373	1.3286
focuspast	0.9996	0.0	0.0143	5.2755	5.2844
negemo	0.9988	0.0	0.0133	2.1777	2.1738
differ	0.9997	0.0	0.0129	4.3281	4.334
netspeak	0.9955	0.0	0.0122	0.9805	0.9727
shehe	0.9873	0.0	0.0116	0.5862	0.5969
percept	0.999	0.0	0.0113	2.4276	2.4274
posemo	0.9994	0.0	0.0112	3.7066	3.7109
tentat	0.9996	0.0	0.0104	3.2786	3.2864
compare	0.9997	0.0	0.0101	4.0943	4.1016
male	0.9969	0.0	0.01	0.9242	0.9186
bio	0.9989	0.0	0.0099	1.6804	1.6789
insight	0.9998	0.0	0.0091	3.3761	3.3813
you	0.9999	0.0	0.0087	4.0579	4.064
quant	0.9996	0.0	0.0084	2.7349	2.7388
focusfuture	0.9998	0.0	0.0081	3.2863	3.2919
power	0.9986	0.0	0.0073	1.599	1.5999
leisure	0.9981	0.0	0.0068	0.599	0.5956
see	0.9986	0.0	0.0064	0.758	0.7554
discrep	0.9998	0.0	0.0063	2.4725	2.4766
achieve	0.9991	0.0	0.0063	1.7073	1.7093
interrog	0.9997	0.0	0.0062	2.2165	2.2209
negate	0.9997	0.0	0.0059	1.897	1.8994
number	0.9952	0.0	0.0057	0.6088	0.6064
motion	0.9997	0.0	0.0057	1.9213	1.9237
health	0.998	0.0	0.0053	0.8045	0.8027
cause	0.9999	0.0	0.005	1.7037	1.7063
certain	0.9999	0.0	0.0049	1.7908	1.794
reward	0.9992	0.0	0.0048	1.3028	1.3039
risk	0.9984	0.0	0.004	0.5806	0.5795
sad	0.9993	0.0	0.0039	0.961	0.9618
anx	0.9983	0.0	0.0035	0.4898	0.4885
family	0.9955	0.0	0.0031	0.3275	0.326
ingest	0.9991	0.0	0.0029	0.4692	0.4692
feel	0.9992	0.0	0.0028	0.8092	0.8102
body	0.9972	0.0	0.0028	0.4164	0.4162
friend	0.9929	0.0	0.0026	0.2506	0.2496
female	0.9984	0.0	0.0022	0.4911	0.4913
hear	0.9998	0.0	0.0021	0.5536	0.5545
anger	0.9988	0.0	0.0017	0.2987	0.2988
home	0.9994	0.0	0.0015	0.2842	0.2841
money	0.9987	0.0	0.0014	0.2422	0.2422
we	0.9996	0.0	0.0014	0.3343	0.3345
they	0.9999	0.0	0.0013	0.3904	0.3911
assent	1.0	0.0	0.001	0.3462	0.3467
relig	0.9973	0.0	0.0009	0.0728	0.0723
death	0.9998	0.0	0.0004	0.0853	0.0854
swear	0.9998	0.0	0.0003	0.0461	0.0461
nonflu	0.9999	0.0	0.0002	0.0498	0.0498
sexual	0.9995	0.0	0.0001	0.0172	0.0172

Per-Category Pearson r — Custom Scorer vs LIWC-22



Section 2 — Divergence Analysis

Top 12 Most Divergent Categories — Custom vs LIWC-22 (dashed = identity line)



Section 3 — LIWC-22 Summary Variables

LIWC-22 Summary Variables — Not Available with External Dictionary

The four LIWC-22 summary variables are not available in this run:

Analytic · Clout · Authentic · Tone

These scores are computed by LIWC-22 from a proprietary regression model trained on the built-in English dictionary. They do not generalise to external dictionary files, so LIWC-22 omits them when a .dicx file is supplied (as here, with the Dutch LIWC-2015 dictionary).

To obtain these variables you would need to run LIWC-22 with its built-in English dictionary (-d LIWC22), which is not appropriate for Dutch text.

See docs/statistical_decisions.md §5 for context.

Section 4 — Category Coverage Differences

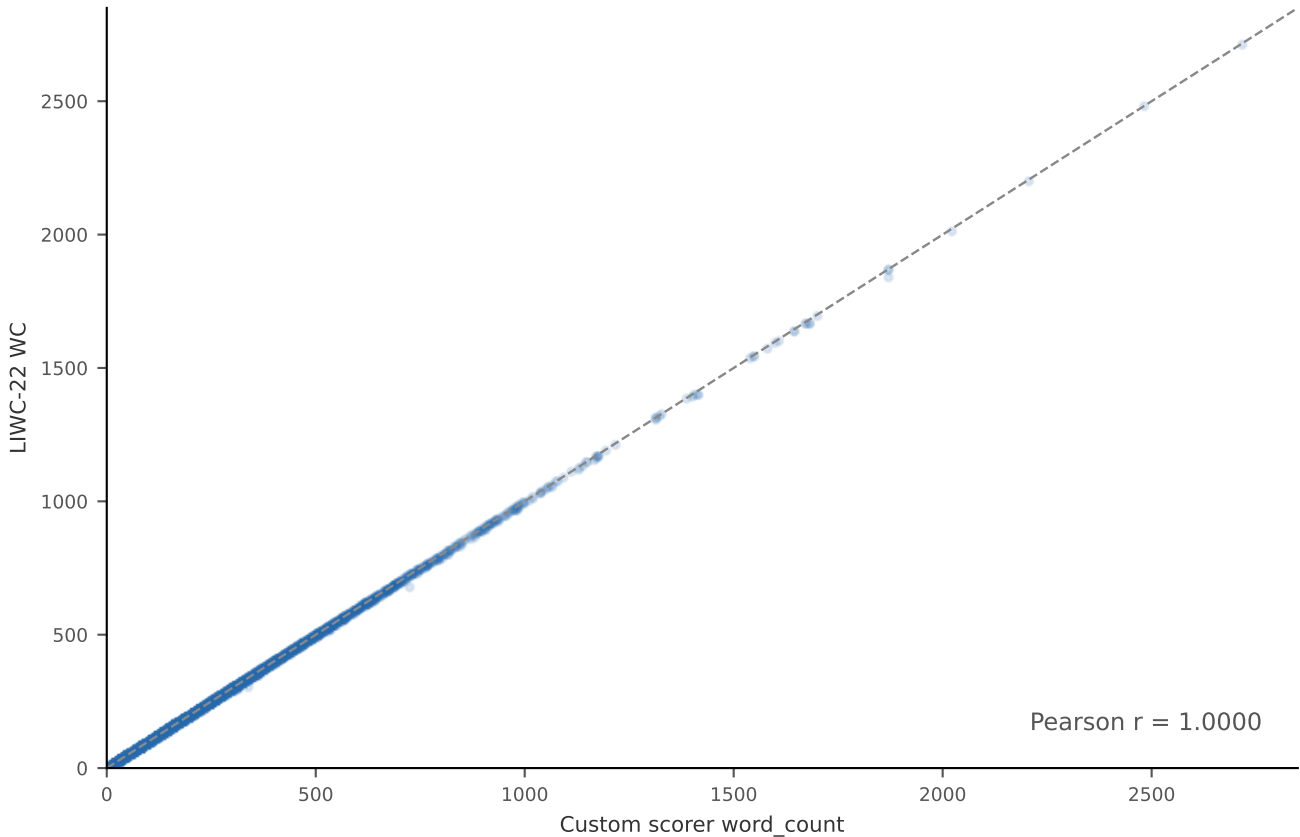
Category Coverage Differences

(custom-only includes fps_dutch fallback; LIWC-22-only includes structural metrics like WC, Emoji, AllPunc)

Custom scorer only	LIWC-22 only
—	—

Section 5 — Diagnostics

Tokenisation Sanity Check — Word Count Agreement (custom scorer re.findall vs LIWC-22 WC; dashed = identity)



8. Pandemic-Period Comparison

Pandemic-Period Comparison

Psycholinguistic markers pre / during / post pandemic
after Yahya & Abdul Rahim (2023), with per-user rank-based statistics

Dataset: Combined Dataset

Section 1 — Diagnostics

Period Definitions

```
pre      : PostDate < 2020-03-11
during   : 2020-03-11 ≤ PostDate < 2022-03-23
post     : PostDate ≥ 2022-03-23
```

Cutoff = WHO pandemic declaration (2020-03-11).

End = end of most Dutch COVID restrictions (2022-03-23).

Dataset Variant × Pandemic Period — messages and unique users
⚠ **Period effects are confounded with dataset variant (old export ≈ pre+during, new export ≈ post); yellow = near-empty cell**

variant	period	messages	users
old	pre	4505	169
old	during	11102	291
old	post	2943	96
new_only	pre	94	15
new_only	during	318	23
new_only	post	25092	520
combined	pre	4910	178
combined	during	12036	307
combined	post	28235	604

Users by Number of Periods Appeared In
Users in >1 period violate between-group independence — see the single-period sensitivity analysis in Section 4

periods_appeared_in	n_users	pct_users
1	816	86.8
2	99	10.5
3	25	2.7

Section 2 — Dictionary Introspection

Yahya & Abdul Rahim (2023) §2.3 Categories vs Available Dictionaries
Only available categories are analyzed; '—' = genuinely absent from that source (omission, not a bug)

yahya_group	liwc2015	liwc22	dic_crosscheck
pronouns	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you
function_words	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant
other_grammar	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb
tense	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent
emotion	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad
informal	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear
summary	—	—	—
absolutist	absolutist	—	—

Section 3 — Main Results (all users)

Kruskal-Wallis Omnibus per Feature (per-user rates)
effect_size = epsilon-squared; p_bh = BH over pooled omnibus + post-hoc family; yellow = significant

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
conj_liwc22	178	307	604	omnibus	7.0511	0.029436	0.0065		0.224076	False
anger_liwc2015	178	307	604	omnibus	7.0534	0.029402	0.0065		0.224076	False
shehe_liwc22	178	307	604	omnibus	7.2031	0.027282	0.0066		0.224076	False
pronoun_liwc22	178	307	604	omnibus	8.6231	0.013413	0.0079		0.224076	False
anger_liwc22	178	307	604	omnibus	7.0119	0.030018	0.0064		0.224076	False
swear_liwc2015	178	307	604	omnibus	7.3713	0.025081	0.0068		0.224076	False
shehe_liwc2015	178	307	604	omnibus	7.1465	0.028064	0.0066		0.224076	False
conj_liwc2015	178	307	604	omnibus	6.9673	0.030695	0.0064		0.224076	False
swear_liwc22	178	307	604	omnibus	7.4067	0.02464	0.0068		0.224076	False
pronoun_liwc2015	178	307	604	omnibus	9.0926	0.010606	0.0084		0.224076	False
prep_liwc2015	178	307	604	omnibus	6.3734	0.041308	0.0059		0.23096	False
prep_liwc22	178	307	604	omnibus	6.4551	0.039655	0.0059		0.23096	False
interrog_liwc2015	178	307	604	omnibus	6.2908	0.04305	0.0058		0.23096	False
interrog_liwc22	178	307	604	omnibus	6.2338	0.044294	0.0057		0.23096	False
ppron_liwc2015	178	307	604	omnibus	5.7635	0.056035	0.0053		0.272706	False
ppron_liwc22	178	307	604	omnibus	5.5873	0.061198	0.0051		0.279217	False
posemo_liwc22	178	307	604	omnibus	5.226	0.073313	0.0048		0.306788	False
posemo_liwc2015	178	307	604	omnibus	5.1634	0.075646	0.0047		0.306788	False
they_liwc2015	178	307	604	omnibus	4.6873	0.095975	0.0043		0.350308	False
they_liwc22	178	307	604	omnibus	4.7452	0.093236	0.0044		0.350308	False
function_liwc22	178	307	604	omnibus	4.3312	0.114679	0.004		0.398645	False
function_liwc2015	178	307	604	omnibus	4.0845	0.129734	0.0038		0.430481	False
anx_liwc22	178	307	604	omnibus	3.55	0.169484	0.0033		0.503942	False
informal_liwc22	178	307	604	omnibus	3.633	0.162595	0.0033		0.503942	False
anx_liwc2015	178	307	604	omnibus	3.5138	0.172583	0.0032		0.503942	False
negemo_liwc22	178	307	604	omnibus	3.3205	0.190094	0.0031		0.518384	False
informal_liwc2015	178	307	604	omnibus	3.3033	0.191731	0.003		0.518384	False
negemo_liwc2015	178	307	604	omnibus	3.1895	0.202962	0.0029		0.51949	False
netspeak_liwc2015	178	307	604	omnibus	3.1561	0.206373	0.0029		0.51949	False
assent_liwc22	178	307	604	omnibus	2.8902	0.235728	0.0027		0.537753	False
assent_liwc2015	178	307	604	omnibus	2.9068	0.233773	0.0027		0.537753	False
netspeak_liwc22	178	307	604	omnibus	2.8942	0.23525	0.0027		0.537753	False
quant_liwc22	178	307	604	omnibus	1.9081	0.385184	0.0018		0.666614	False
adverb_liwc22	178	307	604	omnibus	1.7295	0.421165	0.0016		0.666614	False
i_liwc22	178	307	604	omnibus	1.6605	0.435948	0.0015		0.666614	False
auxverb_liwc22	178	307	604	omnibus	2.0638	0.356322	0.0019		0.666614	False
article_liwc22	178	307	604	omnibus	1.5434	0.462236	0.0014		0.666614	False
adverb_liwc2015	178	307	604	omnibus	1.7753	0.411617	0.0016		0.666614	False
ipron_liwc22	178	307	604	omnibus	1.6195	0.444961	0.0015		0.666614	False
verb_liwc2015	178	307	604	omnibus	1.8679	0.392992	0.0017		0.666614	False
sad_liwc22	178	307	604	omnibus	2.2337	0.32731	0.0021		0.666614	False
focuspresent_liwc2015	178	307	604	omnibus	1.5217	0.467266	0.0014		0.666614	False
auxverb_liwc2015	178	307	604	omnibus	1.9395	0.379173	0.0018		0.666614	False
sad_liwc2015	178	307	604	omnibus	2.2995	0.316719	0.0021		0.666614	False
quant_liwc2015	178	307	604	omnibus	1.9175	0.383371	0.0018		0.666614	False
verb_liwc22	178	307	604	omnibus	2.0223	0.363802	0.0019		0.666614	False
focuspast_liwc22	178	307	604	omnibus	1.9049	0.385798	0.0018		0.666614	False
focuspresent_liwc22	178	307	604	omnibus	1.7793	0.410797	0.0016		0.666614	False
ipron_liwc2015	178	307	604	omnibus	1.5088	0.470301	0.0014		0.666614	False
focuspast_liwc2015	178	307	604	omnibus	1.9709	0.373276	0.0018		0.666614	False
i_liwc2015	178	307	604	omnibus	1.7141	0.424422	0.0016		0.666614	False
absolutist_liwc2015	178	307	604	omnibus	1.4895	0.474848	0.0014		0.666614	False
article_liwc2015	178	307	604	omnibus	1.4049	0.495362	0.0013		0.68229	False
focusfuture_liwc22	178	307	604	omnibus	1.2197	0.543439	0.0011		0.690846	False
filler_liwc22	178	307	604	omnibus	1.212	0.545532	0.0011		0.690846	False
you_liwc22	178	307	604	omnibus	1.2866	0.525566	0.0012		0.690846	False
focusfuture_liwc2015	178	307	604	omnibus	1.1997	0.548891	0.0011		0.690846	False
you_liwc2015	178	307	604	omnibus	1.2262	0.541664	0.0011		0.690846	False
number_liwc22	178	307	604	omnibus	1.1291	0.568627	0.001		0.703555	False
negate_liwc22	178	307	604	omnibus	0.9892	0.609826	0.0009		0.718021	False
filler_liwc2015	178	307	604	omnibus	1.0208	0.600258	0.0009		0.718021	False
negate_liwc2015	178	307	604	omnibus	1.0117	0.602981	0.0009		0.718021	False
compare_liwc2015	178	307	604	omnibus	0.9411	0.624646	0.0009		0.719842	False
compare_liwc22	178	307	604	omnibus	0.9206	0.631094	0.0008		0.719842	False
number_liwc2015	178	307	604	omnibus	0.6825	0.710872	0.0006		0.798364	False
adj_liwc22	178	307	604	omnibus	0.5714	0.7515	0.0005		0.825417	False
adj_liwc2015	178	307	604	omnibus	0.5553	0.757574	0.0005		0.825417	False
nonflu_liwc2015	178	307	604	omnibus	0.364	0.8336	0.0003		0.894894	False
nonflu_liwc22	178	307	604	omnibus	0.3185	0.852768	0.0003		0.902204	False
we_liwc22	178	307	604	omnibus	0.1036	0.949505	0.0001		0.976252	False
we_liwc2015	178	307	604	omnibus	0.1198	0.941854	0.0001		0.976252	False
affect_liwc22	178	307	604	omnibus	0.0157	0.992205	0.0		0.994621	False
affect_liwc2015	178	307	604	omnibus	0.0108	0.994621	0.0		0.994621	False

No features survived omnibus BH correction — no post-hoc tests run

Section 4 — Sensitivity: Single-Period Users Only

Same Design, Restricted to Users Appearing in Exactly One Period
Removes the between-group dependence of multi-period users; compare significance patterns with Section 3

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
netspeak_liwc22	92.0	185.0	539.0	omnibus	13.4567	0.001197	0.0165		0.01361	True
netspeak_liwc2015	92.0	185.0	539.0	omnibus	14.0036	0.00091	0.0172		0.01361	True
pronoun_liwc22	92.0	185.0	539.0	omnibus	13.613	0.001107	0.0167		0.01361	True
pronoun_liwc2015	92.0	185.0	539.0	omnibus	13.9272	0.000946	0.0171		0.01361	True
ppron_liwc2015	92.0	185.0	539.0	omnibus	11.9042	0.0026	0.0146		0.022876	True
ppron_liwc22	92.0	185.0	539.0	omnibus	11.1568	0.003779	0.0137		0.028654	True
informal_liwc22	92.0	185.0	539.0	omnibus	9.1848	0.010128	0.0113		0.058764	False
informal_liwc2015	92.0	185.0	539.0	omnibus	8.3553	0.015335	0.0103		0.077526	False
swear_liwc22	92.0	185.0	539.0	omnibus	7.9828	0.018474	0.0098		0.083829	False
swear_liwc2015	92.0	185.0	539.0	omnibus	7.957	0.018714	0.0098		0.083829	False
i_liwc2015	92.0	185.0	539.0	omnibus	7.8906	0.019345	0.0097		0.083829	False
i_liwc22	92.0	185.0	539.0	omnibus	7.4811	0.023741	0.0092		0.096282	False
function_liwc22	92.0	185.0	539.0	omnibus	6.5625	0.037582	0.0081		0.136798	False
function_liwc2015	92.0	185.0	539.0	omnibus	6.2576	0.043771	0.0077		0.153199	False
absolutist_liwc2015	92.0	185.0	539.0	omnibus	5.7103	0.057547	0.007		0.193954	False
conj_liwc2015	92.0	185.0	539.0	omnibus	5.1331	0.076802	0.0063		0.230457	False
conj_liwc22	92.0	185.0	539.0	omnibus	5.2697	0.071731	0.0065		0.230457	False
interrog_liwc2015	92.0	185.0	539.0	omnibus	5.0891	0.078507	0.0062		0.230457	False
interrog_liwc22	92.0	185.0	539.0	omnibus	5.1009	0.078048	0.0063		0.230457	False
ipron_liwc22	92.0	185.0	539.0	omnibus	4.158	0.125054	0.0051		0.307	False
article_liwc22	92.0	185.0	539.0	omnibus	4.1655	0.124589	0.0051		0.307	False
adj_liwc22	92.0	185.0	539.0	omnibus	4.1883	0.123174	0.0051		0.307	False
shehe_liwc2015	92.0	185.0	539.0	omnibus	4.1506	0.125521	0.0051		0.307	False
shehe_liwc22	92.0	185.0	539.0	omnibus	4.2427	0.119867	0.0052		0.307	False
adj_liwc2015	92.0	185.0	539.0	omnibus	4.2649	0.118549	0.0052		0.307	False
filler_liwc22	92.0	185.0	539.0	omnibus	4.1084	0.128198	0.005		0.307	False
article_liwc2015	92.0	185.0	539.0	omnibus	4.0198	0.134002	0.0049		0.312672	False
assent_liwc22	92.0	185.0	539.0	omnibus	3.8613	0.145057	0.0047		0.314291	False
assent_liwc2015	92.0	185.0	539.0	omnibus	3.8874	0.143176	0.0048		0.314291	False
ipron_liwc2015	92.0	185.0	539.0	omnibus	3.8776	0.143875	0.0048		0.314291	False
filler_liwc2015	92.0	185.0	539.0	omnibus	3.4577	0.177492	0.0042		0.375623	False
number_liwc22	92.0	185.0	539.0	omnibus	3.1347	0.208601	0.0038		0.421838	False
anger_liwc2015	92.0	185.0	539.0	omnibus	2.7948	0.247244	0.0034		0.46299	False
anger_liwc22	92.0	185.0	539.0	omnibus	2.7782	0.249302	0.0034		0.46299	False
number_liwc2015	92.0	185.0	539.0	omnibus	2.8368	0.242104	0.0035		0.46299	False
prep_liwc2015	92.0	185.0	539.0	omnibus	2.423	0.297751	0.003		0.491876	False
focuspast_liwc2015	92.0	185.0	539.0	omnibus	2.4878	0.288254	0.0031		0.491876	False
focusfuture_liwc2015	92.0	185.0	539.0	omnibus	2.3924	0.30234	0.0029		0.491876	False
focusfuture_liwc22	92.0	185.0	539.0	omnibus	2.4431	0.294772	0.003		0.491876	False
focuspast_liwc22	92.0	185.0	539.0	omnibus	2.3901	0.302693	0.0029		0.491876	False
prep_liwc22	92.0	185.0	539.0	omnibus	2.2944	0.317525	0.0028		0.506926	False
nonflu_liwc22	92.0	185.0	539.0	omnibus	2.1424	0.342598	0.0026		0.511088	False
nonflu_liwc2015	92.0	185.0	539.0	omnibus	2.1427	0.342554	0.0026		0.511088	False
negate_liwc22	92.0	185.0	539.0	omnibus	2.2023	0.332491	0.0027		0.511088	False
negate_liwc2015	92.0	185.0	539.0	omnibus	2.2139	0.330561	0.0027		0.511088	False
quant_liwc22	92.0	185.0	539.0	omnibus	1.9415	0.378791	0.0024		0.555967	False
quant_liwc2015	92.0	185.0	539.0	omnibus	1.795	0.407579	0.0022		0.588726	False
you_liwc22	92.0	185.0	539.0	omnibus	1.7358	0.419839	0.0021		0.596958	False
you_liwc2015	92.0	185.0	539.0	omnibus	1.7018	0.427024	0.0021		0.597833	False
anx_liwc22	92.0	185.0	539.0	omnibus	1.4169	0.492415	0.0017		0.653183	False
posemo_liwc22	92.0	185.0	539.0	omnibus	1.491	0.474501	0.0018		0.653183	False
auxverb_liwc22	92.0	185.0	539.0	omnibus	1.4053	0.495271	0.0017		0.653183	False
posemo_liwc2015	92.0	185.0	539.0	omnibus	1.458	0.482391	0.0018		0.653183	False
auxverb_liwc2015	92.0	185.0	539.0	omnibus	1.3425	0.511061	0.0016		0.66438	False
they_liwc22	92.0	185.0	539.0	omnibus	1.2839	0.526267	0.0016		0.664645	False
they_liwc2015	92.0	185.0	539.0	omnibus	1.2578	0.533177	0.0015		0.664645	False
anx_liwc2015	92.0	185.0	539.0	omnibus	1.229	0.540924	0.0015		0.665191	False
adverb_liwc2015	92.0	185.0	539.0	omnibus	1.075	0.584207	0.0013		0.708838	False
adverb_liwc22	92.0	185.0	539.0	omnibus	0.9879	0.610226	0.0012		0.730665	False
negemo_liwc22	92.0	185.0	539.0	omnibus	0.7529	0.686296	0.0009		0.800679	False
negemo_liwc2015	92.0	185.0	539.0	omnibus	0.6432	0.724991	0.0008		0.835116	False
verb_liwc22	92.0	185.0	539.0	omnibus	0.6007	0.740574	0.0007		0.842403	False
we_liwc22	92.0	185.0	539.0	omnibus	0.5711	0.751594	0.0007		0.844384	False
we_liwc2015	92.0	185.0	539.0	omnibus	0.5077	0.775819	0.0006		0.86097	False
verb_liwc2015	92.0	185.0	539.0	omnibus	0.4796	0.786782	0.0006		0.862616	False
affect_liwc2015	92.0	185.0	539.0	omnibus	0.4319	0.805793	0.0005		0.865519	False
affect_liwc22	92.0	185.0	539.0	omnibus	0.4253	0.808451	0.0005		0.865519	False
focuspresent_liwc22	92.0	185.0	539.0	omnibus	0.3995	0.818949	0.0005		0.866562	False
compare_liwc2015	92.0	185.0	539.0	omnibus	0.2867	0.866469	0.0004		0.886659	False
compare_liwc22	92.0	185.0	539.0	omnibus	0.3035	0.859187	0.0004		0.886659	False
focuspresent_liwc2015	92.0	185.0	539.0	omnibus	0.285	0.867172	0.0003		0.886659	False
sad_liwc2015	92.0	185.0	539.0	omnibus	0.0648	0.968108	0.0001		0.971362	False
sad_liwc22	92.0	185.0	539.0	omnibus	0.0581	0.971362	0.0001		0.971362	False